



LIVING GUIDELINE FOR DIAGNOSING AND MANAGING PEDIATRIC CONCUSSION

Reed, N.*, Zemek, R.*, Dawson, J., Ledoux, AA., et al. (2021). Living Guideline for Diagnosing and Managing Pediatric Concussion. Toronto, ON: Ontario Neurotrauma Foundation
<https://doi.org/10.17605/OSF.IO/3VWN9>

* These authors contributed equally.

Last Updated: March 1, 2021



Ontario Neurotrauma Foundation

Fondation ontarienne de neurotraumatologie



TABLE OF CONTENTS

Disclaimer.....	2
<u>Guideline Overview</u>	3
<u>List of Recommendations</u>	4
<u>Recommendations</u>	12
List of <u>Tools</u>	55
<u>References</u>	56

DISCLAIMER:

The recommendations and resources found within the Living Guideline for Diagnosing and Managing Pediatric Concussion are intended to inform and instruct care providers and other stakeholders who deliver services to children and youth who have sustained or are suspected of having sustained a concussion. This guideline is not intended for use with patients or clients over the age of 18 years. It is not for self-diagnosis or treatment. Patient, parents, and/or caregivers may wish to bring their healthcare and other providers' attention to this guideline.

The best available evidence has informed the recommendations in this document, and relevant evidence published after this guideline could influence the recommendations made within. Healthcare professionals should also consider their own judgment, the preferences of their patients, and factors such as the availability of resources in their decisions.

The Ontario Neurotrauma Foundation, the project team, and any developers, contributors, and supporting partners shall not be liable for any damages, claims, liabilities, costs, or obligations arising from the use or misuse of this guideline including loss or damage arising from any claims made by a third party. Also, as the sponsor of this document, the Ontario Neurotrauma Foundation assumes no responsibility or liability whatsoever for changes made to the guideline without its consent. Any changes must be accompanied by the statement: "Adapted from Guideline for Diagnosing and Managing Pediatric Concussion with/ without permission," according to whether or not permission was sought and/or given.

Date of last update: March 2021



Ontario Neurotrauma Foundation

Fondation ontarienne de neurotraumatologie

Ontario Neurotrauma Foundation
90 Eglinton Avenue East, Suite 601
Toronto, Ontario, Canada, M4P 2Y3
Tel: (416) 422-2228
Twitter: @OntNeurotrauma
Email: info@onf.org
Website: www.onf.org

GUIDELINE OVERVIEW

Section A - Concussion Recognition, Initial Medical Assessment, and Management

Domain 1. Concussion Recognition and Directing to Care

Domain 2. Initial Medical Assessment and Management

Domain 3. Medical Follow-Up and Management of Prolonged Symptoms

Domain 4. Medical Clearance for Full-Contact Sport or High-Risk Activity

Domain 5. Sport Concussion Considerations

Domain 15. Telemedicine and Virtual Concussion Care

Section B - Managing Concussion Symptoms

Domain 6. Headache

Domain 7. Sleep

Domain 8. Mental Health

Domain 9. Cognition

Domain 10. Vision, Vestibular, and Oculomotor Function

Domain 11. Fatigue

Domain 12. Return-to-School and Work

Section C - Additional information

Domain 13. Biomarkers - Neuroimaging

Domain 14. Biomarkers - Serologic

LIST OF RECOMMENDATIONS

Domain 1. <u>Concussion Recognition and Directing to Care</u>	
1.1a	<u>School boards, sports organizations, and community centres should provide pre-season concussion education and conduct a review of all concussion policies in effect within the school or sport setting.</u>
1.1b	<u>School boards, sports organizations, and community centres should ensure updated policies are in place to recognize and accommodate a child/adolescent who has sustained a concussion.</u>
1.2	<u>Remove the child/adolescent from the activity immediately if a concussion is suspected for immediate assessment and to avoid another injury.</u>
1.3	<u>Recommend an emergency medical assessment for a child/adolescent with any of the “red flag” symptoms.</u>
1.4	<u>Concussion should be suspected and diagnosed as soon as possible to maintain health and improve outcomes. Concussion can be suspected in the community by healthcare professionals, parents, teachers, coaches, and peers. Those with a suspected concussion should be referred to a physician or nurse practitioner to perform a comprehensive medical assessment to exclude more severe injuries, consider a full differential diagnosis, and confirm the diagnosis of concussion.</u>
Domain 2. <u>Initial Medical Assessment and Management</u>	
2.1	<u>Physicians or nurse practitioners should perform a comprehensive medical assessment on all children/adolescents with a suspected concussion or with acute head or spine trauma.</u>
2.1a	<u>Take a comprehensive clinical history.</u>
2.1b	<u>Note common modifiers that may delay recovery and use a clinical risk score to predict risk of prolonged symptoms.</u>
2.1c	<u>Perform a comprehensive physical examination.</u>
2.1d	<u>Consider CT of the brain or cervical spine only in patients with acute head trauma in whom, after a medical assessment, a structural intracranial or cervical spine injury is suspected; do not conduct routine neuroimaging for the purpose of diagnosing concussion.</u>

2.2	<u>Provide verbal information and written (electronic) handouts regarding the course of recovery and when the child/adolescent can return-to-school/activity/sport/work.</u>
2.3	<u>Recommend graduated return to cognitive and physical activity to promote recovery.</u>
2.3a	<u>Recommend an initial 24-48 hour period of rest with limited physical and cognitive activity.</u>
2.3b	<u>Recommend that low to moderate level physical and cognitive activity be gradually started 24-48 hours after a concussion at a level that does not result in recurrence or exacerbation of symptoms. Activities that pose no/low risk of sustaining a concussion should be resumed even if mild residual symptoms are present or whenever acute symptoms improve sufficiently to permit activity.</u>
2.3c	<u>Recommend that patients avoid activities associated with a risk of contact, fall, or collisions such as high speed and/or contact activities and full-contact sport that may increase the risk of sustaining another concussion during the recovery period.</u>
2.3d	<u>Consider referring select patients (e.g., competitive athletes, those who are not tolerating a graduated return to physical activity, or those who are slow to recover) to a medically supervised interdisciplinary team with the ability to individually assess sub-symptom threshold aerobic exercise tolerance and to prescribe aerobic exercise treatment.</u>
2.4	<u>Provide education and guidance regarding strategies to promote recovery.</u>
2.4a	<u>Advise on the importance of sleep and discuss sleep hygiene.</u>
2.4b	<u>Advise on maintaining social networks and interactions as tolerated beyond a brief initial period of cognitive and physical rest (24-48 hours after injury).</u>
2.4c	<u>Advise on the use of computers, phones, and other devices with screens. Beyond an initial period of cognitive and physical rest (24-48 hours after injury), use of devices with screens may be gradually resumed at a level that does not result in recurrence or exacerbation of symptoms.</u>
2.4d	<u>Advise on avoiding alcohol and other recreational drugs after a concussion.</u>
2.4e	<u>Advise to avoid driving during the first 24-48 hours after a concussion. Advise patients to begin driving when they are feeling improved, can concentrate sufficiently to feel safe behind the wheel, and when the act of driving does not provoke significant concussion symptoms.</u>
2.5	<u>Over-the-counter medications such as acetaminophen and ibuprofen may be recommended to treat acute headache. Advise on limiting the use of these medications to less than 15 days a month and avoiding “around-the-clock” dosing to prevent overuse or rebound headaches (i.e., advise that children/adolescents avoid using over the counter medications at regular scheduled times throughout the day).</u>
2.6	<u>At present, there is limited evidence to support the administration of intravenous medication to treat acute headaches in pediatric concussion patients in the Emergency Department setting.</u>

2.7	After assessment, nearly all children/adolescents with concussion may be safely discharged from clinics and Emergency Departments for observation at home.
2.8	Recommend a medical follow-up in 1-2 weeks to re-assess and monitor clinical status. Recommend an immediate medical follow-up in the presence of any deterioration.
2.9	Consider referral to an interdisciplinary concussion team in the presence of modifiers that may delay recovery.
2.10	Provide post-concussion information and a written medical assessment to the child/adolescent and the parent/caregiver prior to sending the child/adolescent home.
Domain 3. Medical Follow-up and Management of Prolonged Symptoms	
3.1	Perform a repeat medical assessment on all patients presenting with post-concussion symptoms 1-2 weeks following acute injury.
3.1a	Take a focused clinical history based on symptoms described.
3.1b	Examine the child/adolescent and perform a focused physical examination.
3.1c	Recommendation 2.1d: Consider diagnostic brain or cervical spine MRI imaging for those with focal or worrisome symptoms.
3.2	Provide patients with general education and guidance that outlines mental health considerations, non-pharmacological strategies to minimize symptoms including sleep hygiene, activity modifications, limiting triggers, information on screen time, the importance of social interaction, and how to work with the school team to facilitate school success.
3.3	Encourage patients with post-concussion symptoms to engage in cognitive activity and low-risk physical activity as soon as tolerated while staying below their symptom-exacerbation thresholds. Activities that pose no/low risk of sustaining a concussion (no risk of contact, collision, or falling) should be resumed even if mild residual symptoms are present or whenever acute symptoms improve sufficiently to permit activity. Recommendation 2.3d: Consider referring select patients (e.g., competitive athletes, those who are not tolerating a graduated return to physical activity, or those who are slow to recover) to a medically supervised interdisciplinary team with the ability to individually assess sub-symptom threshold aerobic exercise tolerance and to prescribe aerobic exercise treatment.
3.4	Refer to specialized care with an interdisciplinary concussion team if post-concussion symptoms do not gradually resolve by 4 weeks.
3.5	Consider early referral (prior to 4-week post-injury) to an interdisciplinary concussion team in the presence of modifiers that may delay recovery.
3.6	Consider initiating treatment for specific symptoms or concerns while waiting for a referral to an interdisciplinary concussion team or sub-specialist.

3.7	Recommend regular medical follow-up if a child/adolescent is still experiencing post-concussion symptoms or has not completed the return-to-school or return-to-sport/activity stages. Recommend an immediate medical follow-up in the presence of any deterioration.
Domain 4. Medical Clearance for Full-Contact Sport or High-Risk Activity	
4.1	Consider patients for medical clearance to return to full-contact activities and sport/game play if clinical criteria have been met.
4.2	Provide patients with a letter indicating medical clearance to return to all activities when medically cleared.
4.3	Advise medically cleared patients to seek immediate medical attention if he or she develops new concussion-like symptoms or sustains a new suspected concussion.
Domain 5. Sport Concussion Considerations	
5.1	Refer a child/adolescent with multiple concussions or baseline conditions associated with concussion-like symptoms to an interdisciplinary concussion team to help with return to full-contact sports or high-risk activities, or retirement decisions from full-contact sports, or high-risk activities.
5.2	Baseline testing on children/adolescents using concussion assessment tools or tests (or any combination of tests/tools) is not recommended or required for concussion diagnosis or management following an injury.
5.3	Special considerations regarding baseline testing.
5.4	Recommendation 2.3d: Consider referring select patients (e.g., competitive athletes, those who are not tolerating a graduated return to physical activity, or those who are slow to recover) to a medically supervised interdisciplinary team with the ability to individually assess sub-symptom threshold aerobic exercise tolerance and to prescribe aerobic exercise treatment.
Domain 6. Headache	
6.1	Perform a repeat medical assessment on all patients presenting with post-concussion headaches 1-2 weeks following acute injury.
6.1a	Take a focused clinical history.
6.1b	Perform a focused physical examination.
6.1c	Consider diagnostic brain or cervical spine MRI imaging for those with focal or worrisome symptoms.
6.1d	Classify and characterize the headache subtype based on the clinical history and physical examination findings.

6.2	Provide general post-concussion education and guidance on headache management.
6.2a	Advise on non-pharmacological strategies to minimize headaches including sleep hygiene, activity modifications, limiting triggers, and information on screen time.
6.2b	Encourage patients with headaches to engage in cognitive activity and low-risk physical activity as soon as tolerated while staying below their symptom-exacerbation thresholds. Activities that pose no/low risk of sustaining a concussion (no risk of contact, collision, or falling) should be resumed even if mild residual symptoms are present or whenever acute symptoms improve sufficiently to permit activity.
6.2c	Consider suggesting the use of a headache and medication diary in order to monitor symptoms and medications taken. Use clinical judgment and an individualized approach on use or duration of this strategy.
6.2d	Over-the-counter medications such as acetaminophen and ibuprofen may be recommended to treat acute headache. Advise on limiting the use of these medications to less than 15 days a month and avoiding “around-the-clock” dosing to prevent overuse or rebound headaches. I.e., advise that children/adolescents avoid using over the counter medications at regular scheduled times throughout the day.
6.3	Refer patients who have prolonged post-concussion headaches for more than 4 weeks to an interdisciplinary concussion team or to a sub-specialist for further evaluation and management.
6.4	Consider initiating pharmacological therapy to treat and manage prolonged headaches while waiting for the interdisciplinary concussion team or sub-specialist referral.
6.5	Recommend a medical follow-up to reassess clinical status if headaches persist. Recommend an immediate medical follow-up in the presence of any deterioration. Consider early referral (prior to 4-weeks after the acute injury) to an interdisciplinary concussion team in the presence of modifiers that may delay recovery.
Domain 7. Sleep	
7.1	Perform a repeat medical assessment on all patients presenting with post-concussion sleep disturbances 1-2 weeks following acute injury.
7.2	Provide general education and guidance on sleep hygiene that outlines non-pharmacological strategies to improve sleep.
7.2a	Continue to encourage patients with sleep disturbances to engage in sub-symptom threshold cognitive activities and physical activities that pose no/low risk of sustaining a concussion (no risk of contact, collision, or falling) as soon as tolerated.
7.3	Consider managing patients who experience sleep-wake disturbances for more than 4 weeks with cognitive behavioural therapy, treat with daily supplements, and/or refer to an interdisciplinary concussion team.

7.4	Refer patients with prolonged post-concussion sleep disturbances (more than 6 weeks) to a sleep specialist or an interdisciplinary concussion team if the interventions introduced at 4 weeks have been unsuccessful and sleep issues persist.
7.5	Consider prescribing medication on a short-term basis if sleep has not improved after 6 weeks following the acute injury.
7.6	Recommend a medical follow-up to reassess clinical status if sleep disturbances persist. Recommend an immediate medical follow-up in the presence of any deterioration. Consider early referral (before 4 weeks) to an interdisciplinary concussion team in the presence of modifiers that may delay recovery.
Domain 8. Mental Health	
8.1	Assess existing and new mental health symptoms and disorders.
8.2	Assess the child/adolescent's broader environment, including family and caregiver function, mental health, and social connections.
8.3	Treat mental health symptoms or refer to a specialist in pediatric mental health.
Domain 9. Cognition	
9.1	Evaluate a child/adolescent for cognitive symptoms that interfere with daily functioning following the acute injury.
9.2	Manage cognitive symptoms that interfere with daily functioning for more than 4 weeks following acute injury.
Domain 10. Vision, Vestibular, and Oculomotor Function	
10.1	Perform a repeat medical assessment on all patients presenting with dizziness, blurred or double vision, vertigo, difficulty reading, postural imbalance, or headaches elicited by prolonged visual or vestibular stimulation 1-2 weeks following acute injury.
10.2	Screen for oculomotor or vision deficits.
10.3	Screen for benign paroxysmal positional vertigo (BPPV) if the patient reports vertigo or dizziness that occurs for seconds following position changes and consider targeted particle re-positioning manoeuvres.
10.4	Screen for vestibulo-ocular deficits.
10.5	Screen for balance deficits.
10.6	Screen for and consider underlying psychosocial contributors of vestibular, vision, and oculomotor dysfunction.

10.7	Provide general post-concussion education that outlines symptoms of concussion, provides suggestions regarding activity modification and includes academic accommodations to manage visual, vestibular and oculomotor symptoms.
10.8	Encourage patients with post-concussion vestibular, vision, or oculomotor symptoms to engage in cognitive activity and low-risk physical activity as soon as tolerated while staying below their symptom-exacerbation thresholds. Activities that pose no/low risk of sustaining a concussion (no risk of contact, collision, or falling) should be resumed even if mild residual symptoms are present or whenever acute symptoms improve sufficiently to permit activity.
10.9	Refer patients with prolonged post-concussion vestibular functioning, balance or visual dysfunction (more than 4 weeks following the acute injury) to an interdisciplinary concussion team with appropriate experience. Consider early referral (before 4 weeks) to an interdisciplinary concussion team in the presence of modifiers that may delay recovery.
10.10	Recommend a medical follow-up to reassess clinical status if vestibular functioning, balance or visual dysfunction symptoms persist. Recommend an immediate medical follow-up in the presence of any deterioration.
Domain 11. Fatigue	
11.1	Perform a repeat medical assessment on all patients presenting with post-concussion fatigue 1-2 weeks following acute injury.
11.2	Provide patients with post-concussion fatigue with general education and guidance that outlines non-pharmacological strategies to help cope with fatigue symptoms and set expectations.
11.3	Encourage patients with post-concussion fatigue to engage in cognitive activity and low-risk physical activity as soon as tolerated while staying below their symptom-exacerbation thresholds. Activities that pose no/low risk of sustaining a concussion (no risk of contact, collision, or falling) should be resumed even if mild residual symptoms are present or whenever acute symptoms improve sufficiently to permit activity.
11.4	Consider referral to an interdisciplinary concussion team for patients with prolonged post-concussion fatigue (more than 4 weeks following the acute injury) to learn pacing techniques.
11.5	Recommend a medical follow-up to reassess clinical status if fatigue symptoms persist. Recommend an immediate medical follow-up in the presence of any deterioration. Consider early referral (before 4 weeks) to an interdisciplinary concussion team in the presence of modifiers that may delay recovery.
Domain 12. Return-to-School and Work	
12.1	Recommend a stepwise return-to-school plan and monitor once the student is ready to start a graduated return-to-school. Include temporary accommodations based on symptoms and recommendations from the healthcare professional. Modify the return-to-school plan based on ongoing assessment of symptoms.

12.2	Assess for school difficulties using clinical judgment.
12.3	Manage school difficulties.
12.4	Encourage patients with school difficulties to engage in cognitive activity and low-risk physical activity as soon as tolerated while staying below their symptom-exacerbation thresholds. Activities that pose no/low risk of sustaining a concussion (no risk of contact, collision, or falling) should be resumed even if mild residual symptoms are present or whenever acute symptoms improve sufficiently to permit activity.
12.5	Return-to-school and return-to-sport strategies can be performed simultaneously. Recommend that the child/adolescent returns to school full-time at a full academic load, including writing exams without accommodations, before returning to full-contact sport or high-risk activities.
12.6	Prioritize return-to-school before return-to-work.
Domain 13. Biomarkers- Neuroimaging	
13.1	At this stage, advanced neuroimaging biomarkers are not yet ready for clinical implementation/management.
13.2	When conventional MRI is performed in the clinical management of concussion patients, the inclusion of susceptibility-weighted images (SWI) sequences could be considered as it may be useful for detecting small hemorrhages. The clinical significance of small hemorrhages on SWI is not clear at present.
Domain 14. Biomarkers- Serologic	
14.1	The use of serologic biomarkers is not clinically indicated. Presently there is no validated “concussion blood test” that can be used to accurately detect concussion in children/adolescents.
Domain 15. Considerations for Telemedicine and Virtual Concussion Care	
15.1	Virtual medical assessment
15.1a	Obtain informed consent from the patient and/or their parent/caregiver to conduct a remote medical assessment via telemedicine.
15.1b	Take a comprehensive virtual clinical history.
15.1c	See Recommendation 2.b “Note common modifiers that may delay recovery and use a clinical risk score to predict risk of prolonged symptoms.”
15.1d	Considerations for a virtual physical assessment
15.1e	Recommendation 2.1d: “Consider CT of the brain or cervical spine only in patients whom, after a medical assessment, a structural intracranial or cervical spine injury is suspected; do not

	<u>conduct routine neuroimaging for the purpose of diagnosing concussion.</u>
15.2	<u>Provide verbal information and written (electronic) handouts regarding the course of recovery and when the child/adolescent can return to school/activity/sport/driving/work and strategies to promote recovery.</u>
15.3	<u>Provide a written (electronic) medical assessment or clearance letter to the child/adolescent and the parent/caregiver.</u>
15.4	<u>Provide a medical follow-up assessment on children/adolescents with concussion in 1-2 weeks to re-assess and monitor clinical status. A regular medical follow-up is also recommended if a child/adolescent is still experiencing post-concussion symptoms or has not completed the return-to-school or return-to-sport/activity stages. Recommend immediate in-person medical attention in the presence of any deterioration.</u> Please refer to Domain 3: Medical follow-up and Management of Prolonged Symptoms for a full list of clinical guideline recommendations to consider when performing a medical follow-up assessment and managing prolonged concussion symptoms.
15.5	<u>Please refer to Domain 4: Medical Clearance for full-contact sports and high-risk activities for a full list of clinical recommendations to consider when determining if a child/adolescent is ready to return to full-contact sports or high-risk activities. Medical clearance is not required to return to school.</u>
15.6	<u>Recommendations for inter-disciplinary healthcare professionals involved in the care of concussion patients.</u>

RECOMMENDATIONS

Section A: Concussion

Recognition, Initial Medical Assessment, and Management

Introduction:

Concussion is a condition that exists along a clinical-pathological spectrum of traumatic brain injury (TBI) and can result in non-specific physical, cognitive, sleep, and emotional symptoms. To provide a comprehensive medical assessment in a patient with acute head and neck trauma, a more severe form of TBI, cervical spine injury, medical conditions, mental health conditions, and neurological conditions that can present with non-specific neurological symptoms including red flags must be ruled out. This requires a comprehensive clinical history, a comprehensive physical examination, ordering and interpreting necessary diagnostic tests, and initiating referrals to medical specialists and healthcare professionals as needed. Patients diagnosed with a concussion must be provided with education and guidance on how to manage their symptoms and how to make a safe and graduate return-to-school/activity/sport and work.

The majority of children and adolescents with an acute concussion will experience symptom resolution and a return to daily activities within 1-4 weeks'. Those with prolonged symptoms (symptoms that last more than 4 weeks following the acute injury) can benefit from a referral to interdisciplinary teams and experienced healthcare professionals who are optimally trained to evaluate and treat the heterogeneous causes of these symptoms. When providing medical clearance to return to activities with a risk of future concussion, patients should be managed on an individualized basis. Interdisciplinary concussion teams may be recommended to manage recovery in situations in which patients have prolonged symptoms, complex medical histories, repeated concussions, or pre-injury conditions or diagnoses.

Medical and healthcare professionals should work together with patients, their families, and those involved in their lives (teachers, coaches, employers, friends, etc.) using a team and family-centred approach to best manage the needs of the patient after a concussion and promote positive health outcomes. Proper concussion recognition in children and adolescents requires all involved in a child/adolescent's life to be educated on the signs and symptoms of this condition and to collaborate to ensure that youth with suspected concussion are immediately removed from play or activity and directed to the most appropriate care. The following sections provide recommendations specific to concussion recognition, initial medical assessment, and management.



Domain 1: Concussion Recognition and Directing to Care

Introduction:

All school and sport stakeholders including students, athletes, parents, teachers, coaches, officials, and healthcare professionals play an important role in recognizing suspected concussions and supporting the child when returning to school, sport, work, and other activities.

Tool 1.1: [Pediatric Concussion: The Role of School Boards, Community Sports Organizations, and Centres](#)

Tool 1.2: [Concussion Recognition Tool 5](#)

Tool 1.3: [Manage Acute and Prolonged Symptoms Algorithm](#)

Recommendations:

1.1a School boards, sports organizations, and community centres should provide pre-season concussion education and conduct a review of all concussion policies in effect within the school or sport setting.

Level of Evidence: C

1.1b School boards, sports organizations, and community centres should ensure updated policies are in place to recognize and accommodate a child/adolescent who has sustained a concussion.

Tool 1.1: [Pediatric Concussion: The Role of School Boards, Community Sports Organizations, and Centres](#)

Level of Evidence: C

1.2 Remove the child/adolescent from the activity immediately if a concussion is suspected for immediate assessment and to avoid another injury.

Do not leave the child alone and contact the parent/caregiver immediately. Do not let the child/adolescent return-to-sport (practice or game play) or other activities that day. “If in doubt, sit them out.”

A concussion should be suspected:

- In any child/adolescent who sustains a significant impact to the head, face, neck, or body and demonstrates/exhibits *any* of the visual signs of a suspected concussion or reports *any* symptoms of a suspected concussion as detailed in the [Concussion Recognition Tool 5 \(Tool 1.2\)](#).

Premature return to activities and sport can lead to another injury. Another blow to the head may complicate the injury further and have a longer recovery time due to the higher risk of prolonged symptoms. Severe brain swelling or cerebral edema after a concussion is very rare but potentially fatal.

Suggested concussion tools to share with teachers, coaches, parents, peers, and others

- Link: [Parachute Return to School Strategy](#) (handout) (Parachute)
- Link: [Concussion Ed – Parachute Concussion Education](#) (app from Parachute)
- Link: [PAR Concussion Recognition & Response: Concussion symptom recognition tool for coaches and parents](#) (app)
- Link: [CATT: Concussion Resources for School Professionals](#) (handout)

Level of Evidence: B

1.3 Recommend an emergency medical assessment for a child/adolescent with any of the “red flag” symptoms.

If a child/adolescent demonstrates any of the ‘Red Flags’ indicated by the [Concussion Recognition Tool 5](#) (Tool 1.2), a more severe head or spine injury should be suspected and an emergency medical assessment is required. Consider arranging an ambulance service as necessary to facilitate urgent medical assessment at the nearest hospital and execution of the Emergency Action Plan for your organization. When calling an ambulance, describe the specific red flags symptoms over the phone.

Red flag symptoms include:

- Severe or increasing headache
- Neck pain or tenderness
- Double vision
- Weakness or numbness in arms/legs
- Seizure or convulsions
- Loss of consciousness
- Deteriorating conscious state
- Repeated vomiting
- Increasingly restless, agitated or combative state
- Slurred speech

Suggested tools to help identify “Red Flag” symptoms

- Tool 1.2: [Concussion Recognition Tool 5: To help identify concussion in children, adolescents and adults](#) (handout)
- Link: [Concussion Ed – Parachute Concussion Education](#) (app- Parachute)
- Link : [PAR Concussion Recognition & Response](#) (app)

Level of Evidence: B

1.4 Concussion should be suspected and diagnosed as soon as possible to maintain health and improve outcomes. Concussion can be suspected in the community by healthcare professionals, parents, teachers, coaches, and peers. Those with a suspected concussion should be referred to a physician or nurse practitioner to perform a thorough medical assessment to exclude more severe injuries, consider a full differential diagnosis, and confirm the diagnosis of concussion.

Tool 1.3: [Manage Acute and Prolonged Symptoms Algorithm](#)

Symptoms may appear immediately or within several hours after injury

- A concussion is an evolving injury; symptoms may change over time
- See Recommendation 1.3: [Recommend emergency assessment for a child/adolescent with any of the “red flag” symptoms](#)

Suggested tools for the general community to suspect a concussion

- [Concussion Recognition Tool 5: To help identify concussion in children, adolescents and adults](#)

Suggested tools for experienced healthcare professionals to suspect a concussion:

- Link: Sport Concussion Recognition Tool 5 (SCAT5) for Children aged 5-12
[English](#) / [French](#)
- Link: SCAT5 for Athletes aged 13+ [English](#) / [French](#)

Level of Evidence: B



Domain 2: Initial Medical Assessment and Management

Introduction:

A thorough medical assessment by a physician or nurse practitioner should be performed on all children and adolescents with a suspected concussion. This assessment includes a full differential diagnosis (ruling out: a more severe TBI, cervical spine injury, medical conditions, mental health conditions, and neurological conditions that can present with non-specific neurological symptoms including red flags) and a confirmation of the diagnosis of concussion. If a more severe injury or other conditions identified at initial assessment are suspected, emergent referrals should be made to appropriate healthcare professionals. Acute signs and symptoms should be considered in context with the child/adolescent's normal pre-injury performance, especially for those with learning and communication deficits, ADHD, and/or physical disabilities. At the initial assessment, verbal and written information should be shared regarding the course of recovery and about when the child/adolescent can return-to-school/activity/sport and work. Patients and their parents and/or caregivers need to know that most patients recover fully from concussion even though the recovery rate is variable and unpredictable. Providing information reduces anxiety, helps set realistic expectations, promotes recovery, and prevents re-injury.

Some children/adolescents will continue to have symptoms at one month and beyond. In these situations, the healthcare professional should refer the patient to an interdisciplinary concussion team for individualized care that targets specific prolonged symptoms. Findings from the patient's clinical history and initial assessment can identify patients who may have an elevated risk of experiencing prolonged symptoms following concussion. Identifying patients at risk for delayed recovery in the acute stage allows for early supportive care, close monitoring for prolonged symptoms, and an opportunity to consider early referral (before 4-weeks post-acute injury) to an interdisciplinary concussion team. Prolonged rest beyond the first 24-48 hours after a concussion is not recommended and may cause more harm than good. Return to physical and cognitive activity should be gradual and individualized based on activity tolerance and symptom presentation (e.g., the patient is able to engage in an activity without worsening of post-concussion symptoms). Full-contact sport or high-risk activities where there is a risk for repeat concussion should be avoided until 1) post-concussion symptoms have subsided, 2) the child/adolescent has returned to full school activities without accommodations related to post-concussion symptoms, and 3) child/adolescent has medical clearance to return to full-contact sport and high-risk activities following the completion of a return-to-sport protocol. Upon discharge from initial assessment from the Emergency Department or Primary Care Provider (physician or nurse practitioner), families should be provided with written instructions which include red flags to return for urgent re-assessment. Families should be informed that most patients recover fully from concussion even though the recovery rate is variable; this will help set realistic expectations, promote recovery and prevent re-injury.

Tool 1.3: [Manage Acute and Prolonged Symptoms Algorithm](#)

Recommendations:

2.1 Physicians or nurse practitioners should perform a comprehensive medical assessment on all children/adolescents with a suspected concussion or with acute head or spine trauma.

Tool 1.3: [Manage Acute and Prolonged Symptoms Algorithm](#)

Include a clinical history, physical examination, and the evidence-based use of diagnostic tests or imaging as needed.

2.1a Take a comprehensive clinical history.

Details that should be collected in the clinical history include:

- Patient demographics (e.g., age, sex, gender)
- Assess injury mechanism and symptoms at the time of injury
- Assess symptom burden at the time of initial presentation
 - Number of symptoms
 - Severity of symptoms
 - Type of symptoms
- Presence of loss of consciousness, post-traumatic amnesia, and red flags (seizures, neck pain, focal neurological deficits)
- Current post-concussion symptoms (using age-appropriate standardized symptom inventory)
- Review mental health (Domain 8: [Mental Health](#))
- Past medical history (e.g., previous concussions, migraine or non-specific headaches, mental health disorders, coagulopathy). Note the duration until recovery from previous concussions (i.e., within 7-10 days or prolonged).
- Allergies/immunizations
- Ask whether the child/adolescent is taking any substances or medications: Prescribed or over-the-counter medications or supplements, alcohol, or recreational drugs including cannabis. These substances may mask or modify concussion symptoms.
- Ask about school, activities, work, and sports participation

Level of Evidence: B

2.1b Note common modifiers that may delay recovery and use a clinical risk score to predict risk of prolonged symptoms.

Link: [Predicting Persistent Post-Concussive Problems in Pediatrics \(5P\): Score Calculator](#)

Modifiers that may delay recovery:

- Age (increased risk with increased age).
- Sex (female).
- Duration of recovery from a previous concussion.
- High pre-injury symptom burden.
- High symptom burden at initial presentation.
- Clinical evidence of vestibular or oculomotor dysfunction.
- Personal and family history of migraines.
- History of learning or behavioural difficulties.

- Personal and family history of mental health.
- Family socioeconomic status/education.

Level of Evidence: A

2.1c Perform a comprehensive physical examination.

Vital signs (resting heart rate and blood pressure)

- Level of consciousness (GCS)
- Mental status
- A complete neurological examination (cranial nerve, motor, sensory, reflex, cerebellar, gait and balance testing) (Tool 2.1: [Physical Examination](#))
- A cervical spine examination (palpation, range of motion, provocative cervical spine tests) (Tool 2.1: [Physical Examination](#))
- An examination of the visual and vestibular systems

Tools to consider:

- Link: SCAT5 for Children aged 5-12 [English](#) / [French](#)
- Link: SCAT5 for Athletes aged 13+ [English](#) / [French](#)
- Link: [Post-Concussion Symptom Inventory \(PCSI\) Self Assessment](#) (age 5-7, 8-12, age 13-18)
- Link: [CDC Pediatric mTBI Guideline Checklist](#)
- Link: [Heads Up to Health Care Professionals \(CDC\)](#)
- Link: [Acute Concussion Evaluation \(ACE\)](#).

Level of Evidence: B

2.1d Consider CT of the brain or cervical spine only in patients whom, after a medical assessment, a structural intracranial or cervical spine injury is suspected; do not conduct routine neuroimaging for the purpose of diagnosing concussion.

Most children/adolescents who sustain an acute head injury or suspected concussion do not need diagnostic imaging.

Use the following tools, as appropriate, to determine the need for CT imaging in patients with acute head trauma:

- Tool 2.2: [PECARN Management Algorithm for Children after Head Trauma](#)
- Tool 2.3: [CATCH 2 Rule for children after head trauma](#)
- Tool 2.4: [Algorithm for the Management of the Pediatric Patient ≥ 2 Years With Minor Head Trauma](#)

Although validated clinical decision-making rules are highly sensitive, these tools are meant to assist but not replace clinical judgment. CT scans should be used judiciously as the exposure of children/adolescents to the effects of ionizing radiation carries a small increased lifetime risk of cancer. If a structural brain injury is suspected in a patient with acute head trauma undergoing initial medical assessment in the office setting, urgent referral to an Emergency Department should be

arranged.

Diagnostic imaging of the spine should be considered when symptoms are suggestive of structural cervical spine injury. Imaging should be considered in patients with severe neck pain, tenderness or clinical evidence of radiculopathy or myelopathy. The choice of imaging modality (plain radiographs, CT or MRI of the cervical spine) should be guided by the suspected pathology.

Patients with positive traumatic findings observed on diagnostic imaging of the brain or spine should be urgently referred to a neurosurgeon for consultation.

Level of Evidence: A- CT. B- MRI

2.2 Provide verbal information and written (electronic) handouts regarding the course of recovery and when the child/adolescent can return to school/activity/sport/work.

Consider the following anticipatory guidance (verbal reassurance) in order to reduce anxiety, set realistic expectations, promote recovery, and prevent re-injury:

- Most patients recover fully from concussion even though the recovery rate is variable and unpredictable.
- Current symptoms are expected and common.
- The burden and distress parents/caregivers of children/adolescents who have sustained a concussion may experience is common.
- Children typically recover in 1-4 weeks but some children/adolescents will have symptoms at one month and beyond and need to be monitored/seek additional care. Females aged 13-18 years have an increased risk of prolonged recovery.
- Recommendation 2.3 [Recommend graduated return to physical and cognitive activity](#)
- Domain 12: [Return-to-School and Work](#)

Summary of online tools to consider related to lifestyle strategies and expectations:

- Link: [Return to Activity Strategy](#) (CATT)
- Link: [Return to Sport Strategy](#) (CATT)
- Link: [After a Concussion: Return to Sport Strategy](#) (Parachute)
- Link: [Return-to-School Strategy](#) (CATT)
- Link: [HEADS UP Resources for Returning to School](#) (CDC)
- Link: [Return to School Strategy](#) (Parachute)
- Link: Advice for gradually resuming intellectual, physical and sports activities [English/ French](#) (INESSS)

Level of Evidence: B

2.3 Recommend graduated return to cognitive and physical activity to promote recovery.

While most children/adolescents fully recover, the recovery rate can be variable. Return to physical and cognitive activity should be individualized based on activity tolerance and symptom presentation.

2.3a Recommend an initial 24-48 hour period of rest with limited physical and cognitive activity.

Level of Evidence: B- Gradual return to physical activity. C- Gradual return to cognitive activity.

2.3b Recommend that low to moderate level physical and cognitive activity be gradually started 24-48 hours after a concussion at a level that does not result in recurrence or exacerbation of symptoms. Activities that pose no/low risk of sustaining a concussion should be resumed even if mild residual symptoms are present or whenever acute symptoms improve sufficiently to permit activity.

Low to moderate level physical and cognitive activity:

- Avoid activities associated with a risk of contact, fall, or collisions ([Recommendation 2.3c](#)).
- Symptoms should be monitored carefully and activities that make symptoms worse should be avoided
- Child/adolescent should gradually increase exercise/activity that matches an improvement in symptoms
- Emphasize that spreading activities throughout the day helps patients achieve more and that they should avoid doing too much at once. Tool 2.5: [“Four P’s” – Prioritize, Plan, Pace, and Position](#)
- Domain 12: [Return-to-School and Work](#)
- Tool 2.6: [Post-Concussion Information Sheet](#) (includes a list of examples of low-risk activities)
- Link: [CATT Return to Sport Strategy](#) (CATT online; includes examples of low risk activities)

Level of Evidence: B- Gradual return to physical activity. C- Gradual return to cognitive activity.

2.3c Recommend that patients avoid activities associated with a risk of contact, fall, or collisions such as high speed and/or contact activities and full-contact sport that may increase the risk of sustaining another concussion during the recovery period.

Advise/emphasize that returning to full-contact sport or high-risk activities before the child/adolescent has recovered increases the risk of delayed recovery and for sustaining another more severe concussion or more serious injury.

Level of Evidence: B

2.3d Consider referring select patients (e.g., competitive athletes, those who are not tolerating a graduated return to physical activity, or those who are slow to recover) to a medically supervised interdisciplinary team with the ability to individually assess sub-symptom threshold aerobic exercise tolerance and to prescribe aerobic exercise treatment.

Level of Evidence: B

2.4 Provide education and guidance regarding strategies to promote recovery.

2.4a Advise on the importance of sleep and discuss sleep hygiene.

Advise that consistent sleep schedules and duration of sleep may contribute to general recovery from a concussion and alleviate symptoms such as mood, anxiety, pain, fatigue, and cognitive difficulties if these are present.

Summary of tools to consider:

- Tool 2.7: [Strategies to Promote Good Sleep and Alertness](#)
- Link: [Sleep for Youth. CHEO Sleep Hygiene handout](#)
- Link: [Concussion Handbook](#) (Holland Bloorview Kids Rehabilitation Hospital)

Level of Evidence: C

2.4b Advise on maintaining social networks and interactions as tolerated beyond a brief initial period of cognitive and physical rest (24-48 hours after injury).

Beyond a brief initial period of cognitive and physical rest, 24-48 hours after injury, children/adolescents should participate in rewarding social activities, modified as needed, while staying below their symptom-exacerbation thresholds and avoiding risk for re-injury. There is evidence that reducing the risk of social isolation and mental health issues may promote recovery.

- Identify these activities and suggest modifications as appropriate.
- Educate on the principle that participation in rewarding social activities in the presence of residual or prolonged symptoms may have to be limited so that they do not result in a recurrence or exacerbation of symptoms.

Level of Evidence: B

2.4c Advise on the use of computers, phones, and other devices with screens. Beyond an initial period of cognitive and physical rest (24-48 hours after injury), use of devices with screens may be gradually resumed at a level that does not result in recurrence or exacerbation of symptoms.

Advise that computers, phone, and other devices with screens may exacerbate symptoms, especially in the first days after injury. The use of these devices can be increased according to symptom tolerance as the child/adolescent recovers. For sleep hygiene purposes, these devices should not be used in the hour prior to bedtime.

Level of Evidence: C

2.4d Advise on avoiding alcohol and other recreational drugs after a concussion.

Alcohol and recreational drugs may have a negative effect on concussion recovery. Avoiding alcohol or drugs prevents a child/adolescent from self-medicating and resorting to drugs to relieve symptoms. Impaired judgment after a concussion could lead to risky behaviour that causes further harm or may delay the identification of complications.

Level of Evidence: C

2.4e Advise to avoid driving during the first 24-48 hours after a concussion. Advise patients to begin driving when they are feeling improved, can concentrate sufficiently to feel safe behind the wheel, and when the act of driving does not provoke significant concussion symptoms.

Provide verbal information related to when an adolescent should return to driving during recovery from a concussion. Driving is a complex coordinated process that requires vision, balance, reaction time, judgment, cognition, and attention. Concussion may have affected some or all of these skills. Driving impairments have been shown to exist even in asymptomatic patients 48 hours after a concussion. Avoiding driving for at least 24-48 hours after a concussion may potentially prevent motor vehicle accidents and, therefore, injury to the adolescent or to others.

Level of Evidence: C

2.5 Over-the-counter medications such as acetaminophen and ibuprofen may be recommended to treat acute headache. Advise on limiting their use to less than 15 days a month and avoiding “around-the-clock” dosing to prevent overuse or rebound headaches (i.e., advise that children/adolescents avoid using over the counter medications at regular scheduled times throughout the day).

Level of Evidence: B

2.6 At present, there is limited evidence to support the administration of intravenous medication to treat acute headaches in pediatric concussion patients in the Emergency Department setting.

Level of Evidence: B

2.7 After assessment, nearly all children/adolescents with concussion may be safely discharged from clinics and Emergency Departments for observation at home.

The decision to observe in the hospital will depend on clinical judgment. Indicators for longer in-hospital observation (or to return to emergency for re-assessment) may include:

- Worsening symptoms (headache, confusion, irritability)
- Decreased level of consciousness
- Prolonged clinical symptoms (persistent/prolonged vomiting, severe headache, etc.)
- Bleeding disorders
- Multi-system injuries
- Co-morbid symptoms

Other discharge considerations:

- Observe the child/adolescent for a period of time to verify that they do not develop “red flag” symptoms prior to discharge. Use clinical judgment.
- Verify that the child/adolescent has a normal mental status (alertness/behaviour/cognition) and their symptoms are improving prior to discharge.
- Verify that an assessment of clinical risk factors indicating the need for a CT scan was performed or a normal result was obtained if a CT scan was performed prior to discharge. See

[Recommendation 2.1d](#) for more information on when to consider diagnostic brain or cervical spine imaging.

Level of Evidence: B

2.8 Recommend a medical follow-up visit in 1-2 weeks to re-assess and monitor clinical status. Recommend an immediate medical follow-up in the presence of any deterioration.

Those with a confirmed diagnosis of concussion may be managed by a healthcare professional who within their formally designated scope of practice has the capacity to manage ongoing concussion-related symptoms.

Level of Evidence: C

2.9 Consider referral to an interdisciplinary concussion team in the presence of modifiers that may delay recovery.

See Recommendation 2.1b: [Note any modifiers that may delay recovery and use a clinical risk score to predict risk of prolonged symptoms.](#)

Level of Evidence: A

2.10 Provide post-concussion information and a written medical assessment to the child/adolescent and the parent/caregiver prior to sending the child/adolescent home.

Write the discharge note/written medical assessment with the following information:

- The outcome of the initial medical assessment.
- Indicate to the child/adolescent and their parents, teachers, and coaches if they have been medically cleared to participate in full-contact sports or high-risk activities.
- Link: [Canadian Guideline on Concussion in Sport Medical Assessment Letter](#) (Parachute Canada)
- Link: [Montreal Children's Hospital Discharge Instructions](#)

Verbal and written (or electronic) guidance should include:

- Tool 2.6: [Post-Concussion Information Sheet](#)
- An overview of common concussion symptoms.
- Warning signs that should prompt emergency medical assessment.
- Suggestions regarding activity modifications and non-pharmacological strategies to manage symptoms.
- Information on how and when to make a gradual return-to-school and low-risk physical activities. (Domain 12: [Return-to-School and Work](#))
- Recommendation 2.3: [Recommend a graduated return to physical and cognitive activity](#)).
- Information on when a medical follow-up appointment is needed ([Recommendation 2.7](#)).

Examples of patient information handouts to consider:

- Link : Institut national d'excellence en sante et en services sociaux (INESSS- Québec). Advice for gradually resuming intellectual, physical and sports activities. [English](#) / [French](#)
- Link: [CDC: Heads UP Discharge Instructions](#)
- Link: [Concussion & You Handbook: Holland Bloorview Kids Rehabilitation Hospital](#)
- Link: [Understanding and Managing Concussion in Youth: 3rd Edition Concussion Kit: Montreal Children's Hospital](#)

Level of Evidence: B



Domain 3: Medical Follow-up and Management of Prolonged Symptoms

Introduction:

A medical follow-up is recommended if a child/adolescent is experiencing post-concussion symptoms, has not completed the return-to-school or return-to-activity/sport stages, or experiences any deterioration. Post-concussion symptoms and return-to-school/ activity/sport/work status should be reassessed. A medical assessment including clinical history, comprehensive physical examination, and consideration for diagnostic tests or imaging should be considered in patients with red flag symptoms or worrisome clinical findings. Those with a confirmed diagnosis of concussion may be managed by a healthcare professional that, within their formally designated scope of practice, has the capacity to manage ongoing concussion-related symptoms.

Patients who are experiencing clinical improvement in their post-concussion symptoms should continue to be provided with education, reassurance, and guidance on advancing through their return-to-school/activity/sport strategies. At follow-up visits, patients should be provided with an updated medical clearance letter indicating if they are medically cleared to participate in full-contact sport or high-risk activities.

If post-concussion symptoms have not resolved by one-month, or if the child/adolescent's condition/symptoms worsen, consider referral to specialized care with an interdisciplinary concussion team and ensure that the child/adolescent is well supported at school and at home. Additionally, early identification of patients with modifiers that may delay recovery allows for early targeted supportive care, close monitoring for prolonged symptoms, and consideration for early referral (before 4-weeks following acute injury) to an interdisciplinary concussion team.

[Tool 1.3: Manage Acute and Prolonged Symptoms Algorithm](#)

Recommendations:

3.1 Perform a repeat medical assessment on all patients presenting with post-concussion symptoms 1-2 weeks following acute injury.

Include a focused clinical history, focused physical examination, and consideration for the need for diagnostic tests including imaging. See [Tool 1.3: Manage Acute and Prolonged Symptoms Algorithm](#).

3.1a Take a focused clinical history based on symptoms described.

- Domain 6: [Headache](#)
- Domain 7: [Sleep](#)
- Domain 8: [Mental Health](#)
- Domain 9: [Cognition](#)
- Domain 10: [Vision, Vestibular, and Oculomotor Function](#)

- Domain 11: [Fatigue](#)
- Domain 12: [Return-to-School and Work](#)

Consider signs and symptoms in context with the child/adolescent's normal performance, especially for those with pre-existing conditions (e.g., learning and communication deficits, ADHD, and/or physical disabilities) to identify the underlying causes of the prolonged symptoms or concerns and develop a management strategy.

Level of Evidence: B

3.1b Examine the child/adolescent and perform a focused physical examination.

- Tool 2.1: [Physical Examination](#)
- Vital signs (Resting heart rate and blood pressure).
- A complete neurological examination (cranial nerve, motor, sensory, reflex, cerebellar, gait, balance testing) (Tool 2.1: [Physical Examination](#)).
- A cervical spine examination (palpation, range of motion, provocative cervical spine tests) (Tool 2.1: [Physical Examination](#)).
- Review mental health. Perform a post-concussive assessment and a cognitive screen, reassessing for existing and new mental health symptoms such as anxiety and mood.
- Screen the child/adolescent for medication/substances that may mask or modify the symptoms.
- Consider a broad differential diagnosis for children/adolescents with prolonged symptoms.
- Monitor the return-to-activity/sport and return-to-school status.
- Further examination of the child/adolescent should be based on symptoms:
 - Link: [Post-Concussion Symptom Inventory \(PCSI\) Self-Assessment](#) (age 5-7, 8-12, age 13-18)
 - Domain 6: [Headache](#)
 - Domain 7: [Sleep](#)
 - Domain 8: [Mental Health](#)
 - Domain 9: [Cognition](#)
 - Domain 10: [Vision, Vestibular, and Oculomotor Function](#)
 - Domain 11: [Fatigue](#)
 - Domain 12: [Return-to-School and Work](#)

Level of Evidence: B

3.1c Consider diagnostic brain or cervical spine MRI imaging for those with focal or worrisome symptoms.

See [Recommendation 2.1d](#): When to consider diagnostic brain or cervical spine imaging.

Urgent conventional MRI should be considered in concussion patients who present with focal or worrisome symptoms (e.g., deteriorating vision, focal weakness or numbness, altered awareness, prominent behavioural changes, or worsening headaches that are not responding to treatment) and in whom a structural brain injury or abnormality is suspected.

Level of Evidence: B-MRI

3.2 Provide patients with general education and guidance that outlines mental health considerations, non-pharmacological strategies to minimize symptoms including sleep hygiene, activity modifications, limiting triggers, information on screen time, the importance of social interaction, and how to work with the school team to facilitate school success.

- Tool 2.6: [Post-Concussion Information Sheet](#)
- Tool 2.7: [Strategies to Promote Good Sleep and Alertness](#)
- Link: [Sleep for Youth. CHEO Sleep Hygiene handout](#)
- Recommendation 2.3 [Recommend graduated return to cognitive and physical activity](#)
- Domain 12: [Return-to-School and Work](#)
- Recommendation 2.4c: [Advise on the use of computers, phones, and other screen devices](#)

Level of Evidence: C

3.3 Encourage patients with post-concussion symptoms to engage in cognitive activity and low-risk physical activity as soon as tolerated while staying below their symptom-exacerbation thresholds. Activities that pose no/low risk of sustaining a concussion (no risk of contact, collision, or falling) should be resumed even if mild residual symptoms are present or whenever acute symptoms improve sufficiently to permit activity.

Consider referring select patients (e.g., competitive athletes, those who are not tolerating a graduated return to physical activity, or those who are slow to recover) to a medically supervised interdisciplinary team with the ability to individually assess sub-symptom threshold aerobic exercise tolerance and to prescribe aerobic exercise treatment.

See [Recommendation 2.3](#).

See Tool 2.6: [Post-Concussion Information Sheet](#) for examples of low-risk activities.

Level of Evidence: B- Gradual return to physical activity. C- Gradual return to cognitive activity.

3.4 Refer to specialized care with an interdisciplinary concussion team if post-concussion symptoms do not gradually resolve by 4 weeks.

See Tool 1.3: [Manage Acute and Prolonged Symptoms Algorithm](#).

Assessment by an interdisciplinary concussion team can assist in identifying the type of management that is required, along with the medical and health professions on the interdisciplinary concussion team or external to this team who can provide the required management. Not all children/adolescents will require care from all members of the interdisciplinary concussion team and care should be targeted based on identified symptoms and patient needs.

Level of Evidence: B

3.5 Consider early referral (prior to 4-week post-injury) to an interdisciplinary concussion team in the presence of modifiers that may delay recovery.

See Recommendation 2.1b: [Note any modifiers that may delay recovery and use a clinical risk score to predict risk of prolonged symptoms.](#)

Level of Evidence: A

3.6 Consider initiating treatment for specific symptoms or concerns while waiting for a referral to an interdisciplinary concussion team or sub-specialist.

- Domain 6: [Headache](#)
- Domain 7: [Sleep](#)
- Domain 8: [Mental Health](#)
- Domain 9: [Cognition](#)
- Domain 10: [Vision, Vestibular, and Oculomotor Function](#)
- Domain 11: [Fatigue](#)
- Domain 12: [Return-to-School and Work](#)

Level of Evidence: B

3.7 Recommend regular medical follow-up if a child/adolescent is still experiencing post-concussion symptoms or has not completed the return-to-school or return-to-sport/activity stages. Recommend an immediate medical follow-up in the presence of any deterioration.

Level of Evidence: C



Domain 4: Medical Clearance for Full-Contact Sports or High-Risk Activity

Introduction:

Medical clearance to return to full-contact sports or high-risk activities should be made on an individual basis using clinical judgment based on the findings of the medical follow-up. Presently there is no objective test to confirm physiological recovery following concussion. The child/adolescent should have returned to all school activities, including writing exams without accommodations related to their concussion/post-concussion symptoms, before medical clearance is given for returning to full-contact sports or high-risk activities.

Recommendations:

4.1 Consider patients for medical clearance to return to full-contact activities and sport/game play if clinical criteria have been met.

The following clinical criteria should be considered or met before recommending that a child/adolescent returns to full-contact activities and sport/game play:

- Child/adolescent has successfully returned to all school activities including writing exams without symptoms above their previous pre-injury level or requiring accommodations related to their concussion/post-concussion symptoms, (e.g., child/adolescent may have pre-existing accommodations or new accommodations related to something other than their concussion).
- Normal neurological and cervical spine examination.
- Asymptomatic at rest (or back to the pre-injury state in patients with pre-existing conditions such as baseline headaches or mental health conditions).
- Tolerating physical activities and non-contact activities without any symptoms.
- No longer taking any drugs or substances atypical to their pre-injury functioning that could mask symptom presentation.

For children/adolescents with complex medical histories (e.g., repeated concussion, baseline concussion-like symptoms), see [Recommendation 5.1](#) for information regarding returning to full-contact sports or high-risk activities, or retirement from full-contact sports or high-risk activities.

Level of Evidence: B

4.2 Provide patients with a letter indicating medical clearance to return to all activities when medically cleared.

Link: [Canadian Guideline on Concussion in Sport Medical Assessment Letter](#)

Link: Quebec Association of Sport and Exercise Medicine physicians: Medical clearance letter. [English](#) / [French](#)

Level of Evidence: C

4.3 Advise medically cleared patients to seek immediate medical attention if he or she develops new concussion-like symptoms or sustains a new suspected concussion.

Tool 1.2: [Concussion Recognition Tool 5: To help identify concussion in children, adolescents, and adults](#)

Level of Evidence: B



Domain 5: Sport Concussion Considerations

Introduction:

Return to full-contact sport or high-risk activity decisions may be complicated for children/adolescents who have experienced multiple concussions or who have baseline conditions that are associated with concussion-like symptoms. Numerous factors including concussion history, co-morbidities, contraindications, symptom presentation, injury threshold, and sequelae should be considered when making return to sport decisions for children/adolescents who have complex medical histories. The current evidence does not support an added benefit of baseline testing. Mandatory pre-season baseline testing is not recommended.

Link: [Parachute Statement on Baseline Testing](#) (Parachute Canada)

Recommendations:

5.1 Refer a child/adolescent with multiple concussions or baseline conditions associated with concussion-like symptoms to an interdisciplinary concussion team to help with return to full-contact sports, or high-risk activities or retirement decisions from full-contact sports, or high-risk activities.

Return to full-contact sport or high-risk activity decisions can be complicated for children/adolescents with more complex medical histories. The following factors should be taken into consideration in the discussion and decisions made about return-to-sport or retirement:

- Concussion history
- Co-morbidities (e.g., learning and communication deficits, ADHD, physical disabilities, psychiatric disorders)
- Absolute contraindications for return-to-sport and high-risk activities
- Early recurrence or greater frequency of symptoms
- Lower injury threshold
- Increasing recovery time
- Potential short- and long-term sequelae

Some patients may benefit from neuropsychological assessment to determine resolution of cognitive problems. If a post-injury cognitive or neuropsychological assessment is deemed clinically necessary, it is recommended that this assessment be interpreted by a pediatric neuropsychologist.

Level of Evidence: C

5.2 Baseline testing on children/adolescents using concussion assessment tools or tests (or any combination of tests/tools) is not recommended or required for concussion diagnosis or management following an injury.

See the [Parachute Statement on Baseline Testing](#) for more information (Parachute Canada).

“Baseline testing refers to the practice of having an athlete complete certain concussion assessment tools/tests prior to sports participation to provide baseline measurements that can be compared to post-injury values in the event of a suspected concussion. Current evidence does not support a significant added benefit of baseline testing athletes. This includes the Child SCAT5 and the SCAT5 tools, as well as neuropsychological and neurocognitive tests, both computerized or not.” ([Parachute Statement on Baseline Testing](#))

See [Recommendation 5.3](#): Special considerations regarding baseline testing.

Level of Evidence: B

5.3 Special considerations regarding baseline testing.

Please consult the [Parachute Statement on Baseline Testing](#) for more information (Parachute Canada).

“There may be unique athlete populations and sports environments where baseline testing may be considered. These situations should be considered the exception and not the rule.”

- “Clinical neuropsychologists may consider baseline cognitive or neuropsychological testing in select youth athletes (greater than 12 years old) who have pre-existing conditions, such as a history of previous concussion, ADHD, or learning disorders, that may impact the interpretation of post-injury test results.”
- “Certain teams and sporting federations have well-established physician-supervised concussion protocols with dedicated experienced healthcare professionals working directly and continuously with youth athletes (i.e., that are present at training and competition events). In these sport environments, baseline testing may be considered as an optional assessment within the comprehensive concussion protocol as long as the medical teams caring for these athletes include experienced healthcare professionals who have competency-based training and clinical experience to allow them to administer and interpret these tests.”

(Reproduced with permission from Parachute Canada: [Parachute Statement on Baseline Testing](#))

Level of Evidence: B

5.4 [Recommendation 2.3d](#): Consider referring select patients (e.g., competitive athletes, those who are not tolerating a graduated *return* to physical activity, or those who are slow to recover) to a medically supervised interdisciplinary team with the ability to individually assess sub-symptom threshold aerobic exercise tolerance and to prescribe aerobic exercise treatment.

Level of Evidence: B



Domain 15: Telemedicine and Virtual Concussion Care

Introduction:

Preliminary research has demonstrated that telemedicine can be used to provide timely access to safe and cost-effective medical care of carefully selected concussion patients living in underserved regions. Research from the teleneurology, telepsychiatry and teleneuropsychology literature suggests that patients with prolonged concussion symptoms including those with headaches, mental health disorders, and cognitive impairment may also benefit from accessing care via telemedicine. While the COVID-19 pandemic has propelled greater use of virtual care, healthcare professionals using virtual platforms to deliver concussion care should be aware of the limitations of these approaches in order to ensure they are practicing in accordance with regional guidelines and policies outlined by their respective professional regulatory bodies. The future use of telemedicine and virtual care to assist in delivering primary and interdisciplinary concussion care will continue to be refined by future research and experience.

Definitions:

Telemedicine: Telemedicine has been defined as “the delivery of health care services, where distance is a critical factor, by all health care professionals using information and communication technologies for the exchange of valid information for diagnosis, treatment, and prevention of disease and injuries, research and evaluation, and for the continuing education of health care providers, all in the interests of advancing the health of individuals and their communities.” (WHO, 2010). Telemedicine can be delivered synchronously (e.g., in-person videoconferencing) or asynchronously (e.g., store and forward or EConsultation).

Virtual care: Virtual care has been defined as “any interaction between patients and/or members of their circle of care, occurring remotely, using any forms of communication or information technologies, with the aim of facilitating or maximizing the quality and effectiveness of patient care” (Shaw et al, 2018).

The following clinical recommendations and preliminary algorithm have been developed to share evidence and expert opinion on how to guide the medical and interdisciplinary care of pediatric concussion patients using telemedicine and includes suggestions to optimize a virtual medical assessment.

- Tool 15.1: [Considerations for Telemedicine and Virtual Care Algorithm](#)
- Tool 15.2 [Considerations for a virtual physical examination for medical assessment and follow-up of concussion patients](#)

Clinical Recommendations:

Recommendation 15.1 **Virtual medical assessment:**

See [Recommendation 2.1](#) *“Physicians or nurse practitioners should perform a comprehensive medical assessment on all children/adolescents with a suspected concussion or with acute head or spine trauma.*

- *Include a clinical history, physical examination, and the evidence-based use of diagnostic tests or imaging as needed.”*

In-person medical assessments are ideal for all children/adolescents with suspected or diagnosed concussion, however, telemedicine may be considered to assist in the medical assessment for patients who meet the following criteria:

- A previous medical assessment by a physician or nurse practitioner has been performed.
- Access or travel for an in-person medical assessment is limited or difficult
- There is no history of abnormal diagnostic imaging (e.g., intracranial hemorrhage, spine or facial fracture)

Note: All providers should be aware of current public health recommendations when providing care to their patients.

Tool 15.1: [Considerations for Telemedicine and Virtual Care Algorithm](#)

Level of Evidence: C

15.1a Obtain informed consent from the patient and/or their parent/caregiver to conduct a remote medical assessment via telemedicine.

This process should outline the benefits and limitations of performing clinical care via telemedicine as well as taking measures to ensure privacy and confidentiality. Discuss the important limitation of not being able to perform some aspects of the physical examination virtually (e.g., testing of motor or sensory functioning, fundoscopy, etc.) and outline what arrangements will be made to facilitate an urgent in-person assessment as needed to optimize clinical care.

Level of Evidence: C

For other general considerations on how clinicians and patients can be prepared for a virtual visit see the following links:

- Canadian Medical Association. Scaling up virtual care in Canada: CMA a key player in new national task force. February 2020. <https://www.cma.ca/scaling-virtual-care-canada-cma-key-player-new-national-task-force>
- American Medical Association. [TeleHealth Quick Guide](#) September 2020.
- Canadian Medical Association. [Virtual Care Guide for Patients](#). June 2020.

15.1b Take a comprehensive virtual clinical history.

Complete a comprehensive virtual clinical history that addresses the same key components of an in-person medical assessment including patient demographics, injury mechanism, symptoms at the time of injury, symptom burden at the time of presentation, loss of consciousness, post-traumatic amnesia, self-reported red flags, mental health, past medical history, assessment of concussion modifiers, current medications and allergies, school, work, and sports participation.

- See [Recommendation 2.1a](#) for details that should be collected in the clinical history. This recommendation includes a validated age-appropriate symptom inventory that can be considered as a tool to assess current symptoms and severity.

Level of Evidence: C

15.1c See [Recommendation 2.b](#) “Note common modifiers that may delay recovery and use a clinical risk score to predict risk of prolonged symptoms.”

15.1d Considerations for a virtual physical assessment

A virtual physical examination should aim to assess similar aspects of neurological functioning evaluated during an in-person medical assessment with modifications based on the presence of a remote telepresenter or examiner.

- The unassisted virtual physical examination should include assessment of mental status, speech, cranial nerves, coordination, balance, gait, cognitive functioning, oculomotor functioning, and the cervical spine.
- Certain aspects of the physical examination including assessment of motor, sensory and vestibular functioning; palpation and provocative testing of the cervical spine; testing of visual acuity and deep tendon reflexes; as well as fundoscopic and otoscopic examinations cannot be performed virtually.

Tool 15.2: [Considerations for a virtual physical examination for medical assessment and follow-up of concussion patients](#)

Link: [Recommendation 2.1c](#) -recommendations on an in-person medical assessment.

Level of Evidence: C

15.1e [Recommendation 2.1d](#): “Consider CT of the brain or cervical spine only in patients whom, after a medical assessment, a structural intracranial or cervical spine injury is suspected; do not conduct routine neuroimaging for the purpose of diagnosing concussion.”

15.2 Provide verbal information and written (electronic) handouts regarding the course of recovery and when the child/adolescent can return to school/activity/sport/driving/work and strategies to promote recovery.

This should include the following:

- Guidance on the appropriate use of rest and guidance on making a gradual return to symptom-limiting cognitive, school, social, and low-risk physical activities.
 - Recommendation 2.3 Recommend graduated return to cognitive and physical activity to promote recovery.
 - Domain 12: Return to school
- Guidance on sleep (Recommendation 2.4 a)
- Recommendation on social networks and interactions (Recommendation 2.4b)
- Recommendation on driving (Recommendation 2.4e)
- Guidance on the appropriate use of over-the-counter medications for symptom management (Recommendation 2.5)
- Recommendation on screen time and electronic device use (Recommendation 2.4c)
- Recommendation to avoid alcohol and recreational drugs (Recommendation 2.4d)

Where available, provide post-injury education that is appropriate for the patient’s culture and/or preferred language.

Summary of online tools to consider related to lifestyle strategies and expectations:

- Link: [Advice for gradually resuming intellectual, physical and sports activities English/ French \(INESSS\)](#)
- Link: [After a Concussion: Return to Sport Strategy](#) (Parachute Canada)
- Link: [HEADS UP Resources for Returning to School](#) (CDC)
- Link: [Return to Activity Strategy](#) (CATT)
- Link: [Return-to-School Strategy](#) (CATT)
- Link: [Return to Sport Strategy](#) (CATT)
- Link: [Return to School Strategy](#) (Parachute)
- Link: [Post Concussion Information Sheet](#) (Parachute)
- Link: [Post Concussion Information Sheet for First Nations Youth](#) (Parachute)
- Link: [Post Concussion Information Sheet for Nunavut Youth – French](#) (Parachute)
- Link: [Post Concussion Information Sheet for Nunavut Youth – English](#) (Parachute)
- Link: [Post Concussion Information Sheet for Nunavut Youth – Inuktitut](#) (Parachute)
- Link: [Post Concussion Information Sheet for Nunavut Youth – Inuinnaqtun](#) (Parachute)

Level of Evidence: C

15.3 Provide a written (electronic) medical assessment or clearance letter to the child/adolescent and the parent/caregiver.

Link: [Canadian Guideline on Concussion in Sport Medical Assessment Letter](#)

Link: [Canadian Guideline on Concussion in Sport Medical Clearance Letter](#)

Level of Evidence: C

15.4 Provide a medical follow-up assessment on children/adolescents with concussion in 1-2 weeks to re-assess and monitor clinical status. A regular medical follow-up is also recommended if a child/adolescent is still experiencing post-concussion symptoms or has not completed the return-to-

school or return-to-sport/activity stages. Recommend immediate in-person medical attention in the presence of any deterioration.

While in-person medical follow-up is ideal, telemedicine (e.g., real-time in-person videoconferencing) may be considered for follow-up appointments for the following patients with concussion:

- a) Those for whom access or travel for follow-up is limited or unavailable such as those who live in rural and remote communities.
- b) Those who have undergone a previous in-person medical assessment by the treating physician or nurse practitioner
- c) Those whose symptoms are stable, improving, or resolved
- d) Those who do not report any subjective red flags (e.g., blurred or double vision, weakness or numbness, vertigo) and who have a satisfactory virtual physical examination with no objective red flags (e.g., diplopia on extraocular movement testing, positive pronator drift, pain on testing of cervical spine range of motion)
- e) Those who do not require supplemental testing (diagnostic imaging, neuropsychological testing, graded aerobic exercise testing), and/or inter-disciplinary referrals to optimize patient care

Tool 15.1: Considerations for Telemedicine and Virtual Care Algorithm

Level of Evidence: C

Please refer to [Domain 3: Medical follow-up and Management of Prolonged Symptoms](#) for a full list of clinical guideline recommendations to consider when performing a medical follow-up assessment and managing prolonged concussion symptoms.

15.5 Please refer to [Domain 4: Medical Clearance for full-contact sports and high-risk activities](#) for a full list of clinical recommendations to consider when determining if a child/adolescent is ready to return to full-contact sports or high-risk activities. Medical clearance is not required to return to school.

15.6 Recommendations for inter-disciplinary healthcare professionals involved in the care of concussion patients:

a. Neuropsychologists

Virtual care may be considered to assist in the assessment and longitudinal care of concussion patients who develop persistent cognitive and mood-related symptoms or who are having persistent problems in school. Neuropsychologists should be aware of the limitations of performing certain neuropsychological tests via in-person videoconferencing. Specific symptom assessments can be administered virtually or completed before the virtual appointment and sent to the neuropsychologist ahead of time.

Level of Evidence: B

b. Physiotherapists

Virtual care may be considered by physiotherapists to advance vestibular, cervical spine and medically supervised individually tailored sub-symptom threshold aerobic exercise treatment plans in patients who have undergone previous in-person assessment by the treating physiotherapist. Physiotherapists

should recognize that a comprehensive assessment of the cervical spine and vestibular system as well as graded aerobic exercise testing, which are required to provide initial recommendations regarding targeted rehabilitation, cannot be performed virtually. If a physiotherapist is providing virtual care, they must be able to arrange an urgent in-person assessment as needed.

Level of Evidence: C

- See link for general information on telerehabilitation: TR-Telerehab Toolkit: <https://kite-uhn.com/tools/tr-telerehab-toolkit>
- Alberta Physiotherapy: Telerehabilitation Resource Guide https://www.physiotherapyalberta.ca/files/guide_telerehabilitation.pdf

c. Psychiatrists

Telemedicine may be considered to assist in the assessment and longitudinal care of concussion patients who develop persistent psychiatric and sleep-related symptoms and disorders.

Level of Evidence: B

d. Headache neurologists

Telemedicine may be considered to assist in the assessment and longitudinal care of concussion patients who develop persistent headaches.

Level of Evidence: B

e. Occupational therapists

Virtual care may be considered to assist in the assessment and longitudinal care of concussion patients who develop prolonged cognitive and mood-related symptoms and to assist with a successful return to school and other activities of daily living.

Level of Evidence: C

Section B: Managing Concussion Symptoms 1-4 Weeks Following Acute Injury



Domain 6: Headache

Introduction:

Headache is one of the most common symptoms reported by children/adolescents who sustain a concussion. In most cases, headache associated with an acute concussion will resolve spontaneously within 1-4 weeks of injury along with other concussion symptoms. In some cases, headaches can persist beyond this timeframe.

For patients with post-concussion headaches 1-2 weeks following acute injury, a repeat medical assessment must be conducted including a clinical history, physical examination, and the evidence-informed use of diagnostic imaging. The assessment should also include proper classification or characterization of the patient's headache. This assessment will help identify co-morbid medical disorders (e.g., a history of migraine or non-specific headaches or psychiatric disorders) and other factors (e.g., overuse of analgesics) that can contribute to prolonged headaches. Some prolonged post-concussion headaches can be classified according to the [International Classification of Headache Disorders \(ICHD-III\)](#), while others cannot. Proper characterization of prolonged post-concussion headaches can help provide information to execute appropriate interdisciplinary referrals and guide evidence-based management.

Tool 6.1: [Post-Concussion Headache Algorithm](#).

Recommendations:

6.1 Perform a repeat medical assessment on all patients presenting with post-concussion headaches 1-2 weeks following acute injury.

Include a focused history, physical examination, and consideration of diagnostic brain or cervical spine MRI imaging for those with focal or worrisome symptoms.

Tool 6.1: [Post-Concussion Headache Algorithm](#).

Level of Evidence: B

6.1a Take a focused clinical history.

Collect details that help to classify or characterize the headache subtype(s) that are present.

- Headache onset, location, quality or character, severity, and frequency.
- Factors that elicit or worsen headaches (e.g., bright lights, reading, exercise, foods, etc.).
- Factors that alleviate headaches.
- Associated symptoms (e.g., aura, photosensitivity, dizziness, eye strain, neck pain).
- The presence of red flags which may indicate a more severe brain injury or other intracranial pathology (e.g., worsening headaches, repeated vomiting, weakness or numbness of the extremities, visual changes).
- The level of disability associated with the headache (e.g., missed days from school).
- Use of medications or other substances.
- Psychological or social factors or conditions that can be associated with stress and headaches (e.g., mood or anxiety disorders) (Domain 8: [Mental Health](#)).
- Assess how much headaches affect day-to-day activities Link: [pedMIDAS Headache Severity Tool for Children aged 4-18](#).
- Disturbed sleep.
- Personal and family history of headaches and headache disorders (e.g., migraine).
- Future participation in full-contact sport or high-risk activities.

Level of Evidence: B

6.1b Perform a focused physical examination.

- Vital signs (resting heart rate and blood pressure).
- A complete neurological examination (cranial nerve, motor, sensory, reflex, cerebellar, gait and balance testing) Tool 2.1: [Physical Examination](#).
- A cervical spine examination (palpation, range of motion, provocative cervical spine tests) Tool 2.1: [Physical Examination](#).
- With appropriate experience, consider performing an examination of vision, oculomotor and vestibular functioning (Domain 10: [Vision, Vestibular, and Oculomotor Function](#)).

Level of Evidence: B

6.1c Consider diagnostic brain or cervical spine MRI imaging for those with focal or worrisome symptoms.

See [Recommendation 2.1d](#) for more information on when to consider diagnostic brain or cervical spine imaging.

Level of Evidence: A (CT), B (MRI)

6.1d Classify and characterize the headache subtype based on the clinical history and physical examination findings.

Consult the [International Classification of Headache Disorders \(ICHD-III\)*](#) and Tool 6.1: [Post-Concussion Headache Algorithm](#) for more information.

Common prolonged post-concussion headache subtypes include:

- Migraine, tension, or cluster headaches
- Cervicogenic headaches
- Physiological or exercise-induced headaches
- Headaches associated with prolonged visual stimulation
- Occipital neuralgia

* "Reproduced with permission of International Headache Society"

Level of Evidence: B

6.2 Provide general post-concussion education and guidance on headache management.

6.2a Advise on non-pharmacological strategies to minimize headaches including sleep hygiene, activity modifications, limiting triggers, and information on screen time.

Tool 2.7: [Strategies to Promote Good Sleep and Alertness.](#)

Link: [Sleep for Youth. CHEO Sleep Hygiene handout](#)

Recommendation 2.4c: [Advise on the use of computers, phones, and other screen devices.](#)

Level of Evidence: C

6.2b Encourage patients with headaches to engage in cognitive activity and low-risk physical activity as soon as tolerated while staying below their symptom-exacerbation threshold. Activities that pose no/low risk of sustaining a concussion (no risk of contact, collision, or falling) should be resumed even if mild residual symptoms are present or whenever acute symptoms improve sufficiently to permit activity.

See [Recommendation 2.3](#)

See Tool 2.6: [Post-Concussion Information Sheet](#) for examples of low-risk activities.

Level of Evidence: B- Gradual return to physical activity. C- Gradual return to cognitive activity.

6.2c Consider suggesting the use of a headache and medication diary in order to monitor symptoms and medications taken. Use clinical judgment and an individualized approach on use or duration of this strategy.

Link: [Headache and Medication Diary \(Boston Children's Hospital\)](#)

Level of Evidence: C

6.2d Over-the-counter medications such as acetaminophen and ibuprofen may be recommended to treat acute headache. Advise on limiting the use of these medications to less than 15 days a month and avoiding “around-the-clock” dosing to prevent overuse or rebound headaches. I.e., advise that children/adolescents avoid using over the counter medications at regular scheduled times throughout the day.

Level of Evidence: C

6.3 Refer patients who have prolonged post-concussion headaches for more than 4 weeks to an interdisciplinary concussion team or to a sub-specialist for further evaluation and management.

Prolonged headaches in pediatric concussion patients can be difficult to classify and manage and can co-occur with other prolonged post-concussion symptoms (dizziness, neck pain, sleep disturbance, cognitive or mood challenges).

If an interdisciplinary concussion team member is not available:

- Consider appropriate referral to interdisciplinary professionals who have competency-based training and clinical experience to independently manage the identified headaches and headache disorders.

If a child/adolescent with prolonged post-concussion headache has not had a recent vision assessment, refer to an optometrist for an assessment.

Level of Evidence: C

6.4 Consider initiating pharmacological therapy to treat and manage prolonged headaches while waiting for the interdisciplinary concussion team or sub-specialist referral.

For patients with post-traumatic headaches that are migrainous in nature, the use of migraine-specific abortants such as triptan class medications may be used if effective. Due to the risk of developing medication-induced headaches, limit use of abortants to fewer than 6-10 days per month.

Tool 6.2: [General Considerations Regarding Pharmacotherapy](#)

Tool 6.3: [Approved Medications for Pediatric Indications](#)

Prophylactic therapy should be considered:

- If headaches are occurring frequently
- If headaches are disabling
- If acute headache medications are contraindicated, poorly tolerated, or are being used too frequently.

Level of Evidence: B

6.5 Recommend a medical follow-up to reassess clinical status if headaches persist. Recommend an immediate medical follow-up in the presence of any deterioration. Consider early referral (prior to 4-

weeks after the acute injury) to an interdisciplinary concussion team in the presence of modifiers that may delay recovery.

See Recommendation 2.1b: [Note any modifiers that may delay recovery and use a clinical risk score to predict risk of prolonged symptoms.](#)

Level of Evidence: C- Medical follow-up. A- Early referral in the presence of modifiers that may delay recovery



Domain 7: Sleep

Introduction:

Sleep disturbances may occur following a concussion. Symptoms vary between different children/adolescents and within an individual child/adolescent. Examples include having difficulties falling asleep/staying asleep or excessive daytime sleeping. Medication use, headaches and mental health conditions (e.g. anxiety, depression) may also affect sleep. Poor sleep may be related to cognitive complaints and worse post-concussion symptoms. A repeat medical exam to rule out a more severe injury is required for children/adolescents who are experiencing sleep disturbances that last more than 1-2 weeks following a concussion. The presence of sleep disturbances following concussion is a risk factor for a prolonged recovery from concussion. Early identification allows for early targeted supportive care, close monitoring for prolonged symptoms, and consideration for early referral.

Information related to sleep hygiene and non-pharmacological strategies to improve sleep should be shared with families. Children/adolescents with prolonged sleep disturbances should be encouraged to engage in low-risk physical activity and resume sub-symptom cognitive activities. Children/adolescents who are experiencing sleep disturbances more than 4 weeks following a concussion should be referred to a cognitive behavioural therapist or an interdisciplinary concussion team. If sleep disturbances are present for more than 6 weeks following a concussion the child/adolescent may require more specialized care from a sleep specialist.

Tool 7.1: [Managing Post-Concussion Sleep Disturbances Algorithm.](#)

Tool 7.2: [Factors That May Influence the Child/Adolescent's Sleep/Wake Cycle.](#)

Recommendations:

7.1 Perform a repeat medical assessment on all patients presenting with post-concussion sleep disturbances 1-2 weeks following acute injury.

Include a focused history, physical examination, and consider diagnostic brain or cervical spine MRI imaging for those with focal or worrisome symptoms.

- Tool 7.1: [Managing post-concussion sleep disturbances algorithm.](#)

Screen for factors that may influence the child/adolescent's sleep/wake cycle and for sleep-wake disturbances such as insomnia or excessive daytime sleepiness.

- [Tool 7.2: Factors that may influence the child/adolescent's sleep/wake cycle .](#)

Level of Evidence: C

7.2 Provide general education and guidance on sleep hygiene that outlines non-pharmacological strategies to improve sleep.

- Tool 2.7: [Strategies to Promote Good Sleep and Alertness](#)
- Link: [Sleep for Youth. CHEO Sleep Hygiene handout](#)
- Link: [Holland Bloorview Kids Rehabilitation Hospital: Concussion Handbook](#)

Level of Evidence: C

7.2a Continue to encourage patients with sleep disturbances to engage in sub-symptom threshold cognitive activities and physical activities that pose no/low risk of sustaining a concussion (no risk of contact, collision, or falling) as soon as tolerated.

See [Recommendation 2.3](#).

See Tool 2.6: [Post-Concussion Information Sheet](#) for examples of low-risk activities.

Level of Evidence: B- Gradual return to physical activity. C- Gradual return to cognitive activity.

7.3 Consider managing patients who experience sleep-wake disturbances for more than 4 weeks with cognitive behavioural therapy, treat with daily supplements, and/or refer to an interdisciplinary concussion team.

Refer the child/adolescent to a cognitive behavioural specialist.

- The treatment of choice for primary insomnia and insomnia co-morbid to a medical or psychiatric condition is cognitive behavioural therapy (CBT).

Consider suggesting non-pharmacological supplements such as magnesium, melatonin, and zinc to improve sleep and recovery without the use of medication that may have side effects.

- Tool 6.3: [Approved Medications for Pediatric Indications](#)

If CBT is unavailable to the patient or the patient is waiting for CBT treatment:

- Optimize and implement sleep hygiene (Tool 2.7 [Strategies to Promote Good Sleep and Alertness](#))
- Monitor the patient weekly for the first few weeks:
- Re-emphasize that patients with sleep disturbances should continue to engage in sub-symptom threshold cognitive and physical activities that pose no/low risk of sustaining a concussion (no risk of contact, collision, or falling) as tolerated ([Recommendation 2.3](#)).
- Consider referring to an interdisciplinary concussion team

Level of Evidence: C

7.4 Refer patients with prolonged post-concussion sleep disturbances (more than 6 weeks) to a sleep specialist or an interdisciplinary concussion team if the interventions introduced at 4 weeks have been unsuccessful and sleep issues persist.

If sleep issues persist for more than 6 weeks post-acute injury, sleep hygiene can't be optimized, and if poor sleep quality is impacting the ability to return-to-school or ability to recondition:

- Refer to a sleep specialist who has experience with concussion and polysomnography or to an interdisciplinary concussion team that has the expertise to understand sleep disturbances in the context of concussion-related symptoms.

Consider ordering sleep tests to rule out possible sleep-related breathing disorders, nocturnal seizures, periodic limb movements, or narcolepsy.

- Examples of sleep tests include Sleep Study, Multiple Sleep Latency Test, and the Maintenance of Wakefulness Test.

Level of Evidence: C

7.5 Consider prescribing medication on a short-term basis if sleep has not improved after 6 weeks following the acute injury.

Ensure that medications do not result in dependency and that the patient has minimal adverse effects. The aim is to establish a more routine sleep pattern.

- Potential medication options include trazodone 12.5 mg or amitriptyline 5.0 – 10.0 mg
- Tool 6.2: [General Considerations Regarding Pharmacotherapy](#)
- Tool 6.3: [Approved Medications for Pediatric Indications](#)

If sleep disturbances persist after pharmacological treatment refer to a pediatric sleep specialist ideally with experience with concussion and polysomnography.

Level of Evidence: C

7.6 Recommend a medical follow-up to reassess clinical status if sleep disturbances persist. Recommend an immediate medical follow-up in the presence of any deterioration. Consider early referral (before 4 weeks) to an interdisciplinary concussion team in the presence of modifiers that may delay recovery.

See Recommendation 2.1b: [Note any modifiers that may delay recovery and use a clinical risk score to predict risk of prolonged symptoms.](#)

Level of Evidence: C- Medical follow-up. A- Early referral in the presence of modifiers that may delay recovery.



Domain 8: Mental Health

Introduction:

There is evidence that early identification of common mental health disorders and risk factors for mental health disorders may prevent/mitigate additional problems such as learning and behavior problems, school avoidance and exacerbation of pre-existing problems. Having a mental health disorder prior to concussion is a critical risk factor for a post-concussion mental health disorder.

Early identification of family problems or parental mental health disorders permits healthcare professionals to watch for environmental factors that are known to influence recovery from a concussion and identify any associated negative outcomes.

It is important to:

- Assess if there is an association between concussion symptoms and restrictions to activity based on the child/adolescent's mental health.
- Treat and manage the mental health disorder itself and prevent it from becoming a long-term problem.
- Encourage the child/adolescent to remain connected and engage with their peers, friends, and teammates.

Tool 8.1: [Post-Concussion Mental Health Considerations Algorithm](#)

Tool 8.2: [Management of Prolonged Mental Health Disorders Algorithm](#)

Recommendations:

8.1 Assess existing and new mental health symptoms and disorders.

Experienced and trained healthcare professionals should use appropriate screening tools to assess the child/adolescent. These assessments should be considered for children/adolescents with a history of mental health problems or with prolonged post-concussive symptoms.

Use Tool 8.1: [Post-Concussion Mental Health Considerations Algorithm](#) and refer to a mental health specialist using clinical judgment.

Assessment screening tools to consider (direct website links):

- Link: [PHQ-SADS \(somatic\)](#)
- Link: [Severity Measure for Depression- Child Age 11–17 \(adapted from PHQ-9 modified for Adolescents \[PHQ-A\]\)](#)
- Link: [Severity Measure for Generalized Anxiety Disorder-Child Age 11–17 \(adapted from GAD-7\)](#)
- Link: [HEADS-ED Tool](#)—Screening for Pediatric Mental Health (online interactive tool)
- PROMIS Anxiety (pediatric and parent versions are available via [HealthMeasures.net](#))
- PROMIS Depression (pediatric and parent versions are available via [HealthMeasures.net](#))

- PROMIS Profile (25 questions, 37 questions, 49 questions versions are available via [HealthMeasures.net](https://healthmeasures.net))
- Link: [Children’s Somatization Inventory \(CSI\)](#)

Level of Evidence: A

8.2 Assess the child/adolescent’s broader environment, including family and caregiver function, mental health, and social connections.

- Ask about socioeconomic status (caregiver education, family income, occupation)
- Ask about social impacts and life stressors (school setting, friends, teammates)
- Ask the child/adolescent and parents and/or caregivers to complete the following, as appropriate:
 - Link: PROMIS Family Function
 - Link: [Patient Health Questionnaire \(PHQ-9\)](#) (self-administered screen for depression in adults)
 - Link: [Generalized Anxiety Disorder scale \(GAD-7\)](#)

Level of Evidence: B

8.3 Treat mental health symptoms or refer to a specialist in pediatric mental health.

Base the mental health treatment on individual factors, patient preferences, the severity of symptoms, and co-morbidities.

Tools to assist healthcare professionals to make treatment decisions:

- Tool 8.1: [Post-Concussion Mental Health Considerations Algorithm](#)
- Tool 8.2: [Management of Prolonged Mental Health Disorders Algorithm](#)
- Tool 6.2: [General Considerations Regarding Pharmacotherapy](#)
- Tool 6.3: [Approved Medications for Pediatric Indications](#)

Consider referring to a local healthcare professional, specialized pediatric concussion program or to a specialist with experience in pediatric mental health if child/adolescent has prolonged or urgent mental health symptoms. Provide the name of a specialist with experience in pediatric mental health.

For deciding when to refer a child/adolescent to a specialist, use Tool 8.1: [Post-Concussion Mental Health Considerations Algorithm](#).

Level of Evidence: B



Domain 9: Cognition

Introduction:

Prolonged cognitive problems post-concussion that are affecting a child/adolescent's daily functioning (e.g., problems with speech, learning, attention, memory, information processing, etc.) need to be identified and managed appropriately. Identifying the nature and interaction between pre-existing and concussion-related cognitive problems will help to clarify the most appropriate supports based on the child/adolescent's characteristics, including return-to-school and return-to-activity/sport. Experienced school-based educational professionals, where available, can support the healthcare professionals in this area. Examples of school-based educational professionals vary and may include: vice-principals, learning support teachers (LST), guidance counsellors, student success teachers, and school board services such as psychology, social work, speech language pathology (SLP), etc.

Recommendations:

9.1 Evaluate a child/adolescent for cognitive symptoms that interfere with daily functioning following the acute injury.

For symptoms that interfere with daily functioning for more than 4 weeks following acute injury, further evaluation by experienced professionals to assess cognitive problems may be required.

Depending on the nature of the cognitive symptoms, examples of professionals may include:

- Experienced educational professionals.
- Pediatric neuropsychologists.
- Occupational therapists.
- Speech language pathologists.

Other assessments may be required to determine the underlying cause(s) and any pre-existing contributing factors that can be managed:

- Use a risk score to assess any modifiers that may delay recovery ([Recommendation 2.1b](#)).
- Mental health assessment and a closer look at the family (Domain 8: [Mental Health](#)).
- Vision, vestibular, and hearing assessments (Domain 10: [Vision, Vestibular, and Oculomotor Function](#)).
- Physical examination (Tool 2.1: [Physical Examination](#)).
- As per usual pediatric clinical practice, broad clinical history taking is recommended to understand the youth's developmental, medical, social, academic, and family histories. Particular consideration should be given to the interplay between these pre-existing factors and current cognitive profile/presentation/symptoms.

Level of Evidence: B

9.2 Manage cognitive symptoms that interfere with daily functioning for more than 4 weeks following acute injury.

See [Domain 3: Medical Follow-up and Management of Prolonged Symptoms](#)

See Domain 12: [Return-to-School and Work](#) for suggestions to guide an initial discussion about the best pathways for the student in school, employment, sports, social, and home environments. Tools and tests should be used in conjunction with an examination of previous school records such as marks and teacher observations.

Level of Evidence: B



Domain 10: Vision, Vestibular, and Oculomotor Function

Introduction:

Patients with acute head and neck trauma can present with symptoms such as dizziness, blurred or double vision, vertigo, postural imbalance, difficulty focusing, motion sensitivity, and/or headaches during reading. These symptoms may be suggestive of dysfunction within the neurological systems responsible for visual, vestibular, and oculomotor functioning, including balance, and gait. Identification of these deficits can aid in the provision of academic and activity-related accommodations during the acute stage after injury. Most acute concussion patients with these clinical features will experience symptom resolution and return to daily activities within 4 weeks following the acute injury and these patients will only need supportive care and anticipatory guidance. By conducting early screening for impairments in visual, vestibular, and oculomotor functioning as well as balance and gait, appropriate referrals to an interdisciplinary concussion team and sub-specialists can be initiated to provide evidence-based targeted interventions.

A repeat medical assessment on concussion patients with prolonged dizziness, blurred or double vision, vertigo, difficulty reading, postural imbalance, or headaches elicited by prolonged visual or vestibular stimulation is required 1-2 weeks following the acute injury. Although the cause of these prolonged symptoms can be multi-factorial, the assessment often reveals impairments in vestibular functioning, balance, or vision. The repeat medical assessment should include a focused clinical history, focused physical examination, and a consideration for the use of additional diagnostic tests as indicated (i.e., screening vestibular oculomotor assessment, visual field testing, and neuroimaging). The medical assessment must consider conditions such as intraparenchymal hemorrhage, stroke, traumatic cranial neuropathy, or temporal bone fractures.

Tool 10.1: [Post-concussion Vision, Vestibular, and Oculomotor Disturbances Algorithm](#).

Oculomotor or Vision Deficits:

Visual and oculomotor deficits can be due to cranial neuropathies, structural brain injuries, or functional impairments in convergence, accommodation, smooth pursuits, saccades, and vestibulo-ocular reflex functioning. Visual deficits are common symptoms following a concussion. In some patients, these deficits will spontaneously recover and will only need monitoring, supportive care, and anticipatory guidance. However, there is evidence that these deficits may also be associated with increased risk for prolonged symptoms. Identifying these deficits early will allow for early targeted supportive care, management, close monitoring for prolonged, and early referral for further treatment.

Benign Paroxysmal Positional Vertigo:

Benign paroxysmal positional vertigo (BPPV) can be caused by the traumatic displacement of the crystals (otoconia) of the inner ear into one of the semi-circular canals. This displacement results in intermittent brief episodes of vertigo and a characteristic pattern of nystagmus (involuntary eye movement) with head movements that stimulate fluid flow in the affected canal (e.g., laying down, sitting up, rolling in bed, looking up, bending over, rapid horizontal head movements). If the patient reports vertigo or dizziness that occurs for seconds following position changes, a screen for BPPV and consideration for targeted particle re-positioning manoeuvres should be conducted. In patients who continue to experience prolonged vertigo or dizziness after completing 3 particle repositioning manoeuvres,

consideration should be given to a referral to an interdisciplinary concussion team or sub-specialist (i.e., otolaryngology) for further assessment and management.

Vestibulo-ocular Deficits:

The vestibulo-ocular reflex (VOR) enables clear vision with head motion. In many cases, difficulties with clear vision during head motion are reported following a concussion and there is emerging evidence that alterations in VOR function may predict a longer recovery following concussion. Dizziness and/or blurred vision with head motion may be reported and should be further investigated by a healthcare professional with experience in this area. Vestibular rehabilitation has been reported to facilitate recovery when a child/adolescent is experiencing altered gain of the VOR.

Recommendations:

10.1 Perform a repeat medical assessment on all patients presenting with dizziness, blurred or double vision, vertigo, difficulty reading, postural imbalance, or headaches elicited by prolonged visual or vestibular stimulation 1-2 weeks following acute injury.

Depending on the nature of the symptoms, the medical assessment should include a focused history, focused physical examination, and consideration for the need for diagnostic brain or cervical spine MRI imaging for those with focal or worrisome symptoms.

Tool 10.1: [Post-concussion Vision, Vestibular, and Oculomotor Disturbances Algorithm](#)

Tool 2.1: [Physical Examination](#)

[Recommendation 2.1d](#): When to consider diagnostic brain or cervical spine imaging.

Level of Evidence: B

10.2 Screen for oculomotor or vision deficits

Perform an assessment of visual acuity, pupillary function, visual fields, fundoscopy, and extra-ocular movements.

- With appropriate experience, consider an objective assessment of convergence, accommodation, saccades and smooth pursuits.
- Consider additional tests such as automated visual field testing, formal vestibular testing or diagnostic imaging.

Assessment video to consider:

Link: [Visio-vestibular examination](#) (The Children's Hospital of Philadelphia).

Consider referral to an interdisciplinary concussion team or neuro-ophthalmologist, neuro-optometrist, developmental optometrist, occupational therapist, or physiotherapist with competency-based training in vestibular rehabilitation to assess for impairments in convergence, accommodation, saccades and other visual oculomotor disorders.

Level of Evidence: C

10.3 Screen for benign paroxysmal positional vertigo (BPPV) if the patient reports vertigo or dizziness that occurs for seconds following position changes and consider targeted particle re-positioning manoeuvres.

After completing a neurological screen and clearing the cervical spine to move into the test position, perform the Dix-Hallpike Test. If positive for BPPV (i.e., reproduction of vertigo, typically for seconds, in addition to a characteristic pattern of nystagmus for the canal that is being assessed), a Particle Repositioning Manoeuvre may be appropriate.

Consider the Epley Manoeuvre which can be used to treat the anterior and posterior canals in the case of a canalithiasis. There are many subtypes of BPPV that may require further assessment or alternate canalith repositioning manoeuvres and referral to a healthcare professional (often a physiotherapist with competency-based training in vestibular rehabilitation) for treatment. If symptoms are provoked by pressure (i.e., val salva) or accompanied by a change in hearing, referral to an otolaryngologist or neuro-otologist is warranted.

In patients who continue to experience prolonged vertigo or dizziness despite 3 particle repositioning manoeuvres, consider referral to an interdisciplinary concussion team or neuro-otologist or physiotherapist with competency-based training in vestibular rehabilitation. These experienced healthcare professionals should rule out alternative peripheral and central vestibular disorders (e.g., superior semi-circular canal dehiscence (SSCD), vestibular hypofunction) and initiate active management with rehabilitation or referral as appropriate.

Online videos to consider:

- Link: [Dix-Hallpike Test](#) (University of Calgary)
- Link: [Epley Manoeuvre](#) (University of Calgary)

Level of Evidence: C

10.4 Screen for vestibulo-ocular deficits.

With appropriate experience, perform an assessment of the vestibulo-ocular reflex (VOR) such as the head thrust test and dynamic visual acuity.

Consider referral to a physiotherapist with competency-based training in vestibular rehabilitation.

Online videos to consider:

- Link: [Head Thrust Test and Dynamic Visual Acuity \(University of Calgary\)](#)
- Link: [Visio-Vestibular Examination \(The Children's Hospital of Philadelphia\)](#)
- Link: [Dynamic Visual Acuity \(University of Calgary\)](#)

Level of Evidence: B

10.5 Screen for balance deficits.

Assess for prolonged balance deficits and determine which systems (visual reflexes, inner ear, musculoskeletal, nervous system or brain) might be contributing to dizziness, headaches, and balance

problems. Vestibular rehabilitation may improve balance and dizziness. If prolonged impairment is identified, refer to a specialist immediately.

Perform assessment of postural stability and balance.

- Standing balance test (eyes open/closed, tandem stance, single leg stance), Balance Error Scoring System.
- Dynamic balance: Consider the Functional Gait Assessment, BOT.

Consider referral to an interdisciplinary concussion team or physiotherapist with competency-based training in vestibular rehabilitation.

Online videos to consider:

- Link: [Visio-Vestibular Examination](#) (The Children's Hospital of Philadelphia)
- Link: [Balance Error Scoring System](#)
- Link: [Functional Gait Assessment](#) (University of Calgary)

Level of Evidence: C

10.6 Screen for and consider underlying psychosocial contributors of visual, vestibular, and oculomotor dysfunction.

See Domain 8: [Mental Health](#).

Level of Evidence: C

10.7 Provide general post-concussion education that outlines symptoms of concussion, provides suggestions regarding activity modification and includes academic accommodations to manage visual, vestibular and oculomotor symptoms.

Guidance about how to make a gradual return-to-school, cognitive activities, and physical activities:

- Recommendation 2.3: [Recommend graduated return to cognitive and physical activity](#)
- Recommendation 2.4c: [Advise on the use of computers, phones, and other screen devices](#)
- Domain 12: [Return-to-School and Work](#)

Level of Evidence: C

10.8 Encourage patients with post-concussion vestibular, vision or oculomotor symptoms to engage in cognitive activity and low-risk physical activity as soon as tolerated while staying below their symptom-exacerbation thresholds. Activities that pose no/low risk of sustaining a concussion (no risk of contact, collision, or falling) should be resumed even if mild residual symptoms are present or whenever acute symptoms improve sufficiently to permit activity.

See [Recommendation 2.3](#).

See Tool 2.6: [Post-Concussion Information Sheet](#) for examples of low-risk activities.

Level of Evidence: B- Gradual return to physical activity. C- Gradual return to cognitive activity.

10.9 Refer patients with prolonged post-concussion vestibular functioning, balance or visual dysfunction (more than 4 weeks following the acute injury) to an interdisciplinary concussion team with appropriate experience. Consider early referral (before 4 weeks) to an interdisciplinary concussion team in the presence of modifiers that may delay recovery.

Tool 10.1: [Post-Concussion Vision, Vestibular, and Oculomotor Disturbances Algorithm](#).

Assessment by an interdisciplinary concussion team can assist in identifying the type of management that is required, along with the medical and health professions on the interdisciplinary concussion team or external to this team who can provide the required management. Not all children/adolescents will require care from all members of the interdisciplinary concussion team and care should be targeted based on identified symptoms and patient needs.

See Recommendation 2.1b: [Note any modifiers that may delay recovery and use a clinical risk score to predict risk of prolonged symptoms](#).

Level of Evidence: C- Medical follow-up. A- Early referral in the presence of modifiers that may delay recovery

10.10 Recommend a medical follow-up to reassess clinical status if vestibular functioning, balance or visual dysfunction symptoms persist. Recommend an immediate medical follow-up in the presence of any deterioration.

Level of Evidence: C



Domain 11: Fatigue

Introduction:

Physical, mental, or cognitive fatigue is common following a concussion. Fatigue can be defined as weariness or tiredness following physical and/or cognitive/mental exertion. A repeat medical exam is necessary for children/adolescents experiencing fatigue 1-2 weeks following a concussion to rule out a serious medical condition or injury. Children/adolescents experiencing prolonged post-concussion fatigue should be encouraged to participate in low-risk physical and cognitive activities below their symptom exacerbation threshold (at a level that does not bring on symptoms or make symptoms worse). Pacing and energy management techniques should be shared with the child/adolescent (Tool 2.5 [“Four P’s” – Prioritize, Plan, Pace and Position](#)). If a child/adolescent experiences fatigue for more than 4 weeks following the acute injury consider referring to an interdisciplinary concussion team.

Recommendations:

11.1 Perform a repeat medical assessment on all patients presenting with post-concussion fatigue 1-2 weeks following acute injury.

The medical assessment should include a clinical history of symptoms, physical examination, and screen for other causes of fatigue (e.g., mononucleosis, anemia, thyroid dysfunction, mood disorders, pregnancy, etc.).

Level of Evidence: C

11.2 Provide patients with post-concussion fatigue with general education and guidance that outlines non-pharmacological strategies to help cope with fatigue symptoms and set expectations.

Strategies and post-concussion education guidance related to fatigue:

- Emphasize that spreading activities throughout the day helps patients achieve more and that they should avoid doing too much at once. Tool 2.5 [“Four P’s” – Prioritize, Plan, Pace and Position](#)
- Encourage good diet and hydration
- Encourage good sleep hygiene. Tool 2.7: [Strategies to Promote Good Sleep and Alertness](#)
- Link: [Sleep for Youth. CHEO Sleep Hygiene handout](#)
- Avoid daytime napping
- Identify the triggers of fatigue
- Encourage the child/adolescent to plan meaningful goals, record activity achievement, and identify patterns of fatigue by using a notebook or diary.
- Inform that fatigue can be worsened by low mood or stress.
- Inform that deconditioning can compound fatigue.
- Encourage a gradual return-to-school with accommodation (Domain 12: [Return-to-School and Work](#))

Level of Evidence: C

11.3 Encourage patients with post-concussion fatigue to engage in cognitive activity and low-risk physical activity as soon as tolerated while staying below their symptom-exacerbation thresholds. Activities that pose no/low risk of sustaining a concussion (no risk of contact, collision, or falling) should be resumed even if mild residual symptoms are present or whenever acute symptoms improve sufficiently to permit activity.

See [Recommendation 2.3](#).

See Tool 2.6: [Post-Concussion Information Sheet](#) for examples of low-risk activities.

Level of Evidence: B- Gradual return to physical activity. C- Gradual return to cognitive activity.

11.4 Consider referral to an interdisciplinary concussion team for patients with prolonged post-concussion fatigue (more than 4 weeks following the acute injury) to learn pacing techniques.

Level of Evidence: C

11.5 Recommend a medical follow-up to re-assess clinical status if fatigue symptoms persist. Recommend an immediate medical follow-up in the presence of any deterioration. Consider early referral (before 4 weeks) to an interdisciplinary concussion team in the presence of modifiers that may delay recovery.

See Recommendation 2.1b: [Note any modifiers that may delay recovery and use a clinical risk score to predict risk of prolonged symptoms](#).

Level of Evidence: C- Medical follow-up. A- Early referral in the presence of modifiers that may delay recovery



Domain 12: Return-to-School and Work

Introduction:

Parents and/or caregivers need to be made aware that most youth will experience symptom resolution and full return to daily activities following a concussion; however this is highly variable and individual. It is reasonable for a child/adolescent to miss some school after a concussion, regardless of symptoms. However, it is also important not to allow or encourage the child/adolescent to "settle into the habit" of missing school. The school setting provides beneficial contact with peers and social support. A gradual return-to-school and activity is the best way to make sure that the child/adolescent remains symptom-free when he or she fully engages in school and related activities.

Overview:

- The return-to-school process should be coordinated by the school's concussion management team and/or a point person in the school (e.g., guidance counsellor, principal/vice principal, teacher, etc.).
- A key to the initial management of concussion is a gradual return-to-school and activities as symptoms are improving.
- Manage the gradual return to school/activity/sport on a case-by-case basis.
- Making short-term changes to a student's school workload and schedule can help the child/adolescent get back to their regular school routine. Accommodations can be modified as symptoms resolve.
- Recommend an additional assessment or referral to an interdisciplinary concussion team if symptoms worsen or fail to improve 2-4 weeks following the acute injury.

Identifying and managing new or pre-existing school difficulties will:

- Clarify the most appropriate treatment and management options and accommodations based on the child/adolescent's characteristics
- Promotes cognitive recovery and successful reintegration at school or work
- Support/assist return-to-school, engagement in daily activities, return to social engagement, management and treatment of symptoms

Recommendations:

12.1 Recommend a stepwise return-to-school plan and monitor once the student is ready to start a graduated return-to-school. Include temporary accommodations based on symptoms and recommendations from the healthcare professional. Modify the return-to-school plan based on ongoing assessment of symptoms.

This involves collaboration and communication among healthcare professionals, school-based professionals, the child/adolescent, and parents/caregivers.

Summary of tools to consider: These tools are suggestions for initiating a discussion to determine the best pathways for the student in learning environments.

- Tool 12.1: [Concussion Implications and Interventions for the Classroom](#)
- Tool 12.2: [Template: Letter of Accommodation from the Concussion Care Team to the School](#)
- Tool 12.3: [Template Letter of Accommodation from Physician to School](#)
- Tool 12.4: [Sample Letter/Email from School to Parents](#)
- Link: [CATT Student Return to School Strategy](#)
- Link: [CATT Return to School Strategy](#)
- Link: [Heads Up Schools: Helping Students Recover from a Concussion: Classroom Tips for Teachers](#)
- Link: [SCHOOLFirst: Enabling successful return to school for Canadian youth following a concussion](#) (Holland Bloorview Kids Rehabilitation Hospital)
- Link: [Parachute's Protocol for Return to Learn After a Concussion](#) (Parachute Canada)
- Link: [Post-Concussion Academic Accommodation Protocol](#) (University of Oregon)

Level of Evidence: B

12.2 Assess for school difficulties using clinical judgment.

Determine how much school the child/adolescent has missed post-concussion and how much missed workload the child/adolescent is expected to catch up on from missed school days.

Obtain school records to determine what issues may have been present prior to the concussion

School or cognitive difficulties may overlap with vision, vestibular, hearing, mental health, and social/family issues. Please assess.

- Domain 8: [Mental Health](#)
- Domain 9: [Cognition](#)
- Domain 10: [Vision, Vestibular, and Oculomotor Function](#)

Level of Evidence: B

12.3 Manage school difficulties.

On re-evaluation, experienced health professionals (and school-based educational professionals where available) should manage school cognitive difficulties, provide accommodations, and reduce stressors. This should be done in collaboration with the child/adolescent, parents/caregivers, schools and/or employers to support success in the home, school, and community.

Refer to an interdisciplinary concussion team and/or a school-based educational professional (if available) if symptoms interfere with daily functioning more than 4 weeks following a concussion (Domain 9: [Cognition](#)). Refer for a formal evaluation if school difficulties may have been pre-existing.

Use tools to encourage reintegration within the school, employment, sports, social, and home environments. Summary of tools to consider:

- Link: [SCHOOLFirst: Enabling successful return to school for Canadian youth following a concussion](#) page 6 (Holland Bloorview Kids Rehabilitation Hospital)
- Tool 12.1: [Concussion Implications and Interventions for the Classroom](#)

- Tool 12.2: [Template: Letter of Accommodation from the Concussion Care Team to the School](#)
- Tool 12.3: [Template Letter of Accommodation from Physician to School](#)
- Tool 12.4: [Sample Letter/Email from School to Parents](#)

Level of Evidence: C

12.4 Encourage patients with school difficulties to engage in cognitive activity and low-risk physical activity as soon as tolerated while staying below their symptom-exacerbation thresholds. Activities that pose no/low risk of sustaining a concussion (no risk of contact, collision, or falling) should be resumed even if mild residual symptoms are present or whenever acute symptoms improve sufficiently to permit activity.

See [Recommendation 2.3.](#)

Level of Evidence: B- Gradual return to physical activity. C- Gradual return to cognitive activity.

12.5 Return-to-school and return-to-sport strategies can be performed simultaneously. Recommend that the child/adolescent return-to-school full-time at a full academic load, including writing exams without accommodations related to their concussion/post-concussion symptoms, before returning to full-contact sport or high-risk activities.

See Domain 4: [Medical clearance for full-contact sport or high-risk activity.](#)

Level of Evidence: B

12.6 Prioritize return-to-school before return to work.

For teens who work, please consult the “[Guidelines for Concussion/ Mild Traumatic Brain Injury and Persistent Symptoms 3rd Edition For Adults \(18+ years of age\)](#)” for recommendations on how to work with the adolescent’s employer regarding non-academic accommodations so that the adolescent can gradually return-to-work while promoting recovery.

Level of Evidence: B- Need for rest. C- Ideal duration of rest. B- Starting return to activity earlier.

Section C: Additional Information



Domain 13: Biomarkers – Neuroimaging

Recommendations:

13.1 At this stage, advanced neuroimaging biomarkers are not yet ready for clinical implementation/management.

Biomarkers such as functional MRI (fMRI), diffusion tensor imaging (DTI), magnetic resonance spectroscopy (MRS), arterial spin labeling (ASL), cerebrovascular-reactivity mapping (CVR), quantitative susceptibility based susceptibility weighted imaging (qSWI), electroencephalography/event-related potential (EEG/ERP), transcranial magnetic stimulation (TMS), while potentially useful as research tools, are not ready for clinical implementation.

Level of Evidence: C

13.2 When conventional MRI is performed in the clinical management of concussion patients, the inclusion of susceptibility-weighted images (SWI) sequences could be considered as it may be useful for detecting small hemorrhages. The clinical significance of small hemorrhages on SWI is not clear at present.

See [Recommendation 2.1d](#) for more information on when to consider diagnostic brain or cervical spine imaging.

Level of Evidence: C



Domain 14: Biomarkers – Serologic

Recommendations:

14.1 The use of serologic biomarkers is not clinically indicated. Presently there is no validated “concussion blood test” that can be used to accurately detect concussion in children/adolescents.

At this stage, newer serologic and other clinical biomarkers, while potentially useful as research tools, are not ready for clinical implementation/management.

Level of Evidence: C

TOOLS

Tool 1.1: [Pediatric Concussion: The role of school boards, community sports organizations, and centres](#)

Tool 1.2: [Concussion Recognition Tool 5. To help identify concussion in children, adolescents, and adults](#)

Tool 1.3: [Manage Acute and Prolonged Symptoms Algorithm](#)

Tool 2.1: [Physical Examination](#)

Tool 2.2: [PECARN Management Algorithm for Children after Head Trauma](#)

Tool 2.3: [CATCH2 Rule](#)

Tool 2.4: [Algorithm for the Management of the Pediatric Patient \$\geq 2\$ years with Minor Head Trauma](#)

Tool 2.5: [“Four P’s” – Prioritize, Plan, Pace, and Position](#)

Tool 2.6: [Post-Concussion Information Sheet](#)

Tool 2.7: [Strategies to Promote Good Sleep and Alertness](#)

Tool 6.1: [Post-Concussion Headache Algorithm](#)

Tool 6.2: [General Considerations Regarding Pharmacotherapy](#)

Tool 6.3: [Approved Medications for Pediatric Indications](#)

Tool 7.1: [Managing Post-Concussion Sleep Disturbances Algorithm](#)

Tool 7.2: [Factors That May Influence the Child/Adolescent’s Sleep/Wake Cycle](#)

Tool 8.1: [Post-Concussion Mental Health Considerations Algorithm](#)

Tool 8.2: [Management of Prolonged Mental Health Disorders Algorithm](#)

Tool 10.1: [Post-Concussion Vestibular \(balance/dizziness\) and Vision Disturbances Algorithm](#)

Tool 12.1: [Concussion Implications and Interventions for the Classroom](#)

Tool 12.2: [Template: Letter of Accommodation from the Concussion Care Team to the School](#)

Tool 12.3: [Template Letter of Accommodation from Physician to School](#)

Tool 12.4: [Sample Letter/Email from School to Parents](#)

Tool 15.1: [Considerations for Telemedicine and Virtual Care Algorithm](#)

Tool 15.2 [Considerations for a virtual physical examination for medical assessment and follow-up of concussion patients](#)

REFERENCES

Updated: January 4, 2021

Abbassi E, Sirmon-Taylor B. Recovery progression and symptom resolution in sport-related mild traumatic brain injury. *Brain Injury*, (2017). 31(12),1667-1673.

<https://doi.org/10.1080/02699052.2017.1357834>

Abeare CA, Messa I, Zuccato BG, Merker B, Erdodi L. Prevalence of invalid performance on baseline testing for sport-related concussion by age and validity indicator. *JAMA Neurology*, (2018). 1. Shi F,(6),697-703. <https://doi.org/10.1001/jamaneurol.2018.0031>

Aggarwal SS, Ott SD, Padhye NS, et al. Sex, race, ADHD, and prior concussions as predictors of concussion recovery in adolescents. *Brain Inj* [Internet]. 2020 May 11;34(6):811–9. Available from: <https://www.tandfonline.com/doi/full/10.1080/02699052.2020.1740942>

Aggarwal SS, Ott SD, Padhye NS, Meininger JC, Armstrong TS. Clinical and demographic predictors of concussion resolution in adolescents: A retrospective study. *Applied Neuropsychology: Child*, (2019). 8(1),50-60. <https://doi.org/10.1080/21622965.2017.1381099>

Alberts JL, Modic MT, Udeh B, et al. Development and implementation of a multi-disciplinary technology enhanced care pathway for youth and adults with concussion. *J Vis Exp*. 2019 Jan 1;2019(143).

Alberts JL, Thota A, Hirsch J, Ozinga S, Dey T, ... Linder SM. Quantification of the Balance Error Scoring System with Mobile Technology. *Medicine & Science in Sports & Exercise*, (2015). 47(10),2233-2240. <https://doi.org/10.1249/MSS.0000000000000656>

Alkathiry AA, Kontos AP, Furman JM, et al. Vestibulo-Ocular Reflex Function in Adolescents With Sport-Related Concussion: Preliminary Results. *Sports Health* [Internet]. 2019 Nov 14;11(6):479–85. Available from: <http://journals.sagepub.com/doi/10.1177/1941738119865262>

Alsalaheen B, Haines J, Yorke A, Diebold J. King-Devick Test reference values and associations with balance measures in high school American football players. *Scandinavian Journal of Medicine and Science in Sports*, (2016). 26(2),235-239. <https://doi.org/10.1111/sms.12628>

Alsalaheen B, McClafferty A, Haines J, Smith L, Yorke A. Reference values for the balance error scoring system in adolescents. *Brain Injury*, (2016). 30(7),914-918. <https://doi.org/10.3109/02699052.2016.1146965>

Alsalaheen BA, Whitney SL, Marchetti GF, Furman JM, Kontos AP, ... Sparto PJ. Relationship between cognitive assessment and balance measures in adolescents referred for vestibular physical therapy after concussion. *Clinical Journal of Sport Medicine*, (2016). 26(1),46-52. <https://doi.org/10.1097/JSM.0000000000000185>

Anderson SL, Gatens D, Glatts C, et al. Normative Data Set of SWAY Balance Mobile Assessment in Pediatric Athletes. *Clin J Sport Med*. 2019 Sep 1;29(5):413–20.

Anderson V, Davis GA, Takagi M, et al. Trajectories and Predictors of Clinician-Determined Recovery after Child Concussion. *J Neurotrauma* [Internet]. 2020 Jun 15;37(12):1392–400. Available from:

<https://www.liebertpub.com/doi/10.1089/neu.2019.6683>

Anthony CA, Peterson AR. Utilization of a text-messaging robot to assess intraday variation in concussion symptom severity scores. *Clinical Journal of Sport Medicine*, (2015). 1. Shi F,(2),149-152.

<https://doi.org/10.1097/JSM.0000000000000115>

Anzalone AJ, Blueitt D, Case T, McGuffin T, Pollard K, ... Oliver JM. A Positive Vestibular/Ocular Motor Screening (VOMS) Is Associated with Increased Recovery Time after Sports-Related Concussion in Youth and Adolescent Athletes. *American Journal of Sports Medicine*, (2017). 45(2),474-479.

<https://doi.org/10.1177/0363546516668624>

Araujo GC, Antonini TN, Monahan K, Gelfius C, Klamar K, ... Bodin D. The relationship between suboptimal effort and post-concussion symptoms in children and adolescents with mild traumatic brain injury. *Clinical Neuropsychologist*, (2014). 28(5),786-801.

<https://doi.org/10.1080/13854046.2014.896415>

Argyropoulou MI, Alexiou GA, Xydis VG, et al. Pediatric minor head injury imaging practices: results from an ESPR survey. *Neuroradiology* [Internet]. 2020 Feb 11;62(2):251–5. Available from:

<http://link.springer.com/10.1007/s00234-019-02326-6>

Babcock L, Byczkowski T, Wade SL, Ho M, Bazarian JJ. Inability of S100B to predict postconcussion syndrome in children who present to the emergency department with mild traumatic brain injury: A brief report. *Pediatric Emergency Care*, (2013). 29(4),458-461.

<https://doi.org/10.1097/PEC.0b013e31828a202d>

Babcock L, Byczkowski T, Wade SL, Ho M, Mookerjee S, Bazarian JJ. Predicting Postconcussion Syndrome After Mild Traumatic Brain Injury in Children and Adolescents Who Present to the Emergency Department. *JAMA Pediatrics*, (2013). 167(2),156. <https://doi.org/10.1001/jamapediatrics.2013.434>

Babcock L, Kurowski BG, Zhang N, Dexheimer JW, Dyas J, Wade SL. Adolescents with Mild Traumatic Brain Injury Get SMART: An Analysis of a Novel Web-Based Intervention. *Telemedicine and e-Health*, (2017). 23(7),600-607. <https://doi.org/10.1089/tmj.2016.0215>

Babcock L, Yuan W, Leach J, Nash T, Wade S. White matter alterations in youth with acute mild traumatic brain injury. *Journal of Pediatric Rehabilitation Medicine*, (2015). 8(4),285-296.

<https://doi.org/10.3233/PRM-150347>

Babikian T, McArthur D, Asarnow RF. Predictors of 1-month and 1-year neurocognitive functioning from the UCLA longitudinal mild, uncomplicated, pediatric traumatic brain injury study. *Journal of the International Neuropsychological Society*, (2013). 19(2),145-154.

<https://doi.org/10.1017/S135561771200104X>

Babl FE, Dionisio D, Davenport L, Baylis A, Hearps SJC, ... Davis GA. Accuracy of Components of SCAT to Identify Children With Concussion. *Pediatrics*, (2017). 1. Shi F,(2),e20163258.

<https://doi.org/10.1542/peds.2016-3258>

Bailey C, Meyer J, Briskin S, et al. Multidisciplinary Concussion Management. *J Head Trauma Rehabil* [Internet]. 2019;34(6):375–84. Available from: <http://journals.lww.com/00001199-201911000-00002>

Baker DA, Connery AK, Kirk JW, Kirkwood MW. Embedded performance validity indicators within the california verbal learning test, childrens version. *Clinical Neuropsychologist*, (2014). 1. Shi F,(1),116-127.

<https://doi.org/10.1080/13854046.2013.858184>

Baker JG, Leddy JJ, Darling SR, Rieger BP, Mashtare TL, ... Willer BS. Factors Associated with Problems for Adolescents Returning to the Classroom after Sport-Related Concussion. *Clinical Pediatrics*, (2015). 54(10),961-968. <https://doi.org/10.1177/0009922815588820>

Barker T, Russo SA, Barker G, Rice MA, Jeffrey MG, ... Craddock TJA. A case matched study examining the reliability of using ImPACT to assess effects of multiple concussions. *BMC Psychology*, (2017). 5(1),1-9. <https://doi.org/10.1186/s40359-017-0184-1>

Barlow KM, Brooks BL, Esser MJ, et al. Efficacy of melatonin in children with postconcussive symptoms: A randomized clinical trial. *Pediatrics* [Internet]. 2020 Apr;145(4):e20192812. Available from: www.aappublications.org/news

Barlow KM, Crawford S, Brooks BL, Turley B, Mikrogianakis A. The Incidence of Postconcussion Syndrome Remains Stable Following Mild Traumatic Brain Injury in Children. *Pediatric Neurology*, (2015). 53(6),491-497. <https://doi.org/10.1016/j.pediatrneurol.2015.04.011>

Barlow KM, Giregulus KA, Goldstein G, et al. Sleep Parameters and Overnight Urinary Melatonin Production in Children With Persistent Post-concussion Symptoms. *Pediatr Neurol* [Internet]. 2020 Apr 1;105:27–34. Available from: <https://linkinghub.elsevier.com/retrieve/pii/S088789941930921X>

Bazarian JJ, Blyth BJ, He H, Mookerjee S, Jones C, ... Khan J. Classification Accuracy of Serum Apo A-I and S100B for the Diagnosis of Mild Traumatic Brain Injury and Prediction of Abnormal Initial Head Computed Tomography Scan. *Journal of Neurotrauma*, (2013). 30(20),1747-1754. <https://doi.org/10.1089/neu.2013.2853>

Beauchamp MH, Aglipay M, Yeates KO, Désiré N, Keightley M, ... Moore J. Predictors of neuropsychological outcome after pediatric concussion. *Neuropsychology*, (2018). 32(4),495-508. <https://doi.org/10.1037/neu0000419>

Beauchamp MH, Beare R, Ditchfield M, Coleman L, Babl FE, ... Anderson V. Susceptibility weighted imaging and its relationship to outcome after pediatric traumatic brain injury. *Cortex*, (2013). 49(2),591-598. <https://doi.org/10.1016/j.cortex.2012.08.015>

Beebe DW, Powers SW, Slattery EW, Gubanich PJ. Short Sleep and Adolescents' Performance on a Concussion Assessment Battery: An Experimental Sleep Manipulation Study. *Clinical Journal of Sport Medicine*, (2017). 0(0),1. <https://doi.org/10.1097/JSM.0000000000000454>

Bellerose J, Bernier A, Beaudoin C, Gravel J, Beauchamp MH. Long-term brain-injury-specific effects following preschool mild TBI: A study of theory of mind. *Neuropsychology*, (2017). 31(3),229-241. <https://doi.org/10.1037/neu0000341>

Bernard C, McKinlay A, Krieser D, Testa R, Ponsford J. Acute post-concussive symptoms in young children. *Brain Injury*, (2017). 31(11),1414-1421. <https://doi.org/10.1080/02699052.2017.1350999>

Bernard CO, McKinlay A, Krieser D, et al. Postconcussive Symptoms Following Mild TBI and Extracranial Injury: What Are the Contributing Factors? *J Int Neuropsychol Soc* [Internet]. 2020 May 11;26(5):451–63. Available from: https://www.cambridge.org/core/product/identifier/S1355617719001279/type/journal_article

Bernard CO, Ponsford JA, McKinlay A, McKenzie D, Krieser D. Predictors of post-concussive symptoms in young children: Injury versus non-injury related factors. *Journal of the International Neuropsychological Society*, (2016). 22(8),793-803. <https://doi.org/10.1017/S1355617716000709>

Bernard CO, Ponsford JL, McKinlay A, McKenzie D, Krieser D. Do Concussive Symptoms Really Resolve in Young Children? *Journal of Head Trauma Rehabilitation*, (2017). 32(6),413-424.

<https://doi.org/10.1097/HTR.0000000000000298>

Berz K, Divine J, Foss KB, Heyl R, Ford KR, Myer GD. Sex-specific differences in the severity of symptoms and recovery rate following sports-related concussion in young athletes. *Physician and Sportsmedicine*, (2013). 41(2),58-63. <https://doi.org/10.3810/psm.2013.05.2015>

Białyńska A, Salvatore AP. The auditory comprehension changes over time after sport-related concussion can indicate multisensory processing dysfunctions. *Brain and Behavior*, (2017). 7(12),1-8. <https://doi.org/10.1002/brb3.874>

Biederman J, Feinberg L, Chan J, Adeyemo BO, Woodworth KY, ... Faraone S V. Mild traumatic brain injury and attention-deficit hyperactivity disorder in young student athletes. *Journal of Nervous and Mental Disease*, (2015). 203(11),813-819. <https://doi.org/10.1097/NMD.0000000000000375>

Billeck J, Peeler J. The influence of fatiguing exercise on Sport Concussion Assessment Tool (SCAT) scoring in a female pediatric population. *Phys Sportsmed* [Internet]. 2020 Oct 1;48(4):458–62. Available from: <https://doi.org/10.1080/00913847.2020.1746979>

Black AM, Miutz LN, KV VW, et al. Baseline Performance of High School Rugby Players on the Sport Concussion Assessment Tool 5. *J Athl Train* [Internet]. 2020 Feb 1;55(2):116–23. Available from: <https://meridian.allenpress.com/jat/article/55/2/116/433794/Baseline-Performance-of-High-School-Rugby-Players>

Bock S, Grim R, Barron TF, Wagenheim A, Hu YE, ... Deibert E. Factors associated with delayed recovery in athletes with concussion treated at a pediatric neurology concussion clinic. *Child's Nervous System*, (2015). 31(11),2111-2116. <https://doi.org/10.1007/s00381-015-2846-8>

Bogner J, Brenner L, Malec J, et al. Concussive Symptoms following Pediatric Mild Traumatic Brain Injury. *J Head Trauma Rehabil* [Internet]. 2020 Jul;35(4):279–87. Available from: <https://journals.lww.com/10.1097/HTR.0000000000000565>

Bonfield CM, Lam S, Lin Y, Greene S. The impact of attention deficit hyperactivity disorder on recovery from mild traumatic brain injury. *Journal of Neurosurgery: Pediatrics*, (2013). 12(2),97-102. <https://doi.org/10.3171/2013.5.PEDS12424>

Bonow RH, Friedman SD, Perez FA, Ellenbogen RG, Browd SR, ... Rivara FP. Prevalence of Abnormal Magnetic Resonance Imaging Findings in Children with Persistent Symptoms after Pediatric Sports-Related Concussion. *Journal of Neurotrauma*, (2017). 34(19),2706-2712. <https://doi.org/10.1089/neu.2017.4970>

Borland ML, Dalziel SR, Phillips N, et al. Delayed Presentations to Emergency Departments of Children With Head Injury: A PREDICT Study. *Ann Emerg Med* [Internet]. 2019 Jul 1;74(1):1–10. Available from: <https://linkinghub.elsevier.com/retrieve/pii/S0196064418314859>

Bosworth C, Dodd JN. Noncredible effort on the Nonverbal-Medical Symptom Validity Test (NV-MSVT): Impact on cognitive performance in pediatric mild traumatic brain injury. *Appl Neuropsychol Child* [Internet]. 2020 Oct 1;9(4):367–74. Available from: <https://www.tandfonline.com/doi/full/10.1080/21622965.2020.1742717>

Botchway EN, Godfrey C, Anderson V, Catroppa C. A Systematic Review of Sleep-Wake Disturbances in Childhood Traumatic Brain Injury: Relationship with Fatigue, Depression, and Quality of Life. *The Journal of head trauma rehabilitation*, (2019). 34(4),241-256. <https://doi.org/10.1097/HTR.0000000000000446>

Boutis K, Gravel J, Freedman SB, Craig W, Tang K, ... Zemek R. The Diagnosis of Concussion in Pediatric Emergency Departments: A Prospective Multicenter Study. *Journal of Emergency Medicine*, (2018). 54(6),757-765. <https://doi.org/10.1016/j.jemermed.2018.02.041>

Boutis K, Weerdenburg K, Koo E, Schneeweiss S, Zemek R. The diagnosis of concussion in a pediatric emergency department. *Journal of Pediatrics*, (2015). 1. Shi F,(5),1214-1220.e1. <https://doi.org/10.1016/j.jpeds.2015.02.013>

Bozan, Aksel G, Kahraman HA, et al. Comparison of PECARN and CATCH clinical decision rules in children with minor blunt head trauma. *Eur J Trauma Emerg Surg* [Internet]. 2019 Oct 25;45(5):849–55. Available from: <http://link.springer.com/10.1007/s00068-017-0865-8>

Bramley H, Henson A, Lewis MM, Kong L, Stetter C, Silvis M. Sleep Disturbance Following Concussion Is a Risk Factor for a Prolonged Recovery. *Clinical Pediatrics*, (2017). 56(14),1280-1285. <https://doi.org/10.1177/0009922816681603>

Bramley H, Heverley S, Lewis MM, Kong L, Rivera R, Silvis M. Demographics and Treatment of Adolescent Posttraumatic Headache in a Regional Concussion Clinic. *Pediatric Neurology*, (2015). 52(5),493-498. <https://doi.org/10.1016/j.pediatrneurol.2015.01.008>

Bramley H, McFarland C, Lewis MM, Shaffer ML, Cilley R, ... Dias MS. Short-term outcomes of sport- and recreation-related concussion in patients admitted to a pediatric trauma service. *Clinical Pediatrics*, (2014). 53(8),784-790. <https://doi.org/10.1177/0009922814533403>

Bresee N, Aglipay M, Dubrovsky AS, Ledoux A, Momoli F, ... Zemek R. No association between metoclopramide treatment in ED and reduced risk of post-concussion headache. *American Journal of Emergency Medicine*, (2018). 36(12),2225-2231. <https://doi.org/10.1016/j.ajem.2018.04.007>

Bressan S, Clarke CJ, Anderson V, et al. Use of the sport concussion assessment tools in the emergency department to predict persistent post-concussive symptoms in children. *J Paediatr Child Health* [Internet]. 2020 Aug 21;56(8):1249–56. Available from: <https://onlinelibrary.wiley.com/doi/10.1111/jpc.14910>

Brett BL, Smyk N, Solomon G, Baughman BC, Schatz P. Long-term Stability and Reliability of Baseline Cognitive Assessments in High School Athletes Using ImPACT at 1-, 2-, and 3-year Test–Retest Intervals. *Archives of Clinical Neuropsychology*, (2016). 1. Shi F,(August),904-914. <https://doi.org/10.1093/arclin/acw055>

Brett BL, Solomon GS. The influence of validity criteria on Immediate Post-Concussion Assessment and Cognitive Testing (ImPACT) test–retest reliability among high school athletes. *Journal of Clinical and Experimental Neuropsychology*, (2017). 39(3),286-295. <https://doi.org/10.1080/13803395.2016.1224322>

Brett BL, Zuckerman SL, Terry DP, Solomon GS, Iverson GL. Normative Data for the Sway Balance System. *Clinical Journal of Sport Medicine*, (2018).(00),1. <https://doi.org/10.1097/jsm.0000000000000632>

Broglio SP, Lapointe A, O'Connor KL, McCrea M. Head Impact Density: A Model To Explain the Elusive Concussion Threshold. *Journal of Neurotrauma*, (2017). 34(19),2675-2683.

<https://doi.org/10.1089/neu.2016.4767>

Broglio SP, Williams R, Lapointe A, Rettmann A, Moore B, ... Eckner JT. Brain network activation technology does not assist with concussion diagnosis and return to play in football athletes. *Frontiers in Neurology*, (2017). 8(JUN). <https://doi.org/10.3389/fneur.2017.00252>

Brooks BL, Daya H, Khan S, Carlson HL, Mikrogianakis A, Barlow KM. Cognition in the Emergency Department as a Predictor of Recovery after Pediatric Mild Traumatic Brain Injury. *Journal of the International Neuropsychological Society*, (2016). 22(4),379-387.
<https://doi.org/10.1017/S1355617715001368>

Brooks BL, Iverson GL, Atkins JE, Zafonte R, Berkner PD. Sex Differences and Self-Reported Attention Problems During Baseline Concussion Testing. *Applied Neuropsychology: Child*, (2016). 5(2),119-126.
<https://doi.org/10.1080/21622965.2014.1003066>

Brooks BL, Kadoura B, Turley B, Crawford S, Mikrogianakis A, Barlow KM. Perception of recovery after pediatric mild traumatic brain injury is influenced by the “good old days” bias: Tangible implications for clinical practice and outcomes research. *Archives of Clinical Neuropsychology*, (2014). 29(2),186-193.
<https://doi.org/10.1093/arclin/act083>

Brooks BL, Low TA, Daya H, Khan S, Mikrogianakis A, Barlow KM. Test or rest? Computerized cognitive testing in the emergency department after pediatric mild traumatic brain injury does not delay symptom recovery. *Journal of Neurotrauma*, (2016). 33(23),2091-2096. <https://doi.org/10.1089/neu.2015.4301>

Brooks BL, Mannix R, Maxwell B, Zafonte R, Berkner PD, Iverson GL. Multiple Past Concussions in High School Football Players. *American Journal of Sports Medicine*, (2016). 44(12),3243-3251.
<https://doi.org/10.1177/0363546516655095>

Brooks BL, McKay CD, Mrazik M, Barlow KM, Meeuwisse WH, Emery CA. Subjective, but not Objective, Lingering Effects of Multiple Past Concussions in Adolescents. *Journal of Neurotrauma*, (2013). 1. Shi F,(17),1469-1475. <https://doi.org/10.1089/neu.2012.2720>

Brooks BL, Mrazik M, Barlow KM, McKay CD, Meeuwisse WH, Emery CA. Absence of differences between male and female adolescents with prior sport concussion. *Journal of Head Trauma Rehabilitation*, (2014). 29(3),257-264. <https://doi.org/10.1097/HTR.0000000000000016>

Brooks BL, Plourde V, Beauchamp MH, Tang K, Yeates KO, ... Zemek R. Predicting Psychological Distress after Pediatric Concussion. *Journal of Neurotrauma*, (2019). 36(5),679-685.
<https://doi.org/10.1089/neu.2018.5792>

Brooks BL, Sayers PQ, Virani S, Rajaram AA, Tomfohr-Madsen L. Insomnia in adolescents with slow recovery from concussion. *Journal of Neurotrauma*, (2019). 36(16),2391-2399.
<https://doi.org/10.1089/neu.2018.6257>

Brooks BL, Virani S, Khetani A, et al. Functional magnetic resonance imaging study of working memory several years after pediatric concussion. *Brain Inj [Internet]*. 2020 Jun 6;34(7):895–904. Available from: <https://www.tandfonline.com/doi/full/10.1080/02699052.2020.1753240>

Brooks MA, Snedden TR, Mixis B, Hetzel S, McGuine TA. Establishing baseline normative values for the child sport concussion assessment tool. *JAMA Pediatrics*, (2017). 171(7),670-677.
<https://doi.org/10.1001/jamapediatrics.2017.0592>

Brooks TM, Smith MM, Silvis RM, Lerer T, Mulvey CH, ... Smith SR. Symptom-Guided Emergency

Department Discharge Instructions for Children with Concussion. *Pediatric Emergency Care*, (2017). 33(8),553-563. <https://doi.org/10.1097/PEC.0000000000000797>

Brown NJ, Mannix RC, O'Brien MJ, Gostine D, Collins MW, Meehan WP. Effect of Cognitive Activity Level on Duration of Post-Concussion Symptoms. *Pediatrics*, (2014). 133(2),e299-e304. <https://doi.org/10.1542/peds.2013-2125>

Bunt SC, Didehbani N, Tarkenton T, et al. Sex Differences and Reporting of SCAT-5 Concussion Symptoms in Adolescent Athletes. *Clin J Sport Med* [Internet]. 2020 Jan 24; Publish Ahead of Print. Available from: <https://journals.lww.com/10.1097/JSM.0000000000000788>

Caccese JB, DeWolf RM, Kaminski TW, et al. Estimated Age of First Exposure to American Football and Neurocognitive Performance Amongst NCAA Male Student-Athletes: A Cohort Study. *Sport Med*. 2019 Mar 13;49(3):477–87.

Caze II T, Knell GP, Abt J, et al. Management and Treatment of Concussions via Tele-Concussion in a Pediatric Setting: Methodological Approach and Descriptive Analysis. *JMIR Pediatr Parent* [Internet]. 2020 May 6;3(2):e19924. Available from: <https://doi.org/10.2196/preprints.19924>

Chamard E, Lichtenstein JD. A systematic review of neuroimaging findings in children and adolescents with sports-related concussion. *Brain Injury*, (2018). 32(7),816-831. <https://doi.org/10.1080/02699052.2018.1463106>

Champigny CM, Rawana J, Iverson GL, et al. Influence of Anxiety on Baseline Cognitive Testing and Symptom Reporting in Adolescent Student Athletes. *J Neurotrauma* [Internet]. 2020 Dec 15;37(24):2632–8. Available from: <https://www.liebertpub.com/doi/10.1089/neu.2020.7079>

Chan C, Iverson GL, Purtzki J, Wong K, Kwan V, ... Silverberg ND. Safety of Active Rehabilitation for Persistent Symptoms After Pediatric Sport-Related Concussion : A Randomized Controlled Trial. *Archives of Physical Medicine and Rehabilitation*, (2018). 99(2),242-249. <https://doi.org/10.1016/j.apmr.2017.09.108>

Chan S, Kurowski B, Byczkowski T, Timm N. Intravenous migraine therapy in children with posttraumatic headache in the ED. *American Journal of Emergency Medicine*, (2015). 33(5),635-639. <https://doi.org/10.1016/j.ajem.2015.01.053>

Chasle V, Riffaud L, Longuet R, Martineau-Curt M, Collet Y, ... Pladys P. Mild head injury and attention deficit hyperactivity disorder in children. *Child's Nervous System*, (2016). 32(12),2357-2361. <https://doi.org/10.1007/s00381-016-3230-z>

Chern JJ, Sarda S, Howard BM, Jea A, Tubbs RS, ... Boydston W. Utility of surveillance imaging after minor blunt head trauma. *Journal of Neurosurgery: Pediatrics*, (2014). 14(3),306-310. <https://doi.org/10.3171/2014.6.PEDS13682>

Chin EY, Nelson LD, Barr WB, McCrory P, McCrea MA. Reliability and validity of the sport concussion assessment tool-3 (SCAT3) in high school and collegiate athletes. *American Journal of Sports Medicine*, (2016). 44(9),2276-2285. <https://doi.org/10.1177/0363546516648141>

Chrisman SPD, Richardson LP. Prevalence of diagnosed depression in adolescents with history of concussion. *Journal of Adolescent Health*, (2014). 54(5),582-586. <https://doi.org/10.1016/j.jadohealth.2013.10.006>

Chrisman SPD, Whitlock KB, Mendoza JA, et al. Corrigendum: Pilot Randomized Controlled Trial of an

Exercise Program Requiring Minimal In-person Visits for Youth With Persistent Sport-Related Concussion (Frontiers in Neurology, (2019), 10, (623), 10.3389/fneur.2019.00623) [Internet]. Vol. 11, Frontiers in Neurology. Frontiers Media S.A.; 2020. Available from:

<https://www.frontiersin.org/article/10.3389/fneur.2020.00006/full>

Chrisman SPD, Whitlock KB, Somers E, Burton MS, Herring SA, ... Rivara FP. Pilot study of the Sub-Symptom Threshold Exercise Program (SSTEP) for persistent concussion symptoms in youth.

NeuroRehabilitation, (2017). 40(4),493-499. <https://doi.org/10.3233/NRE-161436>

Chung AW, Mannix R, Feldman HA, et al. Longitudinal structural connectomic and rich-club analysis in adolescent mTBI reveals persistent, distributed brain alterations acutely through to one year post-injury.

Sci Rep [Internet]. 2019 Dec 11;9(1):18833. Available from: <http://www.nature.com/articles/s41598-019-54950-0>

Clark R, Stanfill AG. A Systematic Review of Barriers and Facilitators for Concussion Reporting Behavior among Student Athletes. Vol. 26, Journal of Trauma Nursing. Lippincott Williams and Wilkins; 2019. p. 297–311.

Cohen E, Rodean J, Diong C, Hall M, Freedman SB, ... Neuman MI. Low-Value Diagnostic Imaging Use in the Pediatric Emergency Department in the United States and Canada. JAMA Pediatrics, (2019). 173(8).

<https://doi.org/10.1001/jamapediatrics.2019.1439>

Cohrs G, Huhndorf M, Niemczyk N, Volz LJ, Bernsmeier A, ... Knerlich-Lukoschus F. MRI in mild pediatric traumatic brain injury: Diagnostic overkill or useful tool? Child's Nervous System, (2018). 34(7),1345-

1352. <https://doi.org/10.1007/s00381-018-3771-4>

Collings LJ, Cook NE, Porter S, Kusch C, Sun J, ... Panenka WJ. Attention-deficit/hyperactivity disorder is associated with baseline child sport concussion assessment tool third edition scores in child hockey players. Brain Injury, (2017). 31(11),1479-1485. <https://doi.org/10.1080/02699052.2017.1377351>

Concannon LG, Kaufman MS, Herring SA. The million dollar question: When should an athlete retire after concussion? Current sports medicine reports, (2014). 13(6),365-369.

<https://doi.org/10.1249/JSR.0000000000000098>

Conder RL, Conder AA, Register-Mihalik J, Conder LH, Newton S. Preliminary Normative Data on the Penn State University Symbol Cancellation Task With Nonconcussed Adolescents. Applied Neuropsychology: Child, (2015). 4(3),141-147. <https://doi.org/10.1080/21622965.2013.816849>

Connolly EJ, McCormick BF. Mild Traumatic Brain Injury and Psychopathology in Adolescence: Evidence From the Project on Human Development in Chicago Neighborhoods. Journal of Adolescent Health, (2019). 65(1),79-85. <https://doi.org/10.1016/j.jadohealth.2018.12.023>

Cook NE, Huang DS, Silverberg ND, Brooks BL, Maxwell B, ... Iverson GL. Baseline cognitive test performance and concussion-like symptoms among adolescent athletes with ADHD: examining differences based on medication use. Clinical Neuropsychologist, (2017). 31(8),1341-1352.

<https://doi.org/10.1080/13854046.2017.1317031>

Cook NE, Kelshaw PM, Caswell S V, Iverson GL. Children with Attention-Deficit/Hyperactivity Disorder Perform Differently on Pediatric Concussion Assessment. Journal of Pediatrics, (2019). 214,168-174.e1.

<https://doi.org/10.1016/j.jpeds.2019.07.048>

Cook NE, Sapigao RG, Silverberg ND, et al. Attention-Deficit/Hyperactivity Disorder Mimics the Post-

concussion Syndrome in Adolescents. *Front Pediatr* [Internet]. 2020 Feb 5;8. Available from: <https://www.frontiersin.org/article/10.3389/fped.2020.00002/full>

Copley M, Jimenez N, Kroshus E, et al. Disparities in Use of Subspecialty Concussion Care Based on Ethnicity. *J Racial Ethn Heal Disparities* [Internet]. 2020 Jun 2;7(3):571–6. Available from: <http://link.springer.com/10.1007/s40615-019-00686-6>

Corbin-Berrigan LA, Gagnon I. Postconcussion symptoms as a marker of delayed recovery in children and youth who recently sustained a concussion: A brief report. *Clinical Journal of Sport Medicine*, (2017). 27(3),325-327. <https://doi.org/10.1097/JSM.0000000000000355>

Corbin-Berrigan LA, Kowalski K, Faubert J, et al. Could Neurotracker be used as a clinical marker of recovery following pediatric mild traumatic brain injury? An exploratory study. *Brain Inj* [Internet]. 2020 Feb 23;34(3):385–9. Available from: <https://www.tandfonline.com/doi/full/10.1080/02699052.2020.1723699>

Cordingley D, Girardin R, Reimer K, Ritchie L, Leiter J, ... Ellis MJ. Graded aerobic treadmill testing in pediatric sports-related concussion: safety, clinical use, and patient outcomes. *Journal of Neurosurgery: Pediatrics*, (2016). 18(6),693-702. <https://doi.org/10.3171/2016.5.PEDS16139>

Corwin DJ, Arbogast KB, Swann C, et al. Reliability of the visio-vestibular examination for concussion among providers in a pediatric emergency department. *Am J Emerg Med* [Internet]. 2020 Sep;38(9):1847–53. Available from: <https://doi.org/10.1016/j.ajem.2020.06.020>

Corwin DJ, Probert KJ, Zorc JJ, Zonfrillo MR, Wiebe DJ. Use of the vestibular and oculomotor examination for concussion in a pediatric emergency department. *American Journal of Emergency Medicine*, (2019). 37(7),1219-1223. <https://doi.org/10.1016/j.ajem.2018.09.008>

Corwin DJ, Wiebe DJ, Zonfrillo MR, Grady MF, Robinson RL, ... Master CL. Vestibular deficits following youth concussion. *Journal of Pediatrics*, (2015). 166(5),1221-1225. <https://doi.org/10.1016/j.jpeds.2015.01.039>

Corwin DJ, Zonfrillo MR, Master CL, Arbogast KB, Grady MF, ... Wiebe DJ. Characteristics of Prolonged Concussion Recovery in a Pediatric Subspecialty Referral Population. *The Journal of Pediatrics*, (2014). 165(6),1207-1215. <https://doi.org/10.1016/j.jpeds.2014.08.034>

Corwin DJ, Zonfrillo MR, Wiebe DJ, Master CL, Grady MF, Arbogast KB. Vestibular and oculomotor findings in neurologically-normal, non-concussed children. *Brain Injury*, (2018). 32(6),794-799. <https://doi.org/10.1080/02699052.2018.1458150>

Coslick AM, Chin KE, Kalb LG, et al. Participation in Physical Activity at Time of Presentation to a Specialty Concussion Clinic Is Associated With Shorter Time to Recovery. *PM R* [Internet]. 2020 Dec 3;12(12):1195–204. Available from: <https://onlinelibrary.wiley.com/doi/10.1002/pmrj.12443>

Cossette I, Gagné M-È, Ouellet M-C, Fait P, Gagnon I, ... McFadyen BJ. Executive dysfunction following a mild traumatic brain injury revealed in early adolescence with locomotor-cognitive dual-tasks. *Brain Injury*, (2016). 30(13-14),1648-1655. <https://doi.org/10.1080/02699052.2016.1200143>

Covassin T, Crutcher B, Wallace J. Does a 20 minute cognitive task increase concussion symptoms in concussed athletes? *Brain Injury*, (2013). 27(13-14),1589-1594. <https://doi.org/10.3109/02699052.2013.823656>

Covassin T, Elbin R, Crutcher B. The Relationship Between Coping, Neurocognitive Performance, and

Concussion Symptoms in High School and Collegiate Athletes. *The sport psychologist*, (2013). 27(4),372-379. <https://doi.org/10.1123/tsp.27.4.372>

Covassin T, Elbin RJ, Bleecker A, Lipchik A, Kontos AP. Are there differences in neurocognitive function and symptoms between male and female soccer players after concussions? *American Journal of Sports Medicine*, (2013). 41(12),2890-2895. <https://doi.org/10.1177/0363546513509962>

Crichton A, Oakley E, Babl FE, Greenham M, Hearps S, ... Anderson V. Predicting Fatigue 12 Months after Child Traumatic Brain Injury: Child Factors and Postinjury Symptoms. *Journal of the International Neuropsychological Society*, (2018). 24(3),224-236. <https://doi.org/10.1017/S1355617717000893>

Crichton AJ, Ignjatovic V, Babl F, et al. Interleukin-8 Predicts Fatigue at 12 Months Post-injury in Children with Traumatic Brain Injury. *J Neurotrauma* [Internet]. 2019 Dec 27;neu.2018.6083. Available from: <https://www.liebertpub.com/doi/10.1089/neu.2018.6083>

Crowe L, Collie A, Hearps S, Dooley J, Clausen H, ... Anderson V. Cognitive and physical symptoms of concussive injury in children: A detailed longitudinal recovery study. *British Journal of Sports Medicine*, (2016). 50(5),311-316. <https://doi.org/10.1136/bjsports-2015-094663>

Crowe LM, Anderson V, Barton S, Babl FE, Catroppa C. Verbal ability and language outcome following traumatic brain injury in early childhood. *Journal of Head Trauma Rehabilitation*, (2014). 29(3),217-223. <https://doi.org/10.1097/HTR.0b013e31829babfd>

Cushman DM, Borowski L, Hansen C, et al. Gabapentin and Tricyclics in the Treatment of Post-Concussive Headache, a Retrospective Cohort Study. *Headache J Head Face Pain* [Internet]. 2019 Mar 1;59(3):371–82. Available from: <http://doi.wiley.com/10.1111/head.13451>

Custer A, Sufrinko A, Elbin RJ, Covassin T, Collins M, Kontos A. High baseline postconcussion symptom scores and concussion outcomes in athletes. *Journal of Athletic Training*, (2016). 51(2),136-141. <https://doi.org/10.4085/1062-6050-51.2.12>

Darling SR, Leddy JJ, Baker JG, Williams AJ, Surace A, ... Willer B. Evaluation of the Zurich Guidelines and exercise testing for return to play in adolescents following concussion. *Clin J Sport Med*, (2014). 24(2),128-133. <https://doi.org/10.1097/JSM.000000000000026>

Davis GA, Purcell L, Schneider KJ, Yeates KO, Gioia GA, ... Kutcher JS. Sport concussion assessment tool for childrens ages 5 to 12 years. *British Journal of Sports Medicine*, (2017). 51(862-869),bjsports-2017-097492childscat5. <https://doi.org/10.1136/bjsports-2017-097492childscat5>

Davis GA, Purcell L, Schneider KJ, Yeates KO, Gioia GA, ... Kutcher JS. The Child Sport Concussion Assessment Tool 5th Edition (Child SCAT5): Background and rationale. *British journal of sports medicine*, (2017). 51(11),859-861. <https://doi.org/10.1136/bjsports-2017-097492>

De Matteo C, Volterman KA, Breithaupt PG, Claridge EA, Adamich J, Timmons BW. Exertion testing in youth with mild traumatic brain injury/concussion. *Medicine and Science in Sports and Exercise*, (2015). 47(11),2283-2290. <https://doi.org/10.1249/MSS.0000000000000682>

Dégeilh F, Bernier A, Gravel J, Beauchamp MH. Developmental trajectories of adaptive functioning following early mild traumatic brain injury. *Developmental Psychobiology*, (2018). 60(8),1037-1047. <https://doi.org/10.1002/dev.21786>

Del Rossi G, Anania T, Lopez RM. Early Aerobic Exercise for the Treatment of Acute Pediatric Concussions. *J Athl Train* [Internet]. 2020 Jul 1;55(7):649–57. Available from:

<https://meridian.allenpress.com/jat/article/55/7/649/436807/Early-Aerobic-Exercise-for-the-Treatment-of-Acute>

Delefortrie Q, Lejeune F, Kerzmann B, Levy R, Adam JF, ... Hachimi-Idrissi S. Evaluation of the Roche® Elecsys and the Diasorin® Liaison S100 kits in the management of mild head injury in the emergency room. *Clinical Biochemistry*, (2018). 52(August 2017),123-130.

<https://doi.org/10.1016/j.clinbiochem.2017.11.004>

DeMatteo C, Bednar ED, Randall S, et al. Effectiveness of return to activity and return to school protocols for children postconcussion: a systematic review. *BMJ Open Sport Exerc Med* [Internet]. 2020 Feb 24;6(1):e000667. Available from: <https://bmjopensem.bmj.com/lookup/doi/10.1136/bmjsem-2019-000667>

Dematteo C, Greenspoon D, Levac D, Harper JA, Rubinoff M. Evaluating the Nintendo Wii for assessing return to activity readiness in youth with mild traumatic brain injury. *Physical and Occupational Therapy in Pediatrics*, (2014). 34(3),229-244. <https://doi.org/10.3109/01942638.2014.885103>

DeMatteo CA, Lin C-YA, Foster G, et al. Evaluating Adherence to Return to School and Activity Protocols in Children After Concussion. *Clin J Sport Med* [Internet]. 2019 Dec 24;Publish Ah. Available from: <http://dx.doi.org/10.1097/JSM.0000000000000800>

DeMatteo CA, Randall S, Lin CYA, Claridge EA. What comes first: Return to school or return to activity for youth after concussion? Maybe we don't have to choose. *Frontiers in Neurology*, (2019). 10(JUL),1-9. <https://doi.org/10.3389/fneur.2019.00792>

DePadilla PhD L, Miller PhD GF, Everett Jones PhD, MPH, JD S. Characteristics of Schools with Youth Sports Concussion-Related Educational Policies and Practices. *J Sch Health*. 2020;90(7):520–6.

Dillard C, Ditchman N, Nersessova K, Foster N, Wehman P, ... Neblett J. Post-concussion symptoms in mild traumatic brain injury: findings from a paediatric outpatient clinic. *Disability and Rehabilitation*, (2017). 39(6),544-550. <https://doi.org/10.3109/09638288.2016.1152602>

Dobney DM, Grilli L, Kocilowicz H, Beaulieu C, Straub M, ... Gagnon I. Evaluation of an active rehabilitation program for concussion management in children and adolescents. *Brain Injury*, (2017). 31(13-14),1753-1759. <https://doi.org/10.1080/02699052.2017.1346294>

Dobney DM, Grilli L, Kocilowicz H, Beaulieu C, Straub M, ... Gagnon IJ. Is There an Optimal Time to Initiate an Active Rehabilitation Protocol for Concussion Management in Children? A Case Series. *Journal of Head Trauma Rehabilitation*, (2018). 33(3),E11-E17. <https://doi.org/10.1097/HTR.0000000000000339>

Dona O, Noseworthy MD, DeMatteo C, Connolly JF. Fractal analysis of brain blood oxygenation level dependent (BOLD) signals from children with mild traumatic brain injury (mTBI). *PLoS ONE*, (2017). 12(1). <https://doi.org/10.1371/journal.pone.0169647>

Donders J, DeWit C. Parental ratings of daily behavior and child cognitive test performance after pediatric mild traumatic brain injury. *Child Neuropsychology*, (2017). 23(5),554-570. <https://doi.org/10.1080/09297049.2016.1161015>

Donnell Z, Hoffman R, Sarmiento K, Hays C. Concussion attitudes, behaviors, and education among youth ages 12–17: Results from the 2014 YouthStyles survey. *Journal of Safety Research*, (2018). 64,163-169. <https://doi.org/10.1016/j.jsr.2017.12.001>

Dorman JC, Valentine VD, Munce TA, Tjarks BJ, Thompson PA, Bergeron MF. Tracking postural stability

of young concussion patients using dual-task interference. *Journal of Science and Medicine in Sport*, (2015). 18(1),2-7. <https://doi.org/10.1016/j.jsams.2013.11.010>

Dubrovsky AS, Friedman D, Kocilowicz H. Pediatric post-traumatic headaches and peripheral nerve blocks of the scalp: A case series and patient satisfaction survey. *Headache*, (2014). 54(5),878-887. <https://doi.org/10.1111/head.12334>

Durish CL, Yeates KO, Brooks BL. Psychological Resilience as a Predictor of Persistent Post-Concussive Symptoms in Children with Single and Multiple Concussion. *Journal of the International Neuropsychological Society*, (2018). 24(8),759-768. <https://doi.org/10.1017/S1355617718000437>

Eagle SR, Kontos AP, Sinnott A, et al. Utility of a novel perceptual-motor control test for identification of sport-related concussion beyond current clinical assessments. *J Sports Sci [Internet]*. 2020 Aug 2;38(15):1799–805. Available from: <https://www.tandfonline.com/doi/full/10.1080/02640414.2020.1756675>

Eagle SR, Puligilla A, Fazio-Sumrok V, et al. Association of time to initial clinic visit with prolonged recovery in pediatric patients with concussion. *J Neurosurg Pediatr*. 2020 Apr 24;26(2):165–70.

Eckner JT, Seifert T, Pescovitz A, Zeiger M, Kutcher JS. Is migraine headache associated with concussion in athletes? A case-control study. *Clinical Journal of Sport Medicine*, (2017). 27(3),266-270. <https://doi.org/10.1097/JSM.0000000000000346>

Eisenberg MA, Andrea J, Meehan W, Mannix R. Time interval between concussions and symptom duration. *Pediatrics*, (2013). 132(1),8-17. <https://doi.org/10.1542/peds.2013-0432>

Eisenberg MA, Meehan WP, Mannix R. Duration and Course of Post-Concussive Symptoms. *Pediatrics*, (2014). 133(6),999-1006. <https://doi.org/10.1542/peds.2014-0158>

Elbin RJ, D’Amico NR, McCarthy M, et al. How Do ImPACT Quick Test Scores Compare with ImPACT Online Scores in Non-Concussed Adolescent Athletes? *Arch Clin Neuropsychol [Internet]*. 2020 Apr 20;35(3):326–31. Available from: <https://academic.oup.com/acn/article/35/3/326/5733239>

Elbin RJ, Knox J, Kegel N, Schatz P, Lowder HB, ... Kontos AP. Assessing Symptoms in Adolescents Following Sport-Related Concussion: A Comparison of Four Different Approaches. *Applied Neuropsychology: Child*, (2016). 5(4),294-302. <https://doi.org/10.1080/21622965.2015.1077334>

Elbin RJ, Kontos AP, Kegel N, Johnson E, Burkhart S, Schatz P. Individual and combined effects of LD and ADHD on computerized neurocognitive concussion test performance: Evidence for separate norms. *Archives of Clinical Neuropsychology*, (2013). 1. Shi F,(5),476-484. <https://doi.org/10.1093/arclin/act024>

Elbin RJ, Sufrinko A, Schatz P, French J, Henry L, ... Kontos AP. Removal From Play After Concussion and Recovery Time. *Pediatrics*, (2016). 138(3),e20160910-e20160910. <https://doi.org/10.1542/peds.2016-0910>

Ellis M, Krisko C, Selci E, Russell K. Effect of concussion history on symptom burden and recovery following pediatric sports-related concussion. *Journal of Neurosurgery: Pediatrics*, (2018). 21(April),401-408. <https://doi.org/10.3171/2017.9.PEDS17392>.

Ellis MJ, Boles S, Derksen V, et al. Evaluation of a pilot paediatric concussion telemedicine programme for northern communities in Manitoba. *Int J Circumpolar Health [Internet]*. 2019 Jan 1;78(1):1573163. Available from: <https://www.tandfonline.com/doi/full/10.1080/22423982.2019.1573163>

- Ellis MJ, Cordingley D, Girardin R, Ritchie L, Johnston J. Migraine with aura or sports-related concussion: Case report, pathophysiology, and multidisciplinary approach to management. *Current Sports Medicine Reports*, (2017). 16(1),14-18. <https://doi.org/10.1249/JSR.0000000000000323>
- Ellis MJ, Cordingley D, Vis S, Reimer K, Leiter J, Russell K. Vestibulo-ocular dysfunction in pediatric sports-related concussion. *Journal of Neurosurgery: Pediatrics*, (2015). 16(3),248-255. <https://doi.org/10.3171/2015.1.PEDS14524>
- Ellis MJ, Cordingley DM, Vis S, Reimer KM, Leiter J, Russell K. Clinical predictors of vestibulo-ocular dysfunction in pediatric sports-related concussion. *Journal of Neurosurgery: Pediatrics*, (2017). 19(1),38-45. <https://doi.org/10.3171/2016.7.PEDS16310>
- Ellis MJ, Leiter J, Hall T, McDonald PJ, Sawyer S, ... Essig M. Neuroimaging findings in pediatric sports-related concussion. *Journal of Neurosurgery: Pediatrics*, (2015). 16(3),241-247. <https://doi.org/10.3171/2015.1.PEDS14510>
- Ellis MJ, McDonald PJ, Olson A, Koenig J, Russell K. Cervical Spine Dysfunction Following Pediatric Sports-Related Head Trauma. *Journal of Head Trauma Rehabilitation*, (2019). 34(2),103-110. <https://doi.org/10.1097/HTR.0000000000000411>
- Ellis MJ, Ritchie LJ, Koltek M, Hosain S, Cordingley D, ... Russell K. Psychiatric outcomes after pediatric sports-related concussion. *Journal of Neurosurgery: Pediatrics*, (2015). 16(6),709-718. <https://doi.org/10.3171/2015.5.PEDS15220>
- Evans E, Cook NE, Iverson GL, et al. Monitoring Outcome after Hospital-Presenting Milder Spectrum Pediatric Traumatic Brain Injury Using the Glasgow Outcome Scale-Extended, Pediatric Revision. *J Neurotrauma* [Internet]. 2020 Jul 15;37(14):1627–36. Available from: <https://www.liebertpub.com/doi/10.1089/neu.2019.6893>
- Ewing-Cobbs L, Cox CS, Clark AE, Holubkov R, Keenan HT. Persistent postconcussion symptoms after injury. *Pediatrics*, (2018). 142(5). <https://doi.org/10.1542/peds.2018-0939>
- Fabri TL, Wilson KE, Holland N, Hickling A, Murphy J, ... Reed N. Using a dual-task protocol to investigate motor and cognitive performance in healthy children and youth. *Gait and Posture*, (2017). 54,154-159. <https://doi.org/10.1016/j.gaitpost.2017.03.002>
- Fakhran S, Yaeger K, Collins M, Alhilali L. Sex Differences in White Matter Abnormalities after Mild Traumatic Brain Injury: Localization and Correlation with Outcome. *Radiology*, (2014). 272(3),815-823. <https://doi.org/10.1148/radiol.14132512>
- Faris G, Byczkowski T, Ho M, Babcock L. Prediction of Persistent Postconcussion Symptoms in Youth Using a Neuroimaging Decision Rule. *Academic Pediatrics*, (2016). 16(4),336-342. <https://doi.org/10.1016/j.acap.2015.10.007>
- Fehr SD, Nelson LD, Scharer KR, Traudt EA, Veenstra JM, ... Walter KD. Risk Factors for Prolonged Symptoms of Mild Traumatic Brain Injury: A Pediatric Sports Concussion Clinic Cohort. *Clinical Journal of Sport Medicine*, (2019). 29(1),11-17. <https://doi.org/10.1097/JSM.0000000000000494>
- Feiss R, Lutz M, Reiche E, et al. A Systematic Review of the Effectiveness of Concussion Education Programs for Coaches and Parents of Youth Athletes. *Int J Environ Res Public Health* [Internet]. 2020 Apr 13;17(8):2665. Available from: <https://www.mdpi.com/1660-4601/17/8/2665>
- Fino PC, Nussbaum MA, Brolinson PG. Locomotor deficits in recently concussed athletes and matched

- controls during single and dual-task turning gait: Preliminary results. *Journal of NeuroEngineering and Rehabilitation*, (2016). 13(1),1-15. <https://doi.org/10.1186/s12984-016-0177-y>
- Fino PC. A preliminary study of longitudinal differences in local dynamic stability between recently concussed and healthy athletes during single and dual-task gait. *Journal of Biomechanics*, (2016). 49(9),1983-1988. <https://doi.org/10.1016/j.jbiomech.2016.05.004>
- Fremont P, Schneider K, Laroche A, et al. Could a massive open online course be part of the solution to sport-related concussion? Participation and impact among 8368 registrants. *BMJ Open Sport Exerc Med*. 2020;6(1):1–7.
- Furman GR, Lin CC, Bellanca JL, Marchetti GF, Collins MW, Whitney SL. Comparison of the balance accelerometer measure and balance error scoring system in adolescent concussions in sports. *American Journal of Sports Medicine*, (2013). 41(6),1404-1410. <https://doi.org/10.1177/0363546513484446>
- Gabbe BJ, Brooks C, Demmler JC, Macey S, Hyatt MA, Lyons RA. The association between hospitalisation for childhood head injury and academic performance: Evidence from a population e-cohort study. *Journal of Epidemiology and Community Health*, (2014). 68(5),466-470. <https://doi.org/10.1136/jech-2013-203427>
- Gagner C, Landry-Roy C, Bernier A, Gravel J, Beauchamp MH. Behavioral consequences of mild traumatic brain injury in preschoolers. *Psychological Medicine*, (2018). 48(9),1551-1559. <https://doi.org/10.1017/S0033291717003221>
- Gagner C, Tuerk C, De Beaumont L, et al. Brain-Derived Neurotrophic Factor Val66Met Polymorphism and Internalizing Behaviors after Early Mild Traumatic Brain Injury. *J Neurotrauma* [Internet]. 2020 Jan 1;38(1):102–10. Available from: <https://www.liebertpub.com/doi/10.1089/neu.2019.6936>
- Gagnon I, Grilli L, Friedman D, Iverson GL. A pilot study of active rehabilitation for adolescents who are slow to recover from sport-related concussion. *Scandinavian Journal of Medicine and Science in Sports*, (2016). 26(3),299-306. <https://doi.org/10.1111/sms.12441>
- Gagnon I, Teel E, Gioia G, et al. Parent-Child Agreement on Postconcussion Symptoms in the Acute Postinjury Period. *Pediatrics* [Internet]. 2020 Jul;146(1):e20192317. Available from: <http://pediatrics.aappublications.org/lookup/doi/10.1542/peds.2019-2317>
- Galetta KM, Morganroth J, Moehringer N, Mueller B, Hasanaj L, ... Balcer LJ. Adding vision to concussion testing: A prospective study of sideline testing in youth and collegiate athletes. *Journal of Neuro-Ophthalmology*, (2015). 35(3),235-241. <https://doi.org/10.1097/WNO.0000000000000226>
- Gardner RM, Yengo-Kahn A, Bonfield CM, Solomon GS. Comparison of baseline and post-concussion ImPACT test scores in young athletes with stimulant-treated and untreated ADHD. *Physician and Sportsmedicine*, (2017). 1. Shi F,(1),1-10. <https://doi.org/10.1080/00913847.2017.1248221>
- Gauvin-Lepage J, Friedman D, Grilli L, Gagnon I. Effect of Sex on Recovery from Persistent Postconcussion Symptoms in Children and Adolescents Participating in an Active Rehabilitation Intervention. *Journal of Head Trauma Rehabilitation*, (2019). 34(2),96-102. <https://doi.org/10.1097/HTR.0000000000000402>
- Gauvin-Lepage J, Friedman D, Grilli L, Sufrategui M, De Matteo C, ... Gagnon I. Effectiveness of an Exercise-Based Active Rehabilitation Intervention for Youth Who Are Slow to Recover After Concussion. *Clinical Journal of Sport Medicine*, (2018). 00(00),1. <https://doi.org/10.1097/jsm.0000000000000634>

Genc S, Anderson V, Ryan NP, Malpas CB, Catroppa C, ... Silk TJ. Recovery of White Matter following Pediatric Traumatic Brain Injury Depends on Injury Severity. *Journal of Neurotrauma*, (2017). 34(4),798-806. <https://doi.org/10.1089/neu.2016.4584>

Gibson S, Nigrovic LE, O'Brien M, Meehan WP. The effect of recommending cognitive rest on recovery from sport-related concussion. *Brain Injury*, (2013). 27(7-8),839-842. <https://doi.org/10.3109/02699052.2013.775494>

Gidley Larson JC, Flaro L, Peterson RL, Connery AK, Baker DA, Kirkwood MW. The medical symptom validity test measures effort not ability in children: A comparison between mild TBI and fetal alcohol spectrum disorder samples. *Archives of Clinical Neuropsychology*, (2015). 1. Shi F,(3),192-199. <https://doi.org/10.1093/arclin/acv012>

Gillie BL, Fazio-Sumrok V, Eagle SR, et al. Clinical predictors of post-injury anxiety in adolescent patients following concussion. *Appl Neuropsychol Child* [Internet]. 2020;0(0):1–7. Available from: <https://doi.org/10.1080/21622965.2020.1799790>

Gladstone E, Narad ME, Hussain F, et al. Neurocognitive and Quality of Life Improvements Associated With Aerobic Training for Individuals With Persistent Symptoms After Mild Traumatic Brain Injury: Secondary Outcome Analysis of a Pilot Randomized Clinical Trial. *Front Neurol* [Internet]. 2019 Sep 18;10(September):1–9. Available from: <https://www.frontiersin.org/article/10.3389/fneur.2019.01002/full>

Glaviano NR, Benson S, Goodkin HP, Broshek DK, Saliba S. Baseline SCAT2 Assessment of Healthy Youth Student-Athletes: Preliminary Evidence for the Use of the Child-SCAT3 in Children Younger Than 13 Years. *Clinical Journal of Sport Medicine*, (2015). 25(4),373-379. <https://doi.org/10.1097/JSM.0000000000000154>

Gorman M, Hecht S, Samborski A, Lunos S, Elias S, Stovitz SD. SCAT3 assessment of non-head injured and head injured athletes competing in a large international youth soccer tournament. *Applied Neuropsychology: Child*, (2017). 1. Shi F,(4),364-368. <https://doi.org/10.1080/21622965.2016.1210011>

Gornall A, Takagi M, Clarke C, Babl FE, Davis GA, ... Anderson V. Behavioral and Emotional Difficulties after Pediatric Concussion. *Journal of Neurotrauma*, 2019,1-26. <https://doi.org/10.1089/neu.2018.6235>

Grabowski P, Wilson J, Walker A, Enz D, Wang S. Multimodal impairment-based physical therapy for the treatment of patients with post-concussion syndrome: A retrospective analysis on safety and feasibility. *Physical Therapy in Sport*, (2017). 23,22-30. <https://doi.org/10.1016/j.ptsp.2016.06.001>

Gravel J, Boutis K, Tang K, Beauchamp MH, Freedman SB, ... Zemek R. Association between ondansetron use and symptom persistence in children with concussions: A 5P substudy. *Canadian Journal of Emergency Medicine*, (2019). 21(2),204-210. <https://doi.org/10.1017/cem.2018.384>

Gravel J, Ledoux A-A, Tang K, et al. Early versus delayed emergency department presentation following mild Traumatic Brain Injury and the presence of symptom at 1, 4 and 12 weeks in children. *Emerg Med J* [Internet]. 2020 Jun;37(6):338–43. Available from: <https://emj.bmj.com/lookup/doi/10.1136/emmermed-2019-209054>

Graves JM, Moore M, Kehoe L, et al. Family hardship following youth concussion: Beyond the medical bills. *J Pediatr Nurs* [Internet]. 2020 Mar 1;51:15–20. Available from: <https://linkinghub.elsevier.com/retrieve/pii/S0882596319302295>

Green CM, Kirk JW, Connery AK, Baker DA, Kirkwood MW. The use of the Rey 15-Item Test and recognition trial to evaluate noncredible effort after pediatric mild traumatic brain injury. *Journal of Clinical and Experimental Neuropsychology*, (2014). 36(3),261-267.

<https://doi.org/10.1080/13803395.2013.879096>

Green SL, Keightley ML, Lobaugh NJ, Dawson DR, Mihailidis A. Changes in working memory performance in youth following concussion. *Brain Injury*, (2018). 32(2),182-190.

<https://doi.org/10.1080/02699052.2017.1358396>

Greenberg JK, Yan Y, Carpenter CR, Lumba-Brown A, Keller MS, ... Limbrick DD. Development of the CIDSS2 Score for Children with Mild Head Trauma without Intracranial Injury. *Journal of Neurotrauma*, (2018). 35(22),2699-2707. <https://doi.org/10.1089/neu.2017.5324>

Grool AM, Aglipay M, Momoli F, Meehan WP, Freedman SB, ... Zemek R. Association Between Early Participation in Physical Activity Following Acute Concussion and Persistent Postconcussive Symptoms in Children and Adolescents. *Jama*, (2016). 316(23),2504. <https://doi.org/10.1001/jama.2016.17396>

Grubenhoff JA, Currie D, Comstock RD, Juarez-Colunga E, Bajaj L, Kirkwood MW. Psychological Factors Associated with Delayed Symptom Resolution in Children with Concussion. *The Journal of Pediatrics*, (2016). 174(303),27-32.e1. <https://doi.org/10.1016/j.jpeds.2016.03.027>

Grubenhoff JA, Deakynne SJ, Brou L, Bajaj L, Comstock RD, Kirkwood MW. Acute Concussion Symptom Severity and Delayed Symptom Resolution. *Pediatrics*, (2014). 134(1),54-62.

<https://doi.org/10.1542/peds.2013-2988>

Guberman GI, Houde JC, Ptito A, et al. Structural abnormalities in thalamo-prefrontal tracks revealed by high angular resolution diffusion imaging predict working memory scores in concussed children. *Brain Struct Funct* [Internet]. 2020 Jan 1;225(1):441–59. Available from:

<http://link.springer.com/10.1007/s00429-019-02002-8>

Guerriero RM, Kuemmerle K, Pepin MJ, Taylor AM, Wolff R, Meehan WP. The Association Between Premorbid Conditions in School-Aged Children With Prolonged Concussion Recovery. *Journal of Child Neurology*, (2018). 33(2),168-173. <https://doi.org/10.1177/0883073817749655>

Guo X, Edmed SL, Anderson V. Neurocognitive predictors of posttraumatic stress disorder symptoms in children 6 months after traumatic brain injury: A prospective study. *Neuropsychology*, (2017). 31(1),84-92. <https://doi.org/http://dx.doi.org/10.1037/neu0000305>

Gutierrez GM, Conte C, Lightbourne K. The Relationship between Impact Force, Neck Strength, and Neurocognitive Performance in Soccer Heading in Adolescent Females. *Pediatric Exercise Science*, (2014). 26(1),33-40. <https://doi.org/10.1123/pes.2013-0102>

Haarbauer-Krupa JK, Comstock RD, Lionbarger M, Hirsch S, Kavee A, Lowe B. Healthcare professional involvement and RTP compliance in high school athletes with concussion. *Brain Injury*, (2018). 32(11),1337-1344. <https://doi.org/10.1080/02699052.2018.1482426>

Haider MN, Johnson SL, Mannix R, et al. The Buffalo Concussion Bike Test for Concussion Assessment in Adolescents. *Sport Heal A Multidiscip Approach* [Internet]. 2019 Nov 5;11(6):492–7. Available from:

<http://journals.sagepub.com/doi/10.1177/1941738119870189>

Haider MN, Leddy JJ, Pavlesen S, Kluczynski M, Baker JG, ... Willer BS. A systematic review of criteria

used to define recovery from sport-related concussion in youth athletes. British Journal of Sports Medicine, 2018,1179-1190. <https://doi.org/10.1136/bjsports-2016-096551>

Haider MN, Leddy JJ, Wilber CG, Viera KB, Bezherano I, ... Willer BS. The predictive capacity of the buffalo concussion treadmill test after sport-related concussion in adolescents. Frontiers in Neurology, (2019). 10(APR),1-7. <https://doi.org/10.3389/fneur.2019.00395>

Haider MN, Worts PR, Viera KB, et al. Postexercise Slowing on the King-Devick Test and Longer Recovery From Sport-Related Concussion in Adolescents: A Validation Study. J Athl Train [Internet]. 2020 May 1;55(5):482–7. Available from: <https://meridian.allenpress.com/jat/article/55/5/482/436794/Postexercise-Slowing-on-the-KingDevick-Test-and>

Hang BT, Cimpello LB, Pomerantz WJ. Can computerized neuropsychological testing in the pediatric emergency department predict duration of symptoms with sports-related concussions? Clinical Journal of Sport Medicine, (2012). 22(2),175. <https://doi.org/http://dx.doi.org/10.1097/jsm.0b013e31824ee01a>

Hansen C, Cushman D, Anderson N, Chen W, Cheng C, ... Hung M. A Normative Dataset of the Balance Error Scoring System in Children Aged between 5 and 14. Clinical Journal of Sport Medicine, (2016). 26(6),497-501. <https://doi.org/10.1097/JSM.0000000000000285>

Hansen C, Cushman D, Chen W, Bounsanga J, Hung M. Reliability testing of the balance error scoring system in children between the ages of 5 and 14. Clinical Journal of Sport Medicine, (2017). 27(1),64-68. <https://doi.org/10.1097/JSM.0000000000000293>

Hearps SJC, Takagi M, Babl FE, Bressan S, Truss K, ... Anderson V. Validation of a Score to Determine Time to Postconcussive Recovery. Pediatrics, (2017). 139(2),e20162003. <https://doi.org/10.1542/peds.2016-2003>

Henry LC, Elbin R, Collins MW, Marchetti G, Kontos AP. Examining recovery trajectories following sport-related concussion using a multimodal clinical assessment approach. Neurosurgery, (2016). 78(2),232-241. <https://doi.org/10.1227/NEU.0000000000001041>

Heyer GL, Idris SA. Does analgesic overuse contribute to chronic post-traumatic headaches in adolescent concussion patients? Pediatric Neurology, (2014). 50(5),464-468. <https://doi.org/10.1016/j.pediatrneurol.2014.01.040>

Heyer GL, Schaffer CE, Rose SC, Young JA, McNally KA, Fischer AN. Specific Factors Influence Postconcussion Symptom Duration among Youth Referred to a Sports Concussion Clinic. Journal of Pediatrics, (2016). 174,33-38.e2. <https://doi.org/10.1016/j.jpeds.2016.03.014>

Heyer GL, Young JA, Fischer AN. Lightheadedness After Concussion: Not All Dizziness is Vertigo. Clinical Journal of Sport Medicine, (2017). 0(0). <https://doi.org/10.1097/JSM.0000000000000445>

Heyer GL, Young JA, Rose SC, McNally KA, Fischer AN. Post-traumatic headaches correlate with migraine symptoms in youth with concussion. Cephalalgia, (2015). 36(4),309-316. <https://doi.org/10.1177/0333102415590240>

Heyworth BE, Carroll KM, Rizza AJ, McInnis KC, Gill TJ. Treatment of Concussion in High School Athletes: A Proposed Protocol for Athletic and Academic Return to Activity. Orthopaedic Journal of Sports Medicine, (2014). 2(7),2015. <https://doi.org/10.1177/2325967114S00079>

Higgins KL, Denney RL, Maerlender A. Sandbagging on the immediate post-concussion assessment and

cognitive testing (impact) in a high school athlete population. Archives of Clinical Neuropsychology, (2017). 1. Shi F,(3),259-266. <https://doi.org/10.1093/arclin/acw108>

Ho RA, Hall GB, Noseworthy MD, et al. Post-concussive depression: evaluating depressive symptoms following concussion in adolescents and its effects on executive function. Brain Inj [Internet]. 2020 Mar 20;34(4):520–7. Available from:

<https://www.tandfonline.com/doi/full/10.1080/02699052.2020.1725841>

Holmes A, Chen Z, Yahng L, et al. Return to Learn: Academic Effects of Concussion in High School and College Student-Athletes. Front Pediatr. 2020 Mar 4;8.

Hood EA, Klima DW, Chui KK, et al. Pre-season concussion assessment utilizing the King-Devick Test. Res Sport Med [Internet]. 2019 Oct 2;27(4):467–72. Available from:

<https://www.tandfonline.com/doi/full/10.1080/15438627.2018.1535434>

Houston MN, Bay RC, Valovich McLeod TC. The relationship between post-injury measures of cognition, balance, symptom reports and health-related quality-of-life in adolescent athletes with concussion.

Brain Injury, (2016). 30(7),891-898. <https://doi.org/10.3109/02699052.2016.1146960>

Howell D, Osternig L, Van Donkelaar P, Mayr U, Chou LS. Effects of concussion on attention and executive function in adolescents. Medicine and Science in Sports and Exercise, (2013). 45(6),1030-1037.

<https://doi.org/10.1249/MSS.0b013e3182814595>

Howell DR, Beasley M, Vopat L, Meehan WP. The Effect of Prior Concussion History on Dual-Task Gait following a Concussion. Journal of Neurotrauma, (2017). 34(4),838-844.

<https://doi.org/10.1089/neu.2016.4609>

Howell DR, Brilliant A, Berkstresser B, Wang F, Fraser J, Meehan WP. The Association between Dual-Task Gait after Concussion and Prolonged Symptom Duration. Journal of Neurotrauma, (2017). 34(23),3288-

3294. <https://doi.org/10.1089/neu.2017.5191>

Howell DR, Brilliant AN, Storey EP, Podolak OE, Meehan WP, Master CL. Objective Eye Tracking Deficits Following Concussion for Youth Seen in a Sports Medicine Setting. Journal of Child Neurology, (2018).

33(12),794-800. <https://doi.org/10.1177/0883073818789320>

Howell DR, Hanson E, Sugimoto D, Straccioli A, Meehan WP. Assessment of the Postural Stability of Female and Male Athletes. Clinical Journal of Sport Medicine, (2017). 27(5),444-449.

<https://doi.org/10.1097/JSM.0000000000000374>

Howell DR, Kriz P, Mannix RC, Kirchberg T, Master CL, Meehan WP. Concussion Symptom Profiles Among Child, Adolescent, and Young Adult Athletes. Clinical journal of sport medicine, (2019). 29(5),391-397.

<https://doi.org/10.1097/JSM.0000000000000629>

Howell DR, Lugade V, Potter MN, et al. A multifaceted and clinically viable paradigm to quantify postural control impairments among adolescents with concussion. Physiol Meas [Internet]. 2019 Sep

3;40(8):084006. Available from: <https://iopscience.iop.org/article/10.1088/1361-6579/ab3552>

Howell DR, Mannix RC, Quinn B, Taylor JA, Tan CO, Meehan WP. Physical Activity Level and Symptom Duration Are Not Associated after Concussion. American Journal of Sports Medicine, (2015). 44(4),1040-

1046. <https://doi.org/10.1177/0363546515625045>

Howell DR, Myer GD, Grooms D, Diekfuss J, Yuan W, Meehan WP. Examining Motor Tasks of Differing Complexity After Concussion in Adolescents. Archives of Physical Medicine and Rehabilitation, (2019).

100(4),613-619. <https://doi.org/10.1016/j.apmr.2018.07.441>

Howell DR, O'Brien MJ, Beasley MA, Mannix RC, Meehan WP. Initial somatic symptoms are associated with prolonged symptom duration following concussion in adolescents. *Acta Paediatrica, International Journal of Paediatrics*, (2016). 105(9),e426-e432. <https://doi.org/10.1111/apa.13486>

Howell DR, O'Brien MJ, Raghuram A, Shah AS, Meehan WP. Near Point of Convergence and Gait Deficits in Adolescents after Sport-Related Concussion. *Clinical Journal of Sport Medicine*, (2018). 28(3),262-267. <https://doi.org/10.1097/JSM.0000000000000439>

Howell DR, O'Brien MJ, Raghuram A, Shah AS, Meehan WP. Receded near point of convergence and gait are associated after concussion. *British Journal of Sports Medicine*, (2017). 51(11),A9.1-A9. <https://doi.org/10.1136/bjsports-2016-097270.21>

Howell DR, O'Brien MJ, Fraser J, Meehan WP. Continuing Play, Symptom Severity, and Symptom Duration After Concussion in Youth Athletes. *Clinical Journal of Sport Medicine*, (2018). 0(0),1. <https://doi.org/10.1097/jsm.0000000000000570>

Howell DR, Oldham JR, Brilliant AN, Meehan WP. Trouble Falling Asleep After Concussion Is Associated With Higher Symptom Burden Among Children and Adolescents. *Journal of Child Neurology*, (2019). 34(5),256-261. <https://doi.org/10.1177/0883073818824000>

Howell DR, Osternig LR, Chou LS. Adolescents demonstrate greater gait balance control deficits after concussion than young adults. *American Journal of Sports Medicine*, (2015). 43(3),625-632. <https://doi.org/10.1177/0363546514560994>

Howell DR, Osternig LR, Chou LS. Consistency and cost of dual-task gait balance measure in healthy adolescents and young adults. *Gait and Posture*, (2016). 49,176-180. <https://doi.org/10.1016/j.gaitpost.2016.07.008>

Howell DR, Osternig LR, Chou LS. Dual-task effect on gait balance control in adolescents with concussion. *Archives of Physical Medicine and Rehabilitation*, (2013). 94(8),1513-1520. <https://doi.org/10.1016/j.apmr.2013.04.015>

Howell DR, Osternig LR, Chou LS. Return to activity after concussion affects dual-task gait balance control recovery. *Medicine and Science in Sports and Exercise*, (2015). 47(4),673-680. <https://doi.org/10.1249/MSS.0000000000000462>

Howell DR, Osternig LR, Christie AD, Chou LS. Return to physical activity timing and dual-task gait stability are associated 2 months following concussion. *Journal of Head Trauma Rehabilitation*, (2016). 31(4),262-268. <https://doi.org/10.1097/HTR.0000000000000176>

Howell DR, Osternig LR, Koester MC, Chou LS. The effect of cognitive task complexity on gait stability in adolescents following concussion. *Experimental Brain Research*, (2014). 232(6),1773-1782. <https://doi.org/10.1007/s00221-014-3869-1>

Howell DR, Potter MN, Kirkwood MW, Wilson PE, Provance AJ, Wilson JC. Clinical predictors of symptom resolution for children and adolescents with sport-related concussion. *Journal of Neurosurgery: Pediatrics*, (2019). 24(1),54-61. <https://doi.org/10.3171/2018.11.PEDS18626>

Howell DR, Shore BJ, Hanson E, Meehan WP. Evaluation of postural stability in youth athletes: the relationship between two rating systems. *Physician and Sportsmedicine*, (2016). 44(3),304-310. <https://doi.org/10.1080/00913847.2016.1197763>

Howell DR, Straccioli A, Geminiani E, Meehan WP. Dual-task gait differences in female and male adolescents following sport-related concussion. *Gait and Posture*, (2017). 54(November 2016),284-289. <https://doi.org/10.1016/j.gaitpost.2017.03.034>

Howell DR, Wilson JC, Kirkwood MW, Grubenhoff JA. Quality of Life and Symptom Burden 1 Month After Concussion in Children and Adolescents. *Clinical Pediatrics*, (2019). 58(1),42-49. <https://doi.org/10.1177/0009922818806308>

Howell DR, Zemek R, Brilliant AN, Mannix RC, Master CL, Meehan WP. Identifying Persistent Postconcussion Symptom Risk in a Pediatric Sports Medicine Clinic. *American Journal of Sports Medicine*, (2018). 46(13),3254-3261. <https://doi.org/10.1177/0363546518796830>

Hugentobler JA, Gupta R, Slater R, Paterno M V., Riley MA, Quatman-Yates C. Influence of Age on Postconcussive Postural Control Measures and Future Implications for Assessment. *Clinical Journal of Sport Medicine*, (2016). 26(6),510-517. <https://doi.org/10.1097/JSM.0000000000000286>

Hunt AW, Laupacis D, Kawaguchi E, Greenspoon D, Reed N. Key ingredients to an active rehabilitation programme post-concussion: perspectives of youth and parents. *Brain Injury*, (2018). 32(12),1534-1540. <https://doi.org/10.1080/02699052.2018.1502894>

Hunt AW, Panaccia M, Reed N, Keightley M. Concussion-Like Symptoms in Child and Youth Athletes at Baseline: What Is “Typical”? *Journal of Athletic Training*, (2016). 51(10),749-757. <https://doi.org/10.4085/1062-6050-51.11.12>

Iverson GL, Atkins JE, Zafonte R, Berkner PD. Concussion History in Adolescent Athletes with Attention-Deficit Hyperactivity Disorder. *Journal of Neurotrauma*, (2016). 33(23),2077-2080. <https://doi.org/10.1089/neu.2014.3424>

Iverson GL, Jones PJ, Karr JE, et al. Architecture of Physical, Cognitive, and Emotional Symptoms at Preseason Baseline in Adolescent Student Athletes With a History of Mental Health Problems. *Front Neurol* [Internet]. 2020 Mar 20;11. Available from: <https://www.frontiersin.org/article/10.3389/fneur.2020.00175/full>

Iverson GL, Jones PJ, Karr JE, et al. Network Structure of Physical, Cognitive, and Emotional Symptoms at Preseason Baseline in Student Athletes with Attention-Deficit/ Hyperactivity Disorder. *Arch Clin Neuropsychol* [Internet]. 2020 Oct 8;35(7):1109–22. Available from: <https://academic.oup.com/acn/advance-article/doi/10.1093/arclin/acia030/5850528>

Iverson GL, Silverberg ND, Mannix R, Maxwell BA, Atkins JE, ... Berkner PD. Factors associated with concussion-like symptom reporting in high school athletes. *JAMA Pediatrics*, (2015). 169(12),1132-1140. <https://doi.org/10.1001/jamapediatrics.2015.2374>

Iyer KK, Barlow KM, Brooks B, Ofoghi Z, Zalesky A, Cocchi L. Relating brain connectivity with persistent symptoms in pediatric concussion. *Annals of Clinical and Translational Neurology*, (2019). 6(5),954-961. <https://doi.org/10.1002/acn3.764>

Iyer KK, Zalesky A, Barlow KM, et al. Default mode network anatomy and function is linked to pediatric concussion recovery. *Ann Clin Transl Neurol*. 2019 Dec 1;6(12):2544–54.

Iyer KK, Zalesky A, Cocchi L, et al. Neural Correlates of Sleep Recovery following Melatonin Treatment for Pediatric Concussion: A Randomized Controlled Trial. *J Neurotrauma* [Internet]. 2020 Dec 15;37(24):2647–55. Available from: <https://www.liebertpub.com/doi/10.1089/neu.2020.7200>

Jimenez N, Quistberg A, Vavilala MS, Jaffe KM, Rivara FP. Utilization of mental health services after mild pediatric traumatic brain injury. *Pediatrics*, (2017). 139(3). <https://doi.org/10.1542/peds.2016-2462>

Jones KM, Barker-Collo S, Parmar P, Starkey N, Theadom A, ... Feigin VL. Trajectories in health recovery in the 12 months following a mild traumatic brain injury in children: Findings from the BIONIC Study. *Journal of Primary Health Care*, (2018). 10(1),81-89. <https://doi.org/10.1071/HC17038>

Jones KM, Prah P, Starkey N, Theadom A, Barker-Collo S, ... Brown P. Longitudinal patterns of behavior, cognition, and quality of life after mild traumatic brain injury in children: BIONIC study findings. *Brain Injury*, (2019). 33(7),884-893. <https://doi.org/10.1080/02699052.2019.1606445>

Jones KM, Starkey NJ, Theadom A, Gheorghe A, Willix-Payne D, ... Feigin VL. Parent and child ratings of child behaviour following mild traumatic brain injury. *Brain Injury*, (2018). 32(11),1397-1404. <https://doi.org/10.1080/02699052.2018.1496477>

Kaldoja M-L, Kolk A. Does Gender Matter? Differences in Social-Emotional Behavior Among Infants and Toddlers Before and After Mild Traumatic Brain Injury. *Journal of Child Neurology*, (2015). 30(7),860-867. <https://doi.org/10.1177/0883073814544705>

Kanani AN, Hartshorn S. Resumption of physical activity within 7 days of a concussion was associated with lower rates of persistent postconcussive symptoms (PPCS). *Archives of Disease in Childhood: Education and Practice Edition*, (2018). 103(2),110. <https://doi.org/10.1136/archdischild-2017-313141>

Kara S, Crosswell H, Forch K, et al. Less Than Half of Patients Recover Within 2 Weeks of Injury After a Sports-Related Mild Traumatic Brain Injury: A 2-Year Prospective Study. *Clin J Sport Med [Internet]*. 2020 Mar 1;30(2):96–101. Available from: <http://journals.lww.com/10.1097/JSM.0000000000000811>

Katirci Y, Ocak T, Karamercan MA, Kocasaban D, Yurdakul MS, ... Coskun F. Compliance with catch rules in administering computerized tomography scans to children admitted to the emergency department with minor head trauma. *Acta Medica Mediterranea*, (2013). 29(4),717-722. <https://doi.org/> “ “

Keenan HT, Clark AE, Holubkov R, Cox CS, Ewing-Cobbs L. Psychosocial and Executive Function Recovery Trajectories One Year after Pediatric Traumatic Brain Injury: The Influence of Age and Injury Severity. *Journal of Neurotrauma*, (2018). 35(2),286-296. <https://doi.org/10.1089/neu.2017.5265>

Keenan RP, Lovanio K, Lapidus G, et al. Improved Concussion Discharge Instructions in a Pediatric Emergency Department. *Adv Emerg Nurs J*. 2020 Jan 1;42(1):63–70.

Keightley ML, Singh Saluja R, Chen J-K, Gagnon I, Leonard G, ... Ptito A. A Functional Magnetic Resonance Imaging Study of Working Memory in Youth after Sports-Related Concussion: Is It Still Working? *Journal of Neurotrauma*, (2014). 31(5),437-451. <https://doi.org/10.1089/neu.2013.3052>

Kerr ZY, Zuckerman SL, Wasserman EB, Vander Vegt CB, Yengo-Kahn A, ... Dompier TP. Factors associated with post-concussion syndrome in high school student-athletes. *Journal of Science and Medicine in Sport*, (2018). 21(5),447-452. <https://doi.org/10.1016/j.jsams.2017.08.025>

Khetani AM, Brooks BL, Mikrogianakis A, Barlow KM. Incorporating a Computerized Cognitive Battery Into the Emergency Department Care of Pediatric Mild Traumatic Brain Injuries — Is It Feasible ? *Pediatric Emergency Care*, (2018). 34(00),1-6. <https://doi.org/10.1097/PEC.0000000000000959>

Kim S, Connaughton DP. Youth Soccer Parents’ Attitudes and Perceptions About Concussions. *J Adolesc Heal [Internet]*. 2020 Jan;68(1):184–90. Available from: <https://linkinghub.elsevier.com/retrieve/pii/S1054139X20302184>

- King D, Brughelli M, Hume P, Gissane C. Concussions in amateur rugby union identified with the use of a rapid visual screening tool. *Journal of the Neurological Sciences*, (2013). 326(1-2),59-63.
<https://doi.org/10.1016/j.jns.2013.01.012>
- King D, Hume P, Gissane C, Clark T. Use of the King-Devick test for sideline concussion screening in junior rugby league. *Journal of the Neurological Sciences*, (2015). 357(1-2),75-79.
<https://doi.org/10.1016/j.jns.2015.06.069>
- King LA, Horak FB, Mancini M, Pierce D, Priest KC, ... Chapman JC. Instrumenting the balance error scoring system for use with patients reporting persistent balance problems after mild traumatic brain injury. *Archives of Physical Medicine and Rehabilitation*, (2014). 95(2),353-359.
<https://doi.org/10.1016/j.apmr.2013.10.015>
- King R, Grohs MN, Kirton A, Lebel C, Esser MJ, Barlow KM. Microstructural neuroimaging of white matter tracts in persistent post-concussion syndrome: A prospective controlled cohort study. *NeuroImage: Clinical*, (2019). 23(October 2018),101842. <https://doi.org/10.1016/j.nicl.2019.101842>
- King R, Kirton A, Zewdie E, Seeger TA, Ciechanski P, Barlow KM. Longitudinal assessment of cortical excitability in children and adolescents with mild traumatic brain injury and persistent post-concussive symptoms. *Frontiers in Neurology*, (2019). 10(MAY),8-18. <https://doi.org/10.3389/fneur.2019.00451>
- Kirk JW, Hutaff-Lee CF, Connery AK, Baker DA, Kirkwood MW. The Relationship Between the Self-Report BASC-2 Validity Indicators and Performance Validity Test Failure After Pediatric Mild Traumatic Brain Injury. *Assessment*, (2014). 1. Shi F,(5),562-569. <https://doi.org/10.1177/1073191114520626>
- Kirkwood MW, Peterson RL, Connery AK, Baker DA, Forster J. A Pilot Study Investigating Neuropsychological Consultation as an Intervention for Persistent Postconcussive Symptoms in a Pediatric Sample. *Journal of Pediatrics*, (2016). 169,244-249e1.
<https://doi.org/10.1016/j.jpeds.2015.10.014>
- Kirkwood MW, Peterson RL, Connery AK, Baker DA, Grubenhoff JA. Postconcussive Symptom Exaggeration After Pediatric Mild Traumatic Brain Injury. *Pediatrics*, (2014). 1. Shi F,(4),643-650.
<https://doi.org/10.1542/peds.2013-3195>
- Konigs M, Heij HA, van der Sluijs JA, Vermeulen RJ, Goslings JC, ... Oosterlaan J. Pediatric Traumatic Brain Injury and Attention Deficit. *Pediatrics*, (2015). 136(3),534-541. <https://doi.org/10.1542/peds.2015-0437>
- Königs M, Weeda WD, Van Heurn LWE, Vermeulen RJ, Goslings JC, ... Oosterlaan J. Impaired Visual Integration in Children with Traumatic Brain Injury: An Observational Study. *PLoS ONE*, (2015). 10(12),1-15. <https://doi.org/10.1371/journal.pone.0144395>
- Kontos AP, Elbin RJ, Lau B, Simensky S, Freund B, ... Collins MW. Posttraumatic migraine as a predictor of recovery and cognitive impairment after sport-related concussion. *American Journal of Sports Medicine*, (2013). 41(7),1497-1504. <https://doi.org/10.1177/0363546513488751>
- Kontos AP, Jorgensen-Wagers K, Trbovich AM, et al. Association of Time Since Injury to the First Clinic Visit With Recovery Following Concussion. *JAMA Neurol* [Internet]. 2020 Apr 1;77(4):435. Available from: <https://jamanetwork.com/journals/jamaneurology/fullarticle/2757869>
- Kostyun RO, Hafeez I. Protracted Recovery From a Concussion: A Focus on Gender and Treatment Interventions in an Adolescent Population. *Sports Health*, (2015). 7(1),52-57.
<https://doi.org/10.1177/1941738114555075>

Kostyun RO, Milewski MD, Hafeez I. Sleep disturbance and neurocognitive function during the recovery from a sport-related concussion in adolescents. *American Journal of Sports Medicine*, (2015). 43(3),633-640. <https://doi.org/10.1177/0363546514560727>

Kostyun RO, Milewski MD, Hafeez I. Sleep Disturbance and Neurocognitive Function During the Recovery From a Sport-Related Concussion in Adolescents. *The American journal of sports medicine*, (2014). 1. Shi F,1-8. <https://doi.org/10.1177/0363546514560727>

Kraus N, Thompson EC, Krizman J, Cook K, White-Schwoch T, LaBella CR. Auditory biological marker of concussion in children. *Scientific Reports*, (2016). 6(August),1-10. <https://doi.org/10.1038/srep39009>

Kriz PK, Mannix R, Taylor AM, Ruggieri D, Meehan WP. Neurocognitive Deficits of Concussed Adolescent Athletes at Self-reported Symptom Resolution in the Zurich Guidelines Era. *Orthopaedic Journal of Sports Medicine*, (2017). 5(11),1-7. <https://doi.org/10.1177/2325967117737307>

Kriz PK, Stein C, Kent J, Ruggieri D, Dolan E, ... Meehan WP. Physical Maturity and Concussion Symptom Duration among Adolescent Ice Hockey Players. *Journal of Pediatrics*, (2016). 171,234-239e2. <https://doi.org/10.1016/j.jpeds.2015.12.006>

Kroshus E, Bowen D, Opel D, et al. Understanding Decision Making by Families About Youth Football Participation Postconcussion. *Health Promot Pract* [Internet]. 2020 Jul 30;152483992094525. Available from: <http://journals.sagepub.com/doi/10.1177/1524839920945255>

Kuczynski A, Crawford S, Bodell L, Dewey D, Barlow KM. Characteristics of post-traumatic headaches in children following mild traumatic brain injury and their response to treatment: A prospective cohort. *Developmental Medicine and Child Neurology*, (2013). 55(7),636-641. <https://doi.org/10.1111/dmcn.12152>

Kuhn AW, Solomon GS. Supervision and computerized neurocognitive baseline test performance in high school athletes: An initial investigation. *Journal of Athletic Training*, (2014). 49(6),800-805. <https://doi.org/10.4085/1062-6050-49.3.66>

Kundu S, Ghodadra A, Fakhra S, et al. Assessing postconcussive reaction time using transport-based morphometry of diffusion tensor images. In: *American Journal of Neuroradiology* [Internet]. American Society of Neuroradiology; 2019. p. 1117–23. Available from: <http://www.ajnr.org/lookup/doi/10.3174/ajnr.A6087>

Kurowski BG, Hugentobler J, Quatman-Yates C, Taylor J, Gubanich PJ, ... Wade SL. Aerobic exercise for adolescents with prolonged symptoms after mild traumatic brain injury: An exploratory randomized clinical trial. *Journal of Head Trauma Rehabilitation*, (2017). 32(2),79-89. <https://doi.org/10.1097/HTR.0000000000000238>

Kwan V, Plourde V, Yeates KO, et al. Headache long after pediatric concussion: presence, intensity, interference, and association with cognition. *Brain Inj* [Internet]. 2020 Mar 20;34(4):575–82. Available from: <https://www.tandfonline.com/doi/full/10.1080/02699052.2020.1725842>

Lancaster MA, Olson D V., McCrea MA, Nelson LD, LaRoche AA, Muftuler LT. Acute white matter changes following sport-related concussion: A serial diffusion tensor and diffusion kurtosis tensor imaging study. *Human Brain Mapping*, (2016). 37(11),3821-3834. <https://doi.org/10.1002/hbm.23278>

Landry-Roy C, Bernier A, Gravel J, Beauchamp MH. Executive Functions and Their Relation to Sleep Following Mild Traumatic Brain Injury in Preschoolers. *Journal of the International Neuropsychological*

Society, (2018). 24(8),769-780. <https://doi.org/10.1017/S1355617718000401>

Laufer O, Geva A, Ellis JD, et al. Prospective longitudinal investigation shows correlation of event-related potential to mild traumatic brain injury in adolescents. *Brain Inj.* 2020;34(7):871–80.

Lawrence DW, Richards D, Comper P, Hutchison MG. Earlier time to aerobic exercise is associated with faster recovery following acute sport concussion. *PLoS ONE*, (2018). 13(4),1-12.
<https://doi.org/10.1371/journal.pone.0196062>

Lax ID, Panicia M, Agnihotri S, Reed N, Garmaise E, ... Keightley M. Developmental and gender influences on executive function following concussion in youth hockey players. *Brain Injury*, (2015). 29(12),1409-1419. <https://doi.org/10.3109/02699052.2015.1043344>

Le Sage N, Tardif PA, Frenette J, et al. Detection of S-100 β protein in plasma and urine after a mild traumatic brain injury. *Can J Neurol Sci* [Internet]. 2019 Sep 18;46(5):599–602. Available from:
https://www.cambridge.org/core/product/identifier/S0317167119000611/type/journal_article

Leddy JJ, Haider MN, Ellis MJ, Mannix R, Darling SR, ... Willer B. Early Subthreshold Aerobic Exercise for Sport-Related Concussion: A Randomized Clinical Trial. *JAMA Pediatrics*, (2019). 173(4),319-325.
<https://doi.org/10.1001/jamapediatrics.2018.4397>

Leddy JJ, Haider MN, Hinds AL, Darling S, Willer BS. A Preliminary Study of the Effect of Early Aerobic Exercise Treatment for Sport-Related Concussion in Males. *Clinical journal of sport medicine*, (2019). 29(5),353-360. <https://doi.org/10.1097/JSM.0000000000000663>

Ledoux AA, Tang K, Yeates KO, Pusic M V, Boutis K, ... Zemek RL. Natural Progression of Symptom Change and Recovery from Concussion in a Pediatric Population. *JAMA Pediatrics*, (2019). 173(1),1-11.
<https://doi.org/10.1001/jamapediatrics.2018.3820>

Lee YM, Odom MJ, Zuckerman SL, Solomon GS, Sills AK. Does age affect symptom recovery after sports-related concussion? A study of high school and college athletes: Clinical article. *Journal of Neurosurgery: Pediatrics*, (2013). 12(6),537-544. <https://doi.org/10.3171/2013.7.PEDS12572>

Lee YM, Wu A, Zuckerman SL, Stanko KM, LaChaud GY, ... Sills AK. Obesity and neurocognitive recovery after sports-related concussion in athletes: a matched cohort study. *Physician and Sportsmedicine*, (2016). 44(3),217-222. <https://doi.org/10.1080/00913847.2016.1216718>

Lichtenstein JD, Moser RS, Schatz P. Age and test setting affect the prevalence of invalid baseline scores on neurocognitive tests. *American Journal of Sports Medicine*, (2014). 42(2),479-484.
<https://doi.org/10.1177/0363546513509225>

Liller KD, Morris B, Fillion J, Yang Y, Bubu OM. Analysis of baseline computerized neurocognitive testing results among 5-11-year-old male and female children playing sports in recreational leagues in Florida. *International Journal of Environmental Research and Public Health*, (2017). 14(9).
<https://doi.org/10.3390/ijerph14091028>

Lin K, Tung C. Acupuncture for Recovery from Pediatric Sport-Related Concussion. *Medical Acupuncture*, (2016). 28(4),217-222. <https://doi.org/10.1089/acu.2016.1181>

Linden M. The Effects of QEEG-Guided Neurofeedback on Postconcussion Syndrome. *Biofeedback*, (2015). 43(1),42-44. <https://doi.org/10.5298/1081-5937-43.1.08>

Linder SM, Cruickshank J, Zimmerman NM, Figler R, Alberts JL. A technology-enabled electronic incident

report to document and facilitate management of sport concussion: A cohort study of youth and young adults. *Medicine*, (2019). 98(14),e14948. <https://doi.org/10.1097/MD.00000000000014948>

Linder SM, Ozinga SJ, Koop MM, Dey T, Figler R, ... Alberts JL. Cleveland Clinic Postural Stability Index Norms for the Balance Error Scoring System. *Medicine and Science in Sports and Exercise*, (2018). 50(10),1998-2006. <https://doi.org/10.1249/MSS.0000000000001660>

Lindholm EB, D'Cruz R, Fajardo R, et al. Admission of Pediatric Concussion Injury Patients: Is it Necessary? *J Surg Res*. 2019 Dec 1;244:107–10.

Liou YJ, Wei HT, Chen MH, Hsu JW, Huang KL, ... Chen TJ. Risk of Traumatic Brain Injury Among Children, Adolescents, and Young Adults With Attention-Deficit Hyperactivity Disorder in Taiwan. *Journal of Adolescent Health*, (2018). 63(2),233-238. <https://doi.org/10.1016/j.jadohealth.2018.02.012>

Little CE, Emery C, Scott SH, Meeuwisse W, Palacios-Derflingher L, Dukelow SP. Do children and adolescent ice hockey players with and without a history of concussion differ in robotic testing of sensory, motor and cognitive function? *Journal of NeuroEngineering and Rehabilitation*, (2016). 13(1),1-19. <https://doi.org/10.1186/s12984-016-0195-9>

Lovell MR, Solomon GS. Neurocognitive test performance and symptom reporting in cheerleaders with concussions. *Journal of Pediatrics*, (2013). 163(4),1192-1195.e1. <https://doi.org/10.1016/j.jpeds.2013.05.061>

Ludwig R, D'Silva L, Vaduvathirayan P, et al. Sleep Disturbances in the Acute Stage of Concussion are Associated With Poorer Long-Term Recovery: A Systematic Review [Internet]. Vol. 12, PM and R. John Wiley and Sons Inc.; 2020. p. 500–11. Available from: <https://onlinelibrary.wiley.com/doi/abs/10.1002/pmrj.12309>

Lugones M, Parkin G, Bjelosevic S, Takagi M, Clarke C, ... Ignjatovic V. Blood biomarkers in paediatric mild traumatic brain injury: a systematic review. *Neuroscience and Biobehavioral Reviews*, (2018). 87(January),206-217. <https://doi.org/10.1016/j.neubiorev.2018.02.006>

Lumba-Brown A, Harley J, Lucio S, Vaida F, Hilfiker M. Hypertonic Saline as a Therapy for Pediatric Concussive Pain. *Pediatric Emergency Care*, (2014). 30(3),139-145. <https://doi.org/10.1097/PEC.0000000000000084>

Lumba-Brown A, Teramoto M, Bloom OJ, Brody D, Chesnutt J, ... Ghajar J. Concussion Guidelines Step 2: Evidence for Subtype Classification. *Neurosurgery*, (2019). 0(0),1-12. <https://doi.org/10.1093/neuros/nyz332>

Lumba-Brown A, Yeates KO, Sarmiento K, Breiding MJ, Haegerich TM, ... Timmons SD. Centers for Disease Control and Prevention Guideline on the Diagnosis and Management of Mild Traumatic Brain Injury among Children. *JAMA Pediatrics*, (2018). 172(11),1-13. <https://doi.org/10.1001/jamapediatrics.2018.2853>

Lumba-Brown A, Yeates KO, Sarmiento K, Breiding MJ, Haegerich TM, ... Timmons SD. Diagnosis and Management of Mild Traumatic Brain Injury in Children: A Systematic Review. *JAMA Pediatrics*, (2018). 172(11). <https://doi.org/10.1001/jamapediatrics.2018.2847>

Lynch JM, Anderson M, Benton B, Green SS. The gaming of concussions: A unique intervention in postconcussion syndrome. *Journal of Athletic Training*, (2015). 50(3),270-276. <https://doi.org/10.4085/1062-6050-49.3.78>

Lyons TW, Miller KA, Miller AF, Mannix R. Racial and ethnic differences in emergency department utilization and diagnosis for sports-related head injuries. *Frontiers in Neurology*, (2019). 10(JUL),1-8. <https://doi.org/10.3389/fneur.2019.00690>

Mac Donald CL, Barber J, Wright J, Coppel D, De Lacy N, ... Temkin N. Longitudinal Clinical and Neuroimaging Evaluation of Symptomatic Concussion in 10-to 14-year-old Youth Athletes. *Journal of Neurotrauma*, (2019). 36(2),264-274. <https://doi.org/10.1089/neu.2018.5629>

MacDonald J, Duerson D. Reliability of a Computerized Neurocognitive Test in Baseline Concussion Testing of High School Athletes. *Clinical Journal of Sport Medicine*, (2015). 1. Shi F,(4),367-372. <https://doi.org/10.1097/JSM.0000000000000139>

MacDonald J, Patel N, Young J, Stuart E. Returning Adolescents to Driving after Sports-Related Concussions: What Influences Physician Decision-Making. *Journal of Pediatrics*, (2018). 194,177-181. <https://doi.org/10.1016/j.jpeds.2017.10.032>

MacDonald J, Wilson J, Young J, Duerson D, Swisher G, ... Meehan WP. Evaluation of a simple test of reaction time for baseline concussion testing in a population of high school athletes. *Clinical journal of sport medicine : official journal of the Canadian Academy of Sport Medicine*, (2015). 25(1),43-48. <https://doi.org/10.1097/JSM.0000000000000096>

Manikas V, Babl FE, Hearps S, Dooley J, Anderson V. Impact of Exercise on Clinical Symptom Report and Neurocognition after Concussion in Children and Adolescents. *Journal of Neurotrauma*, (2017). 34(11),1932-1938. <https://doi.org/10.1089/neu.2016.4762>

Manning KY, Schranz A, Bartha R, Dekaban GA, Barreira C, ... Menon RS. Multiparametric MRI changes persist beyond recovery in concussed adolescent hockey players. *Neurology*, (2017). 89(21),2157-2166. <https://doi.org/10.1212/WNL.0000000000004669>

Mannix R, Iverson GL, Maxwell B, Atkins JE, Zafonte R, Berkner PD. Multiple prior concussions are associated with symptoms in high school athletes. *Annals of Clinical and Translational Neurology*, (2014). 1(6),433-438. <https://doi.org/10.1002/acn3.70>

Mannix R, Levy R, Zemek R, et al. Fluid Biomarkers of Pediatric Mild Traumatic Brain Injury: A Systematic Review [Internet]. Vol. 37, *Journal of Neurotrauma*. Mary Ann Liebert Inc; 2020. p. 2029–44. Available from: <https://www.liebertpub.com/doi/10.1089/neu.2019.6956>

Mannix R, Zemek R, Yeates KO, Arbogast K, Atabaki S, ... Wisniewski S. Practice patterns in pharmacological and non-pharmacological therapies for children with mild traumatic brain injury: A survey of 15 canadian and United States centers. *Journal of Neurotrauma*, (2019). 36(20),2886-2894. <https://doi.org/10.1089/neu.2018.6290>

Manzano S, Holzinger IB, Kellenberger CJ, Lacroix L, Klima-Lange D, ... Staubli G. Diagnostic performance of S100B protein serum measurement in detecting intracranial injury in children with mild head trauma. *Emergency Medicine Journal*, (2016). 33(1),42-46. <https://doi.org/10.1136/emered-2014-204513>

Marin JR, Rodean J, Hall M, et al. Trends in Use of Advanced Imaging in Pediatric Emergency Departments, 2009-2018. *JAMA Pediatr* [Internet]. 2020 Sep 8;174(9):e202209. Available from: <https://jamanetwork.com/journals/jamapediatrics/fullarticle/2769073>

Marshall CM, Chan N, Tran P, et al. The use of an intensive physical exertion test as a final return to play measure in concussed athletes: a prospective cohort. *Phys Sportsmed* [Internet]. 2019 Apr 3;47(2):158–

66. Available from: <https://www.tandfonline.com/doi/full/10.1080/00913847.2018.1542258>

Martin AK, Petersen AJ, Sesma HW, et al. Concussion symptomology and recovery in children and adolescents with pre-existing anxiety. *J Neurol Neurosurg Psychiatry* [Internet]. 2020 Oct;91(10):1060–6. Available from: <https://jnnp.bmj.com/lookup/doi/10.1136/jnnp-2020-323137>

Master CL, Curry AE, Pfeiffer MR, et al. Characteristics of Concussion in Elementary School-Aged Children: Implications for Clinical Management. *J Pediatr* [Internet]. 2020 Aug;223:128–35. Available from: <https://doi.org/10.1016/j.jpeds.2020.04.001>

Master CL, Master SR, Wiebe DJ, Storey EP, Lockyer JE, ... Grady MF. Vision and Vestibular System Dysfunction Predicts Prolonged Concussion Recovery in Children. *Clinical Journal of Sport Medicine*, (2018). 28(2),139-145. <https://doi.org/10.1097/JSM.0000000000000507>

Master CL, Scheiman M, Gallaway M, Goodman A, Robinson RL, ... Grady MF. Vision Diagnoses Are Common after Concussion in Adolescents. *Clinical Pediatrics*, (2016). 55(3),260-267. <https://doi.org/10.1177/0009922815594367>

Matveev R, Sergio L, Fraser-Thomas J, Macpherson AK. Trends in concussions at Ontario schools prior to and subsequent to the introduction of a concussion policy – An analysis of the Canadian hospitals injury reporting and prevention program from 2009 to 2016. *BMC Public Health*, (2018). 18(1),1-10. <https://doi.org/10.1186/s12889-018-6232-9>

Mautner K, Sussman WI, Axtman M, Al-Farsi Y, Al-Adawi S. Relationship of Attention Deficit Hyperactivity Disorder and Postconcussion Recovery in Youth Athletes. *Clinical Journal of Sport Medicine*, (2015). 25(4),355-360. <https://doi.org/10.1097/JSM.0000000000000151>

Max JE, Friedman K, Wilde EA, Bigler ED, Hanten G, ... Levin HS. Psychiatric disorders in children and adolescents 24 months after mild traumatic brain injury. *Journal of Neuropsychiatry and Clinical Neurosciences*, (2015). 27(2),112-120. <https://doi.org/10.1176/appi.neuropsych.13080190>

Max JE, Pardo D, Hanten G, Schachar RJ, Saunders AE, ... Levin HS. Psychiatric Disorders in Children and Adolescents Six-to-Twelve Months After Mild Traumatic Brain Injury. *The Journal of Neuropsychiatry and Clinical Neurosciences*, (2013). 25(4),272-282. <https://doi.org/10.1176/appi.neuropsych.12040078>

Mayer AR, Hanlon FM, Ling JM. Gray matter abnormalities in pediatric mild traumatic brain injury. *Journal of Neurotrauma*, (2015). 32(10),723-730. <https://doi.org/10.1089/neu.2014.3534>

Mayer AR, Stephenson DD, Dodd AB, et al. Comparison of Methods for Classifying Persistent Post-Concussive Symptoms in Children. *J Neurotrauma* [Internet]. 2020 Jul 1;37(13):1504–11. Available from: <https://www.liebertpub.com/doi/10.1089/neu.2019.6805>

Mayer AR, Stephenson DD, Wertz CJ, et al. Proactive inhibition deficits with normal perfusion after pediatric mild traumatic brain injury. *Hum Brain Mapp* [Internet]. 2019 Dec 15;40(18):5370–81. Available from: <https://onlinelibrary.wiley.com/doi/abs/10.1002/hbm.24778>

McCart M, Glang AE, Slocumb J, Gau J, Beck L, Gomez D. A quasi-experimental study examining the effects of online traumatic brain injury professional development on educator knowledge, application, and efficacy in a practitioner setting. *Disability and Rehabilitation*, (2019). 0(0),1-7. <https://doi.org/10.1080/09638288.2019.1578423>

McCarty CA, Zatzick D, Stein E, Wang J, Hilt R, Rivara FP. Collaborative Care for Adolescents With Persistent Postconcussive Symptoms: A Randomized Trial. *Pediatrics*, (2016). 138(4),e20160459-

e20160459. <https://doi.org/10.1542/peds.2016-0459>

McConnell B, Duffield T, Hall T, Piantino J, Seitz D, ... Williams C. Post-traumatic Headache After Pediatric Traumatic Brain Injury: Prevalence, Risk Factors, and Association With Neurocognitive Outcomes. *Journal of Child Neurology*, (2019). 35(1),63-70. <https://doi.org/10.1177/0883073819876473>

McCormick BF, Connolly EJ, Nelson D V. Mild Traumatic Brain Injury as a Predictor of Classes of Youth Internalizing and Externalizing Psychopathology. *Child Psychiatry Hum Dev* [Internet]. 2020 May 5; Available from: <http://link.springer.com/10.1007/s10578-020-00992-9>

McCrea M, Guskiewicz K, Randolph C, Barr WB, Hammeke TA, ... Kelly JP. Incidence, clinical course, and predictors of prolonged recovery time following sport-related concussion in high school and college athletes. *Journal of the International Neuropsychological Society*, (2013). 19(1),22-33. <https://doi.org/10.1017/S1355617712000872>

McEvoy H, Borsook D, Holmes SA. Clinical features and sex differences in pediatric post-traumatic headache: A retrospective chart review at a Boston area concussion clinic. *Cephalalgia* [Internet]. 2019 Jun 22;0(0):1–11. Available from: <http://journals.sagepub.com/doi/10.1177/0333102419896754>

McEvoy H, Borsook D, Holmes SA. Clinical features and sex differences in pediatric post-traumatic headache: A retrospective chart review at a Boston area concussion clinic. *Cephalalgia*, (2019). 0(0),1-11. <https://doi.org/10.1177/0333102419896754>

McGrath N, Dinn WM, Collins MW, Lovell MR, Elbin RJ, Kontos AP. Post-exertion neurocognitive test failure among student-athletes following concussion. *Brain Injury*, (2013). 27(1),103-113. <https://doi.org/10.3109/02699052.2012.729282>

McGuine TA, Pfaller AY, Post EG, Hetzel SJ, Brooks A, Broglio SP. The influence of athletic trainers on the incidence and management of concussions in high school athletes. *Journal of Athletic Training*, (2018). 53(11),1017-1024. <https://doi.org/10.4085/1062-6050-209-18>

McKay CD, Schneider KJ, Brooks BL, Mrazik M, Emery CA. Baseline Evaluation in Youth Ice Hockey Players: Comparing Methods for Documenting Prior Concussions and Attention or Learning Disorders. *Journal of Orthopaedic & Sports Physical Therapy*, (2014). 44(5),329-335. <https://doi.org/10.2519/jospt.2014.5053>

Mckinlay A, Ligteringen V, Than M. A comparison of concussive symptoms reported by parents for preschool versus school-aged children. *Journal of Head Trauma Rehabilitation*, (2014). 29(3),233-238. <https://doi.org/10.1097/HTR.0b013e3182a2dd7f>

McNally KA, Patrick KE, LaFleur JE, Dykstra JB, Monahan K, Hoskinson KR. Brief cognitive behavioral intervention for children and adolescents with persistent post-concussive symptoms: A pilot study. *Child Neuropsychology*, (2018). 24(3),396-412. <https://doi.org/10.1080/09297049.2017.1280143>

Meehan WP, Mannix R, Monuteaux MC, Stein CJ, Bachur RG. Early symptom burden predicts recovery after sport-related concussion. *Neurology*, (2014). 83(24),2204-2210. <https://doi.org/10.1212/WNL.0000000000001073>

Meehan WP, Mannix RC, O'Brien MJ, Collins MW. The prevalence of undiagnosed concussions in athletes. *Clinical Journal of Sport Medicine*, (2013). 23(5),339-342. <https://doi.org/10.1097/JSM.0b013e318291d3b3>

Meehan WP, Mannix RC, Straccolini A, Elbin RJ, Collins MW. Symptom Severity Predicts Prolonged

Recovery after Sport-Related Concussion, but Age and Amnesia Do Not. *The Journal of Pediatrics*, (2013). 163(3),721-725. <https://doi.org/10.1016/j.jpeds.2013.03.012>

Meier TB, Nelson LD, Huber DL, Bazarian JJ, Hayes RL, McCrea MA. Prospective Assessment of Acute Blood Markers of Brain Injury in Sport-Related Concussion. *Journal of Neurotrauma*, (2017). 1,neu.2017.5046. <https://doi.org/10.1089/neu.2017.5046>

Mihalik JP, Register-Mihalik J, Kerr ZY, Marshall SW, McCrea MC, Guskiewicz KM. Recovery of posttraumatic migraine characteristics in patients after mild traumatic brain injury. *American Journal of Sports Medicine*, (2013). 41(7),1490-1496. <https://doi.org/10.1177/0363546513487982>

Mihindu E, Bhullar I, Tepas J, Kerwin A. Computed tomography of the head in children with mild traumatic brain injury. *American Surgeon*, (2014). 80(9),841-843. <http://www.embase.com/search/results?subaction=viewrecord&from=export&id=L604534421%5Chttp://sfx.library.uu.nl/utrecht?sid=EMBASE&issn=00031348&id=doi:&atitle=Computed+tomography+of+the+head+in+children+with+mild+traumatic+brain+injury&stitle=Am.+Surg.&t>

Miller JH, Gill C, Kuhn EN, Rocque BG, Menendez JY, ... Johnston JM. Predictors of delayed recovery following pediatric sports-related concussion: A case-control study. *Journal of Neurosurgery: Pediatrics*, (2016). 17(4),491-496. <https://doi.org/10.3171/2015.8.PEDS14332>

Moor HM, Eisenhauer RC, Killian KD, Proudfoot N, Henriques AA, ... Reneker JC. The relationship between adherence behaviors and recovery time in adolescents after a sports-related concussion: an observational study. *Int J Sports Phys Ther*, (2015). 10(2),225-233.

Moran R, Covassin T. Risk factors associated with baseline King-Devick performance. *Journal of the Neurological Sciences*, (2017). 383(October),101-104. <https://doi.org/10.1016/j.jns.2017.10.039>

Moran RN, Covassin T. King-Devick test normative reference values and internal consistency in youth football and soccer athletes. *Scandinavian Journal of Medicine and Science in Sports*, (2018). 28(12),2686-2690. <https://doi.org/10.1111/sms.13286>

Morgan CD, Zuckerman SL, King LE, Beaird SE, Sills AK, Solomon GS. Post-concussion syndrome (PCS) in a youth population: defining the diagnostic value and cost-utility of brain imaging. *Child's Nervous System*, (2015). 31(12),2305-2309. <https://doi.org/10.1007/s00381-015-2916-y>

Morgan CD, Zuckerman SL, Lee YM, King L, Beaird S, ... Solomon GS. Predictors of postconcussion syndrome after sports-related concussion in young athletes: a matched case-control study. *Journal of Neurosurgery: Pediatrics*, (2015). 15(June),589-598. <https://doi.org/10.3171/2014.10.PEDS14356.Disclosure>

Moser RS, Olek L, Schatz P. Gender Differences in Symptom Reporting on Baseline Sport Concussion Testing Across the Youth Age Span. *Archives of clinical neuropsychology*, (2019). 34(1),50-59. <https://doi.org/10.1093/arclin/acy007>

Moser RS, Schatz P, Glenn M, Kollias KE, Iverson GL. Examining prescribed rest as treatment for adolescents who are slow to recover from concussion. *Brain Injury*, (2015). 29(1),58-63. <https://doi.org/10.3109/02699052.2014.964771>

Moser RS, Schatz P, Grosner E, Kollias K. One year test–retest reliability of neurocognitive baseline scores in 10- to 12-year olds. *Applied Neuropsychology: Child*, (2017). 6(2),166-171. <https://doi.org/10.1080/21622965.2016.1138310>

Moser RS, Schatz P. Increased Symptom Reporting in Young Athletes Based on History of Previous Concussions. *Developmental Neuropsychology*, (2017). 42(4),276-283.

<https://doi.org/10.1080/87565641.2017.1334785>

Mrazik M, Brooks BL, Jubinville A, Meeuwisse WH, Emery CA. Psychosocial outcomes of sport concussions in youth hockey players. *Archives of Clinical Neuropsychology*, (2016). 31(4),297-304.

<https://doi.org/10.1093/arclin/acw013>

Mucha A, Collins MW, Elbin RJ, Furman JM, Troutman-Enseki C, ... Kontos AP. A brief vestibular/ocular motor screening (VOMS) assessment to evaluate concussions: Preliminary findings. *American Journal of Sports Medicine*, (2014). 42(10),2479-2486. <https://doi.org/10.1177/0363546514543775>

Munia TTK, Haider A, Schneider C, Romanick M, Fazel-Rezai R. A Novel EEG Based Spectral Analysis of Persistent Brain Function Alteration in Athletes with Concussion History. *Scientific Reports*, (2017).

7(1),1-13. <https://doi.org/10.1038/s41598-017-17414-x>

Muñoz-Santanach D, Trenchs Sainz de la Maza V, Maya Gallego S, Cuaresma González A, Luaces Cubells C. Clinical observation: A safe alternative to radiology in infants with mild traumatic brain injury. *Anales de Pediatría (English Edition)*, (2017). 87(3),164-169. <https://doi.org/10.1016/j.anpede.2016.09.010>

Murata NM, Oshiro RS, Furutani T, et al. Hawai'i Concussion Awareness & Management Program (HCAMP): Impact. *Hawai'i J Med public Heal a J Asia Pacific Med Public Heal*. 2019 May;78(5):155–62.

Murdaugh DL, King TZ, Sun B, Jones RA, Ono KE, ... Burns TG. Longitudinal Changes in Resting State Connectivity and White Matter Integrity in Adolescents with Sports-Related Concussion. *Journal of the International Neuropsychological Society*, (2018). 24(8),781-792.

<https://doi.org/10.1017/S1355617718000413>

Murdaugh DL, Ono KE, Morris SO, Burns TG. Effects of Developmental Age on Symptom Reporting and Neurocognitive Performance in Youth After Sports-Related Concussion Compared to Control Athletes. *Journal of Child Neurology*, (2018). 33(7),474-481. <https://doi.org/10.1177/0883073818766815>

Murdaugh DL, Ono KE, Reisner A, Burns TG. Assessment of Sleep Quantity and Sleep Disturbances During Recovery From Sports-Related Concussion in Youth Athletes. *Archives of Physical Medicine and Rehabilitation*, (2018). 99(5),960-966. <https://doi.org/10.1016/j.apmr.2018.01.005>

Murray NG, Ambati VNP, Contreras MM, Salvatore AP, Reed-Jones RJ. Assessment of oculomotor control and balance post-concussion: A preliminary study for a novel approach to concussion management. *Brain Injury*, (2014). 28(4),496-503. <https://doi.org/10.3109/02699052.2014.887144>

Mutch WAC, Ellis MJ, Ryner LN, McDonald PJ, Morissette MP, ... Fisher JA. Patient-specific alterations in CO2 cerebrovascular responsiveness in acute and sub-acute sports-related concussion. *Frontiers in Neurology*, (2018). 9(JAN),1-11. <https://doi.org/10.3389/fneur.2018.00023>

Mutch WAC, Ellis MJ, Ryner LN, Morissette MP, Pries PJ, ... Fisher JA. Longitudinal Brain Magnetic Resonance Imaging CO2 Stress Testing in Individual Adolescent Sports-Related Concussion Patients: A Pilot Study. *Frontiers in Neurology*, (2016). 7(July),1-8. <https://doi.org/10.3389/fneur.2016.00107>

Mutch, W. Alan C. Ellis, Michael J. Ryner, Lawrence N. Graham, Ruth. Dufault, Brenden. Gregson, Brian. Hall, Thomas. Bunge, Martin. Essig M. Brain magnetic resonance imaging CO2 stress testing in adolescent post-concussion syndrome: pCASL findings. *Journal of Neurosurgery*, (2015).

125(September),1-13. <https://doi.org/10.3171/2015.6.JNS15972>.

Nakhjavan-Shahraki B, Yousefifard M, Hajighanbari MJ, Oraii A, Safari S, Hosseini M. Pediatric Emergency Care Applied Research Network (PECARN) prediction rules in identifying high risk children with mild traumatic brain injury. *European Journal of Trauma and Emergency Surgery*, (2017). 43(6),755-762. <https://doi.org/10.1007/s00068-017-0811-9>

Nance ML, Callahan JM, Tharakan SJ, Malamet P, Houseknecht EM, ... Wiebe DJ. Utility of neurocognitive testing of mild traumatic brain injury in children treated and released from the emergency department. *Brain Injury*, (2016). 30(2),184-190. <https://doi.org/10.3109/02699052.2015.1075591>

Nelson LD, Laroche AA, Pfaller AY, Lerner EB, Hammeke TA, ... McCrea MA. Prospective, head-to-head study of three computerized neurocognitive assessment tools (CNTs): Reliability and validity for the assessment of sport-related concussion. *Journal of the International Neuropsychological Society*, (2015). 1. Shi F,(1),24-37. <https://doi.org/10.1017/S1355617715001101>

Nelson LD, Loman MM, LaRoche AA, Furger RE, McCrea MA. Baseline Performance and Psychometric Properties of the Child Sport Concussion Assessment Tool 3 (Child-SCAT3) in 5- to 13-year-old Athletes. *Clinical Journal of Sport Medicine*, (2017). 27(4),381-387. <https://doi.org/10.1097/JSM.0000000000000369>

Nelson LD, Pfaller AY, Rein LE, McCrea MA. Rates and Predictors of Invalid Baseline Test Performance in High School and Collegiate Athletes for 3 Computerized Neurocognitive Tests: NAM,Axon Sports, and ImPACT. *American Journal of Sports Medicine*, (2015). 43(8),2018-2026. <https://doi.org/10.1177/0363546515587714>

Newman JB, Reesman JH, Vaughan CG, Gioia GA. Assessment of processing speed in children with mild tbi: A “first look” at the validity of pediatric ImPACT. *Clinical Neuropsychologist*, (2013). 27(5),779-793. <https://doi.org/10.1080/13854046.2013.789552>

Newsome MR, Li X, Lin X, Wilde EA, Ott S, ... Levin HS. Functional connectivity is altered in concussed adolescent athletes despite medical clearance to return to play: A preliminary report. *Frontiers in Neurology*, (2016). 7(JUL),1-9. <https://doi.org/10.3389/fneur.2016.00116>

Nikles CJ, McKinlay L, Mitchell GK, Carmont SAS, Senior HE, ... Lloyd OT. Aggregated n-of-1 trials of central nervous system stimulants versus placebo for paediatric traumatic brain injury – a pilot study. *Trials*, (2014). 15(1),1-11. <https://doi.org/10.1186/1745-6215-15-54>

Novak Z, Aglipay M, Barrowman N, Yeates KO, Beauchamp MH, ... Zemek RL. Association of persistent postconcussion symptoms with pediatric quality of life. *JAMA Pediatrics*, (2016). 170(12),1-8. <https://doi.org/10.1001/jamapediatrics.2016.2900>

Nowacki R, van Eldik N, Eikens M, Roijen R, Haga N, ... Wennekes M. Evaluation of a follow-up program for mild traumatic brain injury in schoolchildren. *European Journal of Paediatric Neurology*, (2017). 21(2),382-387. <https://doi.org/10.1016/j.ejpn.2016.10.009>

O’Brien MJ, Howell DR, Pepin MJ, Meehan WP. Sport-related concussions: Symptom recurrence after return to exercise. *Orthopaedic Journal of Sports Medicine*, (2017). 5(10),1-6. <https://doi.org/10.1177/2325967117732516>

O’Rourke DJ, Smith RE, Punt S, Coppel DB, Breiger D. Psychosocial correlates of young athletes’ self-reported concussion symptoms during the course of recovery. *Sport, Exercise, and Performance*

Psychology, (2017). 6(3),262-276. <https://doi.org/10.1037/spy0000097>

Oberlander TJ, Olson BL, Weidauer L. Test-Retest Reliability of the King-Devick Test in an Adolescent Population. *Journal of Athletic Training*, (2017). 52(5),439-445. <https://doi.org/10.4085/1062-6050-52.2.12>

Olson A, Ellis MJ, Selci E, et al. Delayed Symptom Onset Following Pediatric Sport-Related Concussion. *Front Neurol*. 2020 Apr 3;11.

Orban SA, Epstein JN, Carr D, et al. Attention-Deficit/Hyperactivity Disorder Status and Sex Moderate Mild Traumatic Brain Injury Symptom Severity in Children and Adolescents. *Clin J Sport Med* [Internet]. 2020 Feb 13;Publish Ah. Available from: <http://dx.doi.org/10.1097/JSM.0000000000000821>

Oris C, Pereira B, Durif J, Simon-Pimmel J, Castellani C, ... Bouvier D. The biomarker s100b and mild traumatic brain injury: A meta-analysis. *Pediatrics*, (2018). 141(6). <https://doi.org/10.1542/peds.2018-0037>

Orr CA, Albaugh MD, Watts R, Garavan H, Andrews T, ... Hudziak JJ. Neuroimaging Biomarkers of a History of Concussion Observed in Asymptomatic Young Athletes. *Journal of Neurotrauma*, (2016). 33(9),803-810. <https://doi.org/10.1089/neu.2014.3721>

Orr R, Bogg T, Fyffe A, Lam LT, Browne GJ. Graded Exercise Testing Predicts Recovery Trajectory of Concussion in Children and Adolescents. *Clinical Journal of Sport Medicine*, (2018). 00(00),1. <https://doi.org/10.1097/jsm.0000000000000683>

Ott S, Schatz P, Solomon G, Ryan JJ. Neurocognitive performance and symptom profiles of spanish-speaking hispanic athletes on the impact test. *Archives of Clinical Neuropsychology*, (2014). 1. Shi F,(2),152-163. <https://doi.org/10.1093/arclin/act091>

Oyegbile TO, Delasobera BE, Zecavati N. Postconcussive Symptoms After Single and Repeated Concussions in 10- to 20-Year-Olds: A Cross-Sectional Study. *Journal of Child Neurology*, (2018). 33(6),383-388. <https://doi.org/10.1177/0883073818759436>

Oyegbile TO, Dougherty A, Tanveer S, et al. High Sleep Disturbance and Longer Concussion Duration in Repeat Concussions. *Behav Sleep Med*. 2020 Mar 3;18(2):241–8.

Ozinga SJ, Linder SM, Koop MM, Dey T, Figler R, ... Alberts JL. Normative performance on the balance error scoring system by youth, high school, and collegiate athletes. *Journal of Athletic Training*, (2018). 53(7),636-645. <https://doi.org/10.4085/1062-6050-129-17>

Paniccia M, Ippolito C, McFarland S, et al. Health-Related Quality of Life in Non-Concussed Children: A Normative Study to Inform Concussion Management. *Dev Neurorehabil* [Internet]. 2020 Nov 16;23(8):534–41. Available from: <https://www.tandfonline.com/doi/full/10.1080/17518423.2020.1736683>

Papa L, Mittal MK, Ramirez J, Ramia M, Kirby S, ... Zonfrillo MR. In Children and Youth with Mild and Moderate Traumatic Brain Injury, Glial Fibrillary Acidic Protein Out-Performs S100β in Detecting Traumatic Intracranial Lesions on Computed Tomography. *Journal of Neurotrauma*, (2016). 33(1),58-64. <https://doi.org/10.1089/neu.2015.3869>

Papa L, Mittal MK, Ramirez J, Silvestri S, Giordano P, ... Zonfrillo MR. Neuronal Biomarker Ubiquitin C-Terminal Hydrolase Detects Traumatic Intracranial Lesions on Computed Tomography in Children and Youth with Mild Traumatic Brain Injury. *Journal of Neurotrauma*, (2017). 34(13),2132-2140.

<https://doi.org/10.1089/neu.2016.4806>

Papa L, Zonfrillo MR, Welch RD, Lewis LM, Braga CF, ... Mittal MK. Evaluating glial and neuronal blood biomarkers GFAP and UCH-L1 as gradients of brain injury in concussive, subconcussive and non-concussive trauma: a prospective cohort study. *BMJ Paediatrics Open*, (2019). 3(1),e000473.

<https://doi.org/10.1136/bmjpo-2019-000473>

Parkin GM, Clarke C, Takagi M, Hearps S, Babl FE, ... Ignjatovic V. Plasma Tumor Necrosis Factor Alpha Is a Predictor of Persisting Symptoms Post-Concussion in Children. *Journal of Neurotrauma*, (2019). 36(11),1768-1775. <https://doi.org/10.1089/neu.2018.6042>

Pasek TA, Locasto LW, Reichard J, Fazio Sumrok VC, Johnson EW, Kontos AP. The headache electronic diary for children with concussion. *Clinical Nurse Specialist*, (2015). 29(2),80-88.

<https://doi.org/10.1097/NUR.000000000000108>

Patsimas T, Howell DR, Potter MN, et al. Concussion-Symptom Rating Correlation Between Pediatric Patients and Their Parents. *J Athl Train* [Internet]. 2020 Oct 1;55(10):1020–6. Available from:

<https://meridian.allenpress.com/jat/article/55/10/1020/436343/ConcussionSymptom-Rating-Correlation-Between>

Pavlou M, Whitney SL, Alkathiry AA, Huett M, Luxon LM, ... Bamiou DE. Visually induced dizziness in children and validation of the Pediatric Visually Induced Dizziness Questionnaire. *Frontiers in Neurology*, (2017). 8(DEC),1-9. <https://doi.org/10.3389/fneur.2017.00656>

Pawlukiewicz A, Yengo-Kahn AM, Solomon G. The effect of pretest exercise on baseline computerized neurocognitive test scores. *Orthopaedic Journal of Sports Medicine*, (2017). 1. Shi F,(10),1-6.

<https://doi.org/10.1177/2325967117734496>

Pearce KL, Sufrinko A, Lau BC, Henry L, Collins MW, Kontos AP. Near point of convergence after a sport-related concussion: Measurement reliability and relationship to neurocognitive impairment and symptoms. *American Journal of Sports Medicine*, (2015). 43(12),3055-3061.

<https://doi.org/10.1177/0363546515606430>

Pearson R, Levyim D, Choe M, Taraman S, Langdon R. Survey of Child Neurologists on Management of Pediatric Post-traumatic Headache. *Journal of Child Neurology*, (2019). 34(12),739-747.

<https://doi.org/10.1177/0883073819856837>

Peltonen K, Vartiainen M, Laitala-Leinonen T, Koskinen S, Luoto T, ... Hokkanen L. Adolescent athletes with learning disability display atypical maturational trajectories on concussion baseline testing: Implications based on a Finnish sample. *Child Neuropsychology*, (2019). 25(3),336-351.

<https://doi.org/10.1080/09297049.2018.1474865>

Peterson RL, Connery AK, Baker DA, Kirkwood MW. Preinjury Emotional-Behavioral Functioning of Children With Lingering Problems After Mild Traumatic Brain Injury. *The Journal of Neuropsychiatry and Clinical Neurosciences*, (2015). 27(4),280-286. <https://doi.org/10.1176/appi.neuropsych.14120373>

Pieper P, Garvan C. Health-related quality-of-life in the first year following a childhood concussion. *Brain Injury*, (2014). 28(1),105-113. <https://doi.org/10.3109/02699052.2013.847208>

Plackett TP, Asturias S, Tadlock M, Wright F, Ton-That H, ... Inaba K. Re-evaluating the need for hospital admission and observation of pediatric traumatic brain injury after a normal head CT. *Journal of*

pediatric surgery, (2015). 50(10),1758-1761. <https://doi.org/10.1016/j.jpedsurg.2015.03.067>

Plourde V, Daya H, Low TA, Barlow KM, Brooks BL. Evaluating anxiety and depression symptoms in children and adolescents with prior mild traumatic brain injury: Agreement between methods and respondents. *Child Neuropsychology*, (2019). 25(1),44-59.
<https://doi.org/10.1080/09297049.2018.1432585>

Plourde V, Kung JY, Gates A, et al. How Perceptions Impact Recovery from Concussion in Childhood and Adolescence: a Systematic Review. *Neuropsychol Rev* [Internet]. 2020 Mar 2;30(1):142–63. Available from: <http://link.springer.com/10.1007/s11065-020-09430-y>

Plourde V, Yeates KO, Brooks BL. Predictors of long-term psychosocial functioning and health-related quality of life in children and adolescents with prior concussions. *Journal of the International Neuropsychological Society*, (2018). 24(6),540-548. <https://doi.org/10.1017/S1355617718000061>

Podolak OE, Chaudhary S, Haarbauer-Krupa J, et al. Characteristics of Diagnosed Concussions in Children Aged 0 to 4 Years Presenting to a Large Pediatric Healthcare Network. *Pediatr Emerg Care* [Internet]. 2020 Jun 16;Publish Ah. Available from: www.pec-online.com

Porter S, Smith-Forrester J, Alhajri N, Kusch C, Sun J, ... Virji-Babul N. The Child Sport Concussion Assessment Tool (Child SCAT3): Normative values and correspondence between child and parent symptom scores in male child athletes. *BMJ Open Sport and Exercise Medicine*, (2017). 3(1),1-6.
<https://doi.org/10.1136/bmjsem-2015-000029>

Povolo CA, Reid JN, Shariff SZ, et al. Concussion in adolescence and the risk of multiple sclerosis: A retrospective cohort study. *Mult Scler J* [Internet]. 2020 Feb 24;135245852090803. Available from: <http://journals.sagepub.com/doi/10.1177/1352458520908037>

Poysophon P, Rao AL. Neurocognitive Deficits Associated With ADHD in Athletes: A Systematic Review. *Sports Health*, (2018). 10(4),317-326. <https://doi.org/10.1177/1941738117751387>

Puffenbarger MS, Ahmad FA, Argent M, Gu H, Samson C, ... Saito JM. Reduction of Computed Tomography Use for Pediatric Closed Head Injury Evaluation at a Nonpediatric Community Emergency Department. *Academic Emergency Medicine*, (2019). 26(7),784-795.
<https://doi.org/10.1111/acem.13666>

Purcell LK, Davis GA, Gioia GA. What factors must be considered in “return to school” following concussion and what strategies or accommodations should be followed? A systematic review. *British Journal of Sports Medicine*, (2019). 53(4),5-18. <https://doi.org/10.1136/bjsports-2017-097853>

Quatman-Yates C, Hugentobler J, Ammon R, Mwase N, Kurowski B, Myer GD. The utility of the balance error scoring system for mild brain injury assessments in children and adolescents. *Physician and Sportsmedicine*, (2014). 42(3),32-38. <https://doi.org/10.3810/psm.2014.09.2073>

Quatman-Yates CC, Bonnette S, Hugentobler JA, Médé B, Kiefer AW, ... Riley MA. Postconcussion Postural Sway Variability Changes in Youth: The Benefit of Structural Variability Analyses. *Pediatric Physical Therapy*, (2015). 27(4),316-327. <https://doi.org/10.1097/PEP.0000000000000193>

Rabinowitz AR, Li X, McCauley SR, Wilde EA, Barnes A, ... Levin HS. Prevalence and Predictors of Poor Recovery from Mild Traumatic Brain Injury. *Journal of Neurotrauma*, (2015). 32(19),1488-1496.
<https://doi.org/10.1089/neu.2014.3555>

Raghuram A, Cotter SA, Gowrisankaran S, et al. Postconcussion: Receded Near Point of Convergence is

not Diagnostic of Convergence Insufficiency. Am J Ophthalmol [Internet]. 2019 Oct 1;206:235–44. Available from: <https://linkinghub.elsevier.com/retrieve/pii/S0002939419301825>

Raji CA, Wang MB, Nguyen NN, et al. Connectome mapping with edge density imaging differentiates pediatric mild traumatic brain injury from typically developing controls: proof of concept. Pediatr Radiol [Internet]. 2020 Oct 30;50(11):1594–601. Available from: <http://link.springer.com/10.1007/s00247-020-04743-9>

Ransom DM, Burns AR, Youngstrom EA, Vaughan CG, Sady MD, Gioia GA. Applying an Evidence-Based Assessment Model to Identify Students at Risk for Perceived Academic Problems following Concussion. Journal of the International Neuropsychological Society, (2016). 22(10),1038-1049. <https://doi.org/10.1017/S1355617716000916>

Ransom DM, Vaughan CG, Pratson L, Sady MD, McGill CA, Gioia GA. Academic effects of concussion in children and adolescents. Pediatrics, (2015). 135(6),1043-1050. <https://doi.org/10.1542/peds.2014-3434>

Reches A, Kutcher J, Elbin RJ, Or-Ly H, Sadeh B, ... Kontos AP. Preliminary investigation of Brain Network Activation (BNA) and its clinical utility in sport-related concussion. Brain Injury, (2017). 31(2),237-246. <https://doi.org/10.1080/02699052.2016.1231343>

Reddy CC, Collins M, Lovell M, Kontos AP. Efficacy of amantadine treatment on symptoms and neurocognitive performance among adolescents following sports-related concussion. Journal of Head Trauma Rehabilitation, (2013). 28(4),260-265. <https://doi.org/10.1097/HTR.0b013e318257fbc6>

Reed N, Taha T, Monette G, Keightley M. A Preliminary Exploration of Concussion and Strength Performance in Youth Ice Hockey Players. International Journal of Sports Medicine, (2016). 37(9),708-713. <https://doi.org/10.1055/s-0042-104199>

Reesman J, Pineda J, Carver J, Brice PJ, Andrew Zabel T, Schatz P. Utility of the ImPACT test with deaf adolescents. Clinical Neuropsychologist, (2016). 30(2),318-327. <https://doi.org/10.1080/13854046.2016.1142613>

Register-Mihalik JK, De Maio VJ, Tibbo-Valeriotte HL, Wooten JD. Characteristics of Pediatric and Adolescent Concussion Clinic Patients With Postconcussion Amnesia. Clinical Journal of Sport Medicine, (2014). 27599,1. <https://doi.org/10.1097/JSM.0000000000000161>

Renaud MI, Klees C, van Haastregt JCM, et al. Process evaluation of ‘Brains Ahead!’: an intervention for children and adolescents with mild traumatic brain injury within a randomized controlled trial. Clin Rehabil [Internet]. 2020 May 5;34(5):688–97. Available from: <http://journals.sagepub.com/doi/10.1177/0269215520911439>

Renaud MI, Lambregts SAM, van de Port IGL, et al. Predictors of activities and participation six months after mild traumatic brain injury in children and adolescents. Eur J Paediatr Neurol [Internet]. 2020 Mar 1;25:145–56. Available from: <https://linkinghub.elsevier.com/retrieve/pii/S1090379819304179>

Reynolds E, Fazio VC, Sandel N, Schatz P, Henry LC. Cognitive Development and the Immediate Postconcussion Assessment and Cognitive Testing: A Case for Separate Norms in Preadolescents. Applied Neuropsychology: Child, (2016). 5(4),283-293. <https://doi.org/10.1080/21622965.2015.1057637>

Rhine T, Babcock L, Zhang N, Leach J, Wade SL. Are UCH-L1 and GFAP promising biomarkers for children with mild traumatic brain injury? Brain Injury, (2016). 30(10),1231-1238. <https://doi.org/10.1080/02699052.2016.1178396>

Rhine T, Quatman-Yates C, Clark RA. A longitudinal examination of postural impairments in children with mild traumatic brain injury: Implications for acute testing. *Journal of Head Trauma Rehabilitation*, (2017). 32(2),E18-E23. <https://doi.org/10.1097/HTR.000000000000192>

Rhine TD, Byczkowski TL, Clark RA, Babcock L. Investigating the Feasibility and Utility of Bedside Balance Technology Acutely after Pediatric Concussion: A Pilot Study. *Clinical Journal of Sport Medicine*, (2016). 26(3),221-225. <https://doi.org/10.1097/JSM.0000000000000234>

Rieger BP, Lewandowski LJ, Callahan JM, Spenceley L, Truckenmiller A, ... Miller LA. A prospective study of symptoms and neurocognitive outcomes in youth with concussion vs orthopaedic injuries. *Brain Injury*, (2013). 27(2),169-178. <https://doi.org/10.3109/02699052.2012.729290>

Rocheftort C, Walters-Stewart C, Aglipay M, Barrowman N, Zemek R, Sveistrup H. Balance markers in adolescents at 1 month postconcussion. *Orthopaedic Journal of Sports Medicine*, (2017). 5(3),1-7. <https://doi.org/10.1177/2325967117695507>

Rocheftort C, Walters-Stewart C, Aglipay M, Barrowman N, Zemek R, Sveistrup H. Self-reported balance status is not a reliable indicator of balance performance in adolescents at one-month post-concussion. *Journal of Science and Medicine in Sport*, (2017). 20(11),970-975. <https://doi.org/10.1016/j.jsams.2017.04.008>

Root JM, McNamara B, Ledda M, et al. Pediatric Patient Compliance With Recommendations for Acute Concussion Management. *Clin Pediatr (Phila)* [Internet]. 2019 Jun;58(7):731–7. Available from: <http://journals.sagepub.com/doi/10.1177/0009922819839230>

Root JM, Sady MD, Gai J, et al. Effect of Cognitive and Physical Rest on Persistent Postconcussive Symptoms following a Pediatric Head Injury. *J Pediatr* [Internet]. 2020 Dec;227:184-190.e4. Available from: <https://linkinghub.elsevier.com/retrieve/pii/S0022347620309069>

Root JM, Zuckerbraun NS, Wang L, Winger DG, Brent D, ... Hickey RW. History of Somatization Is Associated with Prolonged Recovery from Concussion. *Journal of Pediatrics*, (2016). 174,39-44.e1. <https://doi.org/10.1016/j.jpeds.2016.03.020>

Rose SC, Schaffer CE, Young JA, McNally KA, Fischer AN, Heyer GL. Utilization of conventional neuroimaging following youth concussion. *Brain Injury*, (2017). 31(2),260-266. <https://doi.org/10.1080/02699052.2016.1235285>

Roumpf SK, Welch JL. Can S100B Serum Biomarker Testing Reduce Head Computed Tomography Scanning in Children With Mild Traumatic Brain Injury ? *Annals of Emergency Medicine*, (2019). 73(5),456-458. <https://doi.org/10.1016/j.annemergmed.2018.10.012>

Rozbacher A, Selci E, Leiter J, Ellis M, Russell K. The Effect of Concussion or Mild Traumatic Brain Injury on School Grades, National Examination Scores, and School Attendance: A Systematic Review. *Journal of Neurotrauma*, (2017). 34(14),2195-2203. <https://doi.org/10.1089/neu.2016.4765>

Russell K, Hutchison MG, Selci E, Leiter J, Chateau D, Ellis MJ. Academic outcomes in high-school students after a concussion: A retrospective population-based analysis. *PLoS ONE*, (2016). 11(10),1-12. <https://doi.org/10.1371/journal.pone.0165116>

Russell K, Selci E, Black B, Cochrane K, Ellis M. Academic outcomes following adolescent sport-related concussion or fracture injury: A prospective cohort study. *PLoS ONE*, (2019). 14(4),1-14. <https://doi.org/10.1371/journal.pone.0215900>

- Russell K, Selci E, Black B, Ellis MJ. Health-related quality of life following adolescent sports-related concussion or fracture: A prospective cohort study. *Journal of Neurosurgery: Pediatrics*, (2019). 23(4),455-464. <https://doi.org/10.3171/2018.8.PEDS18356>
- Russell K, Selci E, Chu S, Fineblit S, Ritchie L, Ellis MJ. Longitudinal Assessment of Health-Related Quality of Life following Adolescent Sports-Related Concussion. *Journal of Neurotrauma*, (2017). 34(13),2147-2153. <https://doi.org/10.1089/neu.2016.4704>
- Ryan NP, Catroppa C, Beare R, Coleman L, Ditchfield M, ... Anderson VA. Predictors of longitudinal outcome and recovery of pragmatic language and its relation to externalizing behaviour after pediatric traumatic brain injury. *Brain and Language*, (2015). 142,86-95. <https://doi.org/10.1016/j.bandl.2015.01.007>
- Ryan NP, van Bijnen L, Catroppa C, Beauchamp MH, Crossley L, ... Anderson V. Longitudinal outcome and recovery of social problems after pediatric traumatic brain injury (TBI): Contribution of brain insult and family environment. *International Journal of Developmental Neuroscience*, (2016). 49,23-30. <https://doi.org/10.1016/j.ijdevneu.2015.12.004>
- Sady MD, Vaughan CG, Gioia GA. Psychometric characteristics of the postconcussion symptom inventory in children and adolescents. *Archives of Clinical Neuropsychology*, (2014). 29(4),348-363. <https://doi.org/10.1093/arclin/acu014>
- Saleem GT, Slomine BS, Suskauer SJ. Sleep Symptoms Predict School Attendance After Pediatric Concussion. *Clin Pediatr (Phila)* [Internet]. 2020 Jun 16;59(6):580–7. Available from: <http://journals.sagepub.com/doi/10.1177/0009922820913960>
- Salinas CM, Dean P, LoGalbo A, Dougherty M, Field M, Webbe FM. Attention-Deficit Hyperactivity Disorder Status and Baseline Neurocognitive Performance in High School Athletes. *Applied Neuropsychology: Child*, (2016). 5(4),264-272. <https://doi.org/10.1080/21622965.2015.1052814>
- Salmon DM, McGowan J, Sullivan SJ, et al. What they know and who they are telling: Concussion knowledge and disclosure behaviour in New Zealand adolescent rugby union players. *J Sports Sci* [Internet]. 2020 Jul 17;38(14):1585–94. Available from: <https://www.tandfonline.com/doi/full/10.1080/02640414.2020.1749409>
- Saluja RS, Chen JK, Gagnon JJ, Keightley M, Ptito A. Navigational memory functional magnetic resonance imaging: A test for concussion in children. *Journal of Neurotrauma*, (2015). 32(10),712-722. <https://doi.org/10.1089/neu.2014.3470>
- Sambasivan K, Grilli L, Gagnon I. Balance and mobility in clinically recovered children and adolescents after a mild traumatic brain injury. *Journal of Pediatric Rehabilitation Medicine*, (2015). 8(4),335-344. <https://doi.org/10.3233/PRM-150351>
- Sandel NK, Schatz P, Goldberg KB, Lazar M. Sex-Based Differences in Cognitive Deficits and Symptom Reporting among Acutely Concussed Adolescent Lacrosse and Soccer Players. *American Journal of Sports Medicine*, (2017). 45(4),937-944. <https://doi.org/10.1177/0363546516677246>
- Sandel NK, Worts PR, Burkhart S, Henry L. Comparison of baseline IMPACT performance in amateur motocross riders to football and basketball athletes. *Brain Injury*, (2018). 32(4),493-497. <https://doi.org/10.1080/02699052.2018.1429020>
- Santo A, Lynall RC, Guskiewicz KM, Mihalik JP. Clinical utility of the sport concussion assessment tool 3

(SCAT3) tandem-gait test in high school athletes. *Journal of Athletic Training*, (2017). 52(12),1096-1100. <https://doi.org/10.4085/1062-6050-52.11.26>

Sarmiento K, Daugherty J, DePadilla L. Youth and high school sports coaches' experience with and attitudes about concussion and access to athletic trainers by sport type and age of athlete coached. *J Safety Res* [Internet]. 2019 Jun 1;69:217–25. Available from: <https://linkinghub.elsevier.com/retrieve/pii/S0022437519300490>

Sarmiento K, Donnell Z, Bell E, et al. A qualitative study of barriers and opportunities for concussion communication and management among parents of youth sports athletes. *J Concussion* [Internet]. 2019 Jan 9;3:205970021986186. Available from: <http://journals.sagepub.com/doi/10.1177/2059700219861863>

Sarmiento K, Donnell Z, Bell E, Hoffman R. From the CDC: A qualitative study of middle and high school professionals' experiences and views on concussion: Identifying opportunities to support the return to school process. *Journal of Safety Research*, (2019). 68,223-229. <https://doi.org/10.1016/j.jsr.2018.10.010>

Satchell EK, Friedman SD, Bompadre V, et al. Use of diffusion tensor imaging in the evaluation of pediatric concussions. *Musculoskelet Sci Pract* [Internet]. 2019 Jul 1;42:162–5. Available from: <https://linkinghub.elsevier.com/retrieve/pii/S2468781218302480>

Schatz P, Corcoran M, Kontos AP, et al. Youth Soccer Parents' Perceptions of Long-Term Effects of Concussion. *Dev Neuropsychol* [Internet]. 2020 Apr 2;45(3):110–7. Available from: <https://www.tandfonline.com/doi/full/10.1080/87565641.2020.1766464>

Schatz P, Kelley T, Ott SD, Solomon GS, Elbin RJ, ... Moser RS. Utility of repeated assessment after invalid baseline neurocognitive test performance. *Journal of Athletic Training*, (2014). 49(5),659-664. <https://doi.org/10.4085/1062-6050-49.3.37>

Schatz P, Robertshaw S. Comparing post-concussive neurocognitive test data to normative data presents risks for under-classifying “above average” athletes. *Archives of Clinical Neuropsychology*, (2014). 29(7),625-632. <https://doi.org/10.1093/arclin/acu041>

Scheiman MM, Talasan H, Lynn Mitchell G, Alvarez TL. Objective Assessment of Vergence after Treatment of Concussion-Related CI: A Pilot Study. *Optometry and Vision Science*, (2017). 94(1),74-88. <https://doi.org/10.1097/OPX.0000000000000936>

Schilling S, Mansour A, Sullivan L, et al. Symptom burden and profiles in concussed children with and without prolonged recovery. *Int J Environ Res Public Health*. 2020 Jan 1;17(1).

Schmidt AT, Li X, Hanten GR, McCauley SR, Faber J, Levin HS. A longitudinal investigation of sleep quality in adolescents and young adults after mild traumatic brain injury. *Cognitive and Behavioral Neurology*, (2015). 28(2),53-62. <https://doi.org/10.1097/WNN.0000000000000056>

Schmidt J, Hayward KS, Brown KE, Zwicker JG, Ponsford J, ... Boyd LA. Imaging in pediatric concussion: A systematic review. *Pediatrics*, (2018). 141(5). <https://doi.org/10.1542/peds.2017-3406>

Schneider KJ, Meeuwisse WH, Nettel-Aguirre A, Barlow K, Boyd L, ... Emery CA. Cervicovestibular rehabilitation in sport-related concussion: a randomised controlled trial. *British journal of sports medicine*, (2014). 48(17),1294-1298. <https://doi.org/10.1136/bjsports-2013-093267>

Seeger TA, Orr S, Bodell L, Lockyer L, Rajapakse T, Barlow KM. Occipital nerve blocks for pediatric

posttraumatic headache: A case series. *Journal of Child Neurology*, (2015). 30(9),1142-1146.
<https://doi.org/10.1177/0883073814553973>

Segev S, Shorer M, Rassovsky Y, Peleg TP, Apter A, Fennig S. The contribution of posttraumatic stress disorder and mild traumatic brain injury to persistent post concussive symptoms following motor vehicle accidents. *Neuropsychology*, (2016). 30(7),800-810. <https://doi.org/10.1037/neu0000299>

Seidman DH, Burlingame J, Yousif LR, Donahue XP, Krier J, ... Shaw MK. Evaluation of the King-Devick test as a concussion screening tool in high school football players. *Journal of the Neurological Sciences*, (2015). 356(1-2),97-101. <https://doi.org/10.1016/j.jns.2015.06.021>

Seiger A, Goldwater E, Deibert E. Does mechanism of injury play a role in recovery from concussion? *Journal of Head Trauma Rehabilitation*, (2015). 30(3),E52-E56.
<https://doi.org/10.1097/HTR.0000000000000051>

Shapiro JS, Silk T, Takagi M, et al. Examining Microstructural White Matter Differences between Children with Typical and Those with Delayed Recovery Two Weeks Post-Concussion. *J Neurotrauma* [Internet]. 2020 Jun 1;37(11):1300–5. Available from: <https://www.liebertpub.com/doi/10.1089/neu.2019.6768>

Sharma B, Allison D, Tucker P, et al. Cognitive and neural effects of exercise following traumatic brain injury: A systematic review of randomized and controlled clinical trials. *Brain Inj* [Internet]. 2020 Jan 28;34(2):149–59. Available from: <https://doi.org/10.1080/02699052.2019.1683892>

Shaw L, Morozova M, Abu-Arafeh I. Chronic post-traumatic headache in children and adolescents: systematic review of prevalence and headache features. *Pain management*, (2018). 8(1),57-64.
<https://doi.org/10.2217/pmt-2017-0019>

Shendell DG, Gonzalez L, Listwan TA, Pancella J, Blackborow M, Boyd J. Developing and Piloting a School-Based Online Adolescent Student-Athlete Concussion Surveillance System. *Journal of School Health*, (2019). 89(7),527-535. <https://doi.org/10.1111/josh.12775>

Shorer M, Segev S, Rassovsky Y, et al. Efficacy of Psychological Intervention for Children with Concurrent Posttraumatic Stress Disorder and Mild Traumatic Brain Injury. *J Trauma Stress* [Internet]. 2020 Jun 11;33(3):330–7. Available from: <https://onlinelibrary.wiley.com/doi/10.1002/its.22512>

Silverberg ND, Iverson GL, McCrea M, Apps JN, Hammeke TA, Thomas DG. Activity-related symptom exacerbations after pediatric concussion. *JAMA Pediatrics*, (2016). 170(10),946-953.
<https://doi.org/10.1001/jamapediatrics.2016.1187>

Sinnott AM, Elbin RJ, Collins MW, et al. Persistent vestibular-ocular impairment following concussion in adolescents. *J Sci Med Sport* [Internet]. 2019 Dec 1;22(12):1292–7. Available from: <https://linkinghub.elsevier.com/retrieve/pii/S1440244018312982>

Sinopoli KJ, Chen J-K, Wells G, Fait P, Ptito A, ... Keightley M. Imaging “Brain Strain” in Youth Athletes with Mild Traumatic Brain Injury during Dual-Task Performance. *Journal of Neurotrauma*, (2014). 31(22),1843-1859. <https://doi.org/10.1089/neu.2014.3326>

Smolyansky V, Morettin CE, Hitzman SA, Beckerman S. Test-retest reliability of the King-Devick test in elite junior olympic athletes. *Optometry & Visual Performance*, (2016). 4(3),147-154.

Snedden TR, Brooks MA, Hetzel S, McGuine T. Normative Values of the Sport Concussion Assessment Tool 3 (SCAT3) in High School Athletes. *Clinical Journal of Sport Medicine*, (2017). 1. Shi F,(5),462-467.

<https://doi.org/http://dx.doi.org/10.1097/jsm.0000000000000389>

Snook ML, Henry LC, Sanfilippo JS, Zeleznik AJ, Kontos AP. Association of concussion with abnormal menstrual patterns in adolescent and young women. *JAMA Pediatrics*, (2017). 171(9),879-886.

<https://doi.org/10.1001/jamapediatrics.2017.1140>

Snyder AR, Bauer RM. A Normative Study of the Sport Concussion Assessment Tool (SCAT2) in Children and Adolescents. *The Clinical Neuropsychologist*, (2014). 28(7),1091-1103.

<https://doi.org/10.1080/13854046.2014.952667>

Starkey NJ, Jones K, Case R, Theadom A, Barker-Collo S, Feigin V. Post-concussive symptoms after a mild traumatic brain injury during childhood and adolescence. *Brain Injury*, (2018). 32(5),617-626.

<https://doi.org/10.1080/02699052.2018.1439533>

Stazyk K, DeMatteo C, Moll S, Missiuna C. Depression in youth recovering from concussion: Correlates and predictors. *Brain Injury*, (2017). 31(5),631-638. <https://doi.org/10.1080/02699052.2017.1283533>

Stein E, Howard W, Rowhani-Rahbar A, Rivara FP, Zatzick D, McCarty CA. Longitudinal trajectories of post-concussive and depressive symptoms in adolescents with prolonged recovery from concussion. *Brain Injury*, (2017). 31(13-14),1736-1744. <https://doi.org/10.1080/02699052.2017.1380843>

Stephens JA, Liu P, Lu H, Suskauer SJ. Cerebral Blood Flow after Mild Traumatic Brain Injury: Associations between Symptoms and Post-Injury Perfusion. *Journal of Neurotrauma*, (2018). 35(2),241-248.

<https://doi.org/10.1089/neu.2017.5237>

Stockbridge MD, Doran A, King K, Newman RS. The effects of concussion on rapid picture naming in children. *Brain Injury*, (2018). 32(4),506-514. <https://doi.org/10.1080/02699052.2018.1429660>

Stockbridge MD, Newman RS, Zukowski A, et al. Language profiles in children with concussion. *Brain Inj* [Internet]. 2020 Mar 20;34(4):567–74. Available from:

<https://www.tandfonline.com/doi/full/10.1080/02699052.2020.1725836>

Storey EP, Master SR, Lockyer JE, Podolak OE, Grady MF, Master CL. Near Point of Convergence after Concussion in Children. *Optometry and Vision Science*, (2017). 94(1),96-100.

<https://doi.org/10.1097/OPX.0000000000000910>

Storey EP, Wiebe DJ, D'Alonzo BA, Nixon-Cave K, Jackson-Coty J, ... Master CL. Vestibular rehabilitation is associated with visuovestibular improvement in pediatric concussion. *Journal of Neurologic Physical Therapy*, (2018). 42(3),134-141. <https://doi.org/10.1097/NPT.0000000000000228>

Stuart EA, Duerson DH, Rodenberg RE, Ravindran R, MacDonald JP. Return to Drive Counseling After Sports-Related Concussion. *Pediatric Quality and Safety*, (2016). 1(2),e006.

<https://doi.org/10.1097/pq9.000000000000006>

Studer M, Goeggel Simonetti B, Heinks T, Steinlin M, Leichtle A, ... Joeris A. Acute S100B in serum is associated with cognitive symptoms and memory performance 4 months after paediatric mild traumatic brain injury. *Brain Injury*, (2015). 29(13-14),1667-1673.

<https://doi.org/10.3109/02699052.2015.1075250>

Studer M, Goeggel Simonetti B, Joeris A, Margelisch K, Steinlin M, ... Heinks T. Post-concussive symptoms and neuropsychological performance in the post-acute period following pediatric mild traumatic brain injury. *Journal of the International Neuropsychological Society*, (2014). 20(10),982-993.

<https://doi.org/10.1017/S1355617714000927>

Stumph J, Young J, Singichetti B, Yi H, Valasek A, ... Fischer A. Effect of Exercise Recommendation on Adolescents With Concussion. *Journal of Child Neurology*, 2019.

<https://doi.org/10.1177/0883073819877790>

Sufrinko A, McAllister-Deitrick J, Elbin R, Collins MW, Kontos AP. Family History of Migraine Associated with Posttraumatic Migraine Symptoms Following Sport-Related Concussion. *Journal of Head Trauma Rehabilitation*, (2018). 33(1),7-14. <https://doi.org/10.1097/HTR.0000000000000315>

Sufrinko AM, Kontos AP, Apps JN, McCrear M, Hickey RW, ... Thomas DG. The Effectiveness of Prescribed Rest Depends on Initial Presentation After Concussion. *Journal of Pediatrics*, (2017). 185,167-172.

<https://doi.org/10.1016/j.jpeds.2017.02.072>

Sufrinko AM, Marchetti GF, Cohen PE, Elbin RJ, Re V, Kontos AP. Using Acute Performance on a Comprehensive Neurocognitive, Vestibular, and Ocular Motor Assessment Battery to Predict Recovery Duration after Sport-Related Concussions. *American Journal of Sports Medicine*, (2017). 45(5),1187-1194. <https://doi.org/10.1177/0363546516685061>

Sufrinko AM, Mucha A, Covassin T, Marchetti G, Elbin R, ... Director R. Sex Differences in Vestibular/Ocular and Neurocognitive outcomes following Sport-related Concussion HHS Public Access. *Clin J Sport Med*, (2017). 27(2),133-138. <https://doi.org/10.1097/JSM.0000000000000324>

Sufrinko AM, Mucha A, Covassin T, Marchetti G, Elbin RJ, ... Kontos AP. Sex differences in vestibular/ocular and neurocognitive outcomes after sport-related concussion. *Clinical Journal of Sport Medicine*, (2017). 27(2),133-138. <https://doi.org/10.1097/JSM.0000000000000324>

Sullivan L, Newton A, Shi J, et al. Concussion-Related Health Care Utilization Among Medicaid Insured Children in Ohio: Rural and Urban Differences. *J Rural Heal*. 2020;00(1):1–8.

Sussman WI, Mautner K, Mason RA, Bonecutter K, Shealy AK. Sphenopalatine ganglion block for management of refractory chronic posttraumatic headaches after a sport-related concussion. *Clinical Journal of Sport Medicine*, (2017). 27(2),e6-e8. <https://doi.org/10.1097/JSM.0000000000000325>

Sutton M, Chan V, Escobar M, et al. Neck Injury Comorbidity in Concussion-Related Emergency Department Visits: A Population-Based Study of Sex Differences Across the Life Span. *J Women's Heal [Internet]*. 2019 Apr 1;28(4):473–82. Available from:

<https://www.liebertpub.com/doi/10.1089/jwh.2018.7282>

Swanson MW, Weise KK, Dreer LE, Johnston J, Davis RD, ... Swanson E. Academic Difficulty and Vision Symptoms in Children with Concussion. *Optometry and Vision Science*, (2017). 94(1),60-67.

<https://doi.org/10.1097/OPX.0000000000000977>

Takagi M, Hearps SJC, Babl FE, Anderson N, Bressan S, ... Anderson V. Does a computerized neuropsychological test predict prolonged recovery in concussed children presenting to the ED? *Child Neuropsychology*, (2019). 00(00),1-15. <https://doi.org/10.1080/09297049.2019.1639653>

Tamura K, Furutani T, Oshiro R, et al. Concussion recovery timeline of high school athletes using a stepwise return-to-play protocol: Age and sex effects. *J Athl Train [Internet]*. 2020 Jan 1;55(1):6–10.

Available from: <https://meridian.allenpress.com/jat/article/55/1/6/433840/Concussion-Recovery-Timeline-of-High-School>

Tanveer S, Zecavati N, Delasobera EB, Oyegbile TO. Gender Differences in Concussion and Postinjury Cognitive Findings in an Older and Younger Pediatric Population. *Pediatric Neurology*, (2017). 70,44-49.

<https://doi.org/10.1016/j.pediatrneurol.2017.02.001>

Tator C, Starkes J, Dolansky G, et al. Rowan's Rugby Fatality Prompts Canada's First Concussion Legislation [Internet]. Vol. 46, Canadian Journal of Neurological Sciences. Cambridge University Press; 2019. p. 280–2. Available from:

https://www.cambridge.org/core/product/identifier/S0317167119000155/type/journal_article

Taubman B, Rosen F, McHugh J, Grady MF, Elci OU. The Timing of Cognitive and Physical Rest and Recovery in Concussion. *Journal of Child Neurology*, (2016). 31(14),1555-1560.

<https://doi.org/10.1177/0883073816664835>

Taylor KM, Kioumourtzoglou MA, Clover J, Coull BA, Dennerlein JT, ... Weisskopf MG. Concussion History and Cognitive Function in a Large Cohort of Adolescent Athletes. *American Journal of Sports Medicine*, (2018). 46(13),3262-3270. <https://doi.org/10.1177/0363546518798801>

Teel EF, Marshall SW, Shankar V, McCrea M, Guskiewicz KM. Predicting Recovery Patterns After Sport-Related Concussion. *Journal of Athletic Training*, (2017). 52(3),288-298. <https://doi.org/10.4085/1062-6050-52.1.12>

Teel EF, Zemek RL, Tang K, Gioia G, Vaughan C, ... Gagnon IJ. The stability of retrospective pre-injury symptom ratings following pediatric concussion. *Frontiers in Neurology*, (2019). 10(JUN).

<https://doi.org/10.3389/fneur.2019.00672>

Teh Z, Takagi M, Hearps SJC, et al. Acute cognitive postconcussive symptoms follow longer recovery trajectories than somatic postconcussive symptoms in young children. *Brain Inj*. 2020 Feb 23;34(3):350–6.

Terry DP, Reddi PJ, Cook NE, et al. Acute Effects of Concussion in Youth With Pre-existing Migraines. *Clin J Sport Med* [Internet]. 2019 Nov 26;Publish Ah. Available from:

<http://dx.doi.org/10.1097/JSM.0000000000000791>

Terwilliger VK, Pratson L, Vaughan CG, Gioia GA. Additional Post-Concussion Impact Exposure May Affect Recovery in Adolescent Athletes. *Journal of Neurotrauma*, (2016). 33(8),761-765.

<https://doi.org/10.1089/neu.2015.4082>

Tham SW, Fales J, Palermo TM. Subjective and Objective Assessment of Sleep in Adolescents with Mild Traumatic Brain Injury. *Journal of Neurotrauma*, (2015). 32(11),847-852.

<https://doi.org/10.1089/neu.2014.3559>

Theadom A, Starkey N, Jones K, Cropley M, Parmar P, ... Feigin VL. Sleep difficulties and their impact on recovery following mild traumatic brain injury in children. *Brain Injury*, (2016). 30(10),1243-1248.

<https://doi.org/10.1080/02699052.2016.1183171>

Thomas DG, Apps JN, Hoffmann RG, McCrea M, Hammeke T. Benefits of strict rest after acute concussion: A randomized controlled trial. *Pediatrics*, (2015). 135(2),213-223.

<https://doi.org/10.1542/peds.2014-0966>

Tjarks BJ, Dorman JC, Valentine VD, Munce TA, Thompson PA, ... Bergeron MF. Comparison and utility of King-Devick and ImPACT® composite scores in adolescent concussion patients. *Journal of the Neurological Sciences*, (2013). 334(1-2),148-153. <https://doi.org/10.1016/j.jns.2013.08.015>

Tomczyk CP, Mormile M, Wittenberg MS, Langdon JL, Hunt TN. An examination of adolescent athletes and nonathletes on baseline neuropsychological test scores. *Journal of Athletic Training*, (2018).

53(4),404-409. <https://doi.org/10.4085/1062-6050-84-17>

Truss K, Godfrey C, Takagi M, Babl FE, Bressan S, ... Anderson V. Trajectories and Risk Factors for Post-Traumatic Stress Symptoms following Pediatric Concussion. *Journal of Neurotrauma*, (2017). 34(14),2272-2279. <https://doi.org/10.1089/neu.2016.4842>

Truss K, Hearps SJC, Babl FE, et al. Trajectories and Risk Factors for Pediatric Postconcussive Symptom Recovery. *Neurosurgery* [Internet]. 2020 Dec 15;88(1):36–45. Available from: <https://academic.oup.com/neurosurgery/article/88/1/36/5876906>

Tsushima WT, Geling O, Arnold M, Oshiro R. Effects of Two Concussions on the Neuropsychological Functioning and Symptom Reporting of High School Athletes. *Applied Neuropsychology: Child*, (2016). 5(1),9-13. <https://doi.org/10.1080/21622965.2014.902762>

Tsushima WT, Shirakawa N, Geling O. Neurocognitive functioning and symptom reporting of high school athletes following a single concussion. *Applied Neuropsychology: Child*, (2013). 2(1),13-16. <https://doi.org/10.1080/09084282.2011.643967>

Tsushima WT, Siu AM, Pearce AM, Zhang G, Oshiro RS. Two-year Test-Retest Reliability of ImpACT in High School Athletes. *Archives of Clinical Neuropsychology*, (2016). 1. Shi F,(1),105-111. <https://doi.org/10.1093/arclin/acv066>

Tsushima WT, Tsushima VG, Murata NM. ImpACT Normative Data of Ethnically Diverse Adolescent Athletes. *Clinical Journal of Sport Medicine*, (2018). 38(2),1. <https://doi.org/10.1097/jsm.0000000000000567>

Tsushima WT, Tsushima VG, Oshiro RO, Murata NM. Role of Native Language in Immediate Post-Concussion Assessment and Cognitive Testing (ImpACT) of Youth Athletes. *Archives of Clinical Neuropsychology*, (2017). 32(4),450-455. <https://doi.org/10.1093/arclin/acx025>

Tylicka M, Matuszczak E, Dębek W, Hermanowicz A, Ostrowska H. Circulating proteasome activity following mild head injury in children. *Child's Nervous System*, (2014). 30(7),1191-1196. <https://doi.org/10.1007/s00381-014-2409-4>

Urban KJ, Barlow KM, Jimenez JJ, Goodyear BG, Dunn JF. Functional Near-Infrared Spectroscopy Reveals Reduced Interhemispheric Cortical Communication after Pediatric Concussion. *Journal of Neurotrauma*, (2015). 32(11),833-840. <https://doi.org/10.1089/neu.2014.3577>

Urban KJ, Riggs L, Wells GD, Keightley M, Chen J-K, ... Sinopoli KJ. Cortical Thickness Changes and Their Relationship to Dual-Task Performance following Mild Traumatic Brain Injury in Youth. *Journal of Neurotrauma*, (2017). 34(4),816-823. <https://doi.org/10.1089/neu.2016.4502>

Van Beek L, Ghesquière P, Lagae L, De Smedt B. Mathematical Difficulties and White Matter Abnormalities in Subacute Pediatric Mild Traumatic Brain Injury. *Journal of Neurotrauma*, (2015). 32(20),1567-1578. <https://doi.org/10.1089/neu.2014.3809>

Van Beek L, Vanderauwera J, Ghesquière P, Lagae L, De Smedt B. Longitudinal changes in mathematical abilities and white matter following paediatric mild traumatic brain injury. *Brain Injury*, (2015). 29(13-14),1701-1710. <https://doi.org/10.3109/02699052.2015.1075172>

van Markus-Doornbosch F, Peeters E, van der Pas S, Vlieland TV, Meesters J. Physical activity after mild traumatic brain injury: What are the relationships with fatigue and sleep quality? *European Journal of Paediatric Neurology*, (2019). 23(1),53-60. <https://doi.org/10.1016/j.ejpn.2018.11.002>

Vasiukova OR, Akhlebinina MI, Manzhurtsev A V., et al. The diffusion-tensor imaging reveals alterations in water diffusion parameters in acute pediatric concussion. *Acta Neurol Belg* [Internet]. 2020 Apr 3; Available from: <http://link.springer.com/10.1007/s13760-020-01347-w>

Vassilyadi M, Macartney G, Barrowman N, Anderson P, Dube K. Symptom experience and quality of life in children after sport-related head injuries: A cross-sectional study. *Pediatric Neurosurgery*, (2015). 50(4),196-203. <https://doi.org/10.1159/000431232>

Vaughan CG, Gerst EH, Sady MD, Newman JB, Gioia GA. The relation between testing environment and baseline performance in child and adolescent concussion assessment. *American Journal of Sports Medicine*, (2014). 1. Shi F,(7),1716-1723. <https://doi.org/10.1177/0363546514531732>

Vernau BT, Grady MF, Goodman A, Wiebe DJ, Basta L, ... Master CL. Oculomotor and neurocognitive assessment of youth ice hockey players: Baseline associations and observations after concussion. *Developmental Neuropsychology*, (2015). 1. Shi F,(1),7-11. <https://doi.org/10.1080/87565641.2014.971955>

Wang Y, Nelson LD, LaRoche AA, Pfaller AY, Nencka AS, ... McCrea MA. Cerebral Blood Flow Alterations in Acute Sport-Related Concussion. *Journal of Neurotrauma*, (2016). 33(13),1227-1236. <https://doi.org/10.1089/neu.2015.4072>

Wang Y, West JD, Bailey JN, Westfall DR, Xiao H, ... McDonald BC. Decreased cerebral blood flow in chronic pediatric mild TBI: An MRI perfusion study. *Developmental Neuropsychology*, (2015). 40(1),40-44. <https://doi.org/10.1080/87565641.2014.979927>

Ware AL, Shukla A, Goodrich-Hunsaker NJ, et al. Post-acute white matter microstructure predicts post-acute and chronic post-concussive symptom severity following mild traumatic brain injury in children. *NeuroImage Clin* [Internet]. 2020 Jan 1;25:102106. Available from: <https://linkinghub.elsevier.com/retrieve/pii/S221315821930453X>

Wasserman EB, Bazarian JJ, Mapstone M, Block R, Van Wijngaarden E. Academic dysfunction after a concussion among US high school and college students. *American Journal of Public Health*, (2016). 106(7),1247-1253. <https://doi.org/10.2105/AJPH.2016.303154>

Watson J, Brown C, Hirst RB, et al. Commonality in Abnormality: “abnormal” Neuropsychological Scores Are Common in Baseline Testing of Youth Athletes. *Arch Clin Neuropsychol* [Internet]. 2020 Jul 24;35(5):469–81. Available from: <https://academic.oup.com/acn/article/35/5/469/5809554>

Weise KK, Swanson MW, Penix K, Hale MH, Ferguson D. King-Devick and Pre-season Visual Function in Adolescent Athletes. *Optometry and Vision Science*, (2017). 94(1),89-95. <https://doi.org/10.1097/OPX.0000000000000938>

Wendling-Keim DS, Kičignig A, Dietz HG, Lehner M. Ambulatory or inpatient management of mild TBI in children: a post-concussion analysis. *Pediatric Surgery International*, (2017). 33(2),249-261. <https://doi.org/10.1007/s00383-016-4021-z>

Westfall DR, West JD, Bailey JN, Arnold TW, Kersey PA, ... McDonald BC. Increased brain activation during working memory processing after pediatric mild traumatic brain injury (mTBI). *Journal of Pediatric Rehabilitation Medicine*, (2015). 8(4),297-308. <https://doi.org/10.3233/PRM-150348>

Willer BS, Haider MN, Bezherano I, Wilber CG, Mannix R, ... Leddy JJ. Comparison of Rest to Aerobic Exercise and Placebo-like Treatment of Acute Sport-Related Concussion in Male and Female

Adolescents. Archives of Physical Medicine and Rehabilitation, 2019,1-9.

<https://doi.org/10.1016/j.apmr.2019.07.003>

Wilmoth K, Curcio N, Tarkenton T, et al. Utility of Brief Psychological Measures for Prediction of Prolonged Symptom Clearance in Concussed Student Athletes. Arch Clin Neuropsychol [Internet]. 2019 Nov 26; Available from: [https://academic.oup.com/acn/advance-](https://academic.oup.com/acn/advance-article/doi/10.1093/arclin/acz061/5640307)

[article/doi/10.1093/arclin/acz061/5640307](https://academic.oup.com/acn/advance-article/doi/10.1093/arclin/acz061/5640307)

Wiseman-Hakes C, Gosselin N, Sharma B, Langer L, Gagnon I. A Longitudinal Investigation of Sleep and Daytime Wakefulness in Children and Youth With Concussion. ASN Neuro, (2019). 11.

<https://doi.org/10.1177/1759091418822405>

Womble MN, Reynolds E, Schatz P, Shah KM, Kontos AP. Test-Retest reliability of computerized neurocognitive testing in youth ice hockey players. Archives of Clinical Neuropsychology, (2016). 1. Shi F,(4),305-312. <https://doi.org/10.1093/arclin/acw011>

Wu T, Merkley TL, Wilde EA, Barnes A, Li X, ... Levin HS. A preliminary report of cerebral white matter microstructural changes associated with adolescent sports concussion acutely and subacutely using diffusion tensor imaging. Brain Imaging and Behavior, 2017,1-12. <https://doi.org/10.1007/s11682-017-9752-5>

Yang MN, Clements-Nolle K, Parrish B, Yang W. Adolescent concussion and mental health outcomes: A population-based study. American Journal of Health Behavior, (2019). 43(2),258-265.

<https://doi.org/10.5993/AJHB.43.2.3>

Yeates KO, Tang K, Barrowman N, Freedman SB, Gravel J, ... Zemek R. Derivation and Initial Validation of Clinical Phenotypes of Children Presenting with Concussion Acutely in the Emergency Department: Latent Class Analysis of a Multi-Center, Prospective Cohort, Observational Study. Journal of Neurotrauma, (2019). 36(11),1758-1767. <https://doi.org/10.1089/neu.2018.6009>

Yengo-Kahn AM, Solomon G. Are psychotropic medications associated with differences in baseline neurocognitive assessment scores for young athletes? A pilot study. Physician and Sportsmedicine, (2015). 43(3),227-235. <https://doi.org/10.1080/00913847.2015.1071638>

Yue JK, Upadhyayula PS, Avalos LN, et al. Pediatric Traumatic Brain Injury in the United States: Rural-Urban Disparities and Considerations. Brain Sci [Internet]. 2020 Feb 28;10(3):135. Available from:

<https://www.mdpi.com/2076-3425/10/3/135>

Zalneraitis BH, Yengo-Kahn AM, Pawlukiewicz AJ, Solomon GS. Self-reported history of seizure and baseline neurocognitive test performance in student-athletes: an initial investigation. Physician and Sportsmedicine, (2017). 1. Shi F,(4),470-474. <https://doi.org/10.1080/00913847.2017.1372035>

Zamani A, Ryan NP, Wright DK, et al. The Impact of Traumatic Injury to the Immature Human Brain: A Scoping Review with Insights from Advanced Structural Neuroimaging [Internet]. Vol. 37, Journal of Neurotrauma. Mary Ann Liebert Inc.; 2020. p. 724–38. Available from:

<https://www.liebertpub.com/doi/10.1089/neu.2019.6895>

Zaremski JL, Herman DC, Clugston JR, Hurley RW, Ahn AH. Occipital neuralgia as a sequela of sports concussion: A case series and review of the literature. Current Sports Medicine Reports, (2015). 14(1),16-19. <https://doi.org/10.1249/JSR.0000000000000121>

Zemek R, Barrowman N, Freedman SB, Gravel J, Gagnon I, ... Moore J. Clinical risk score for persistent

postconcussion symptoms among children with acute concussion in the ED. JAMA – Journal of the American Medical Association, (2016). 315(10),1014-1025. <https://doi.org/10.1001/jama.2016.1203>

Zemek R, Barrowman N, Freedman SB, Gravel J, Gagnon I, ... Moore J. Clinical risk score for persistent postconcussion symptoms among children with acute concussion in the ED. Journal of the American Medical Association, (2016). 315(10),1014-1025. <https://doi.org/10.1001/jama.2016.1203>

Zewdie E, Ciechanski P, Kuo HC, et al. Safety and tolerability of transcranial magnetic and direct current stimulation in children: Prospective single center evidence from 3.5 million stimulations. Brain Stimul [Internet]. 2020 May 1;13(3):565–75. Available from: <https://linkinghub.elsevier.com/retrieve/pii/S1935861X19304966>

Zhou G, Brodsky JR. Objective vestibular testing of children with dizziness and balance complaints following sports-related concussions. Otolaryngology – Head and Neck Surgery (United States), (2015). 152(6),1133-1139. <https://doi.org/10.1177/0194599815576720>

Zonfrillo MR, Durbin DR, Koepsell TD, Wang J, Temkin NR, ... Rivara FP. Prevalence of and Risk Factors for Poor Functioning after Isolated Mild Traumatic Brain Injury in Children. Journal of Neurotrauma, (2014). 31(8),722-727. <https://doi.org/10.1089/neu.2013.3088>

Zou L, Li H, Jiang Z, et al. Modified decision-making rule supported by scheduled telephone follow-up reduces head computed tomography utilization in children with mild traumatic brain injury: A cohort study. Medicine (Baltimore). 2020 May 1;99(18):e20088.

Zuckerbraun NS, Atabaki S, Collins MW, Thomas D, Gioia GA. Use of Modified Acute Concussion Evaluation Tools in the Emergency Department. Pediatrics, (2014). 133(4),635-642. <https://doi.org/10.1542/peds.2013-2600>

Zuckerman SL, Apple RP, Odom MJ, Lee YM, Solomon GS, Sills AK. Effect of sex on symptoms and return to baseline in sport-related concussion. Journal of Neurosurgery: Pediatrics, (2014). 13(1),72-81. <https://doi.org/10.3171/2013.9.PEDS13257>

Zynda AJ, Sabatino MJ, Ellis HB, et al. Continued play following sport-related concussion in United States youth soccer. Int J Exerc Sci [Internet]. 2020;13(6):87–100. Available from: <http://www.intjexersci.com>



TOOL 1.1: Pediatric Concussion: The Role of School Boards, Community Sports Organizations, and Centres

School boards, community sports organizations, and centres must ensure updated policies are in place to recognize and accommodate a child/adolescent who has sustained a concussion.

- Publicize these policies widely among schools, staff, and sports organizations.
- Schools and their staff should be ready to promote the child/adolescent's safe and effective return-to-school.
- Immediate removal from play/activity is the first step in mitigating prolonged recovery.
- Allow the child/adolescent:
 - Enough time away from school and sport to begin cognitive recovery
 - Accommodations to support cognitive deficits such as additional time for homework and/or exams during recovery

Links to toolkits that are useful for developing school board concussion policies:

- [Ontario Ministry of Education School Board Policies for Concussion](#)
- [SCHOOLFirst Handbook: Enabling successful return to school for Canadian youth following a concussion \(Holland Bloorview Kids Rehabilitation Hospital\)](#)
- [Pan-Canadian School Concussion Protocol Template](#): Concussion Protocol Harmonization Project

Links to tools that are useful for informing school policies:

- [An Educator's Guide to Concussions in the Classroom, 2nd Edition](#): Nationwide Children's Hospital, Columbus, OH.
- [Concussion Awareness Training Tool: CATT Online](#)
- [SCHOOLFirst Handbook: Enabling successful return to school for Canadian youth following a concussion \(Holland Bloorview Kids Rehabilitation Hospital\)](#)
- [Parachute Concussion Series: Concussion Guide for Teachers](#)
- Advice for gradually resuming activities after a concussion: Institut national d'excellence en sante et en services sociaux (INESSS – Québec)
[English handout](#) / [French handout](#)
- [Canadian Guideline on Concussion in Sport Medical Assessment Letter](#)

Link to toolkits that are useful for developing sports organization policies:

- [A roadmap for implementing concussion management policies and protocols in sport \(Canadian Academy of Sport and Exercise Medicine\)](#)
- Advice for gradually resuming activities after a concussion: Institut national d'excellence en sante et en services sociaux (INESSS – Québec)
[English handout](#) / [French handout](#)
- [Parachute: Canadian Guideline for Concussion in Sport](#)

Links to tools that are useful for informing a policy for sports organizations:

- [Parachute Concussion Series – After a Concussion: Return-to-Sport strategy](#)
- [2016 Berlin Consensus in Sport Group Statement](#)
- [HEADS UP Concussion and Helmet Safety App \(Centers for Disease Control and Prevention\)](#)
- [Concussion Ed – Parachute Concussion Education](#)
- Advice for gradually resuming activities after a concussion: Institut national d'excellence en sante et en services sociaux (INESSS – Québec)
[English handout](#) / [French handout](#)
- [Canadian Guideline on Concussion in Sport Pre-Season Education Sheet \(Parachute\)](#)
- [Concussion recognition tool 5: To help identify concussion in children, adolescents and adults](#)
- [Canadian Guideline on Concussion in Sport Medical Assessment Letter](#)



TOOL 1.2: Concussion Recognition Tool 5

CONCUSSION RECOGNITION TOOL 5[©]

To help identify concussion in children, adolescents and adults



FIFA[®]

Supported by



RECOGNISE & REMOVE

Head impacts can be associated with serious and potentially fatal brain injuries. The Concussion Recognition Tool 5 (CRT5) is to be used for the identification of suspected concussion. It is not designed to diagnose concussion.

STEP 1: RED FLAGS – CALL AN AMBULANCE

If there is concern after an injury including whether ANY of the following signs are observed or complaints are reported then the player should be safely and immediately removed from play/game/activity. If no licensed healthcare professional is available, call an ambulance for urgent medical assessment:

- Neck pain or tenderness
- Double vision
- Weakness or tingling/ burning in arms or legs
- Severe or increasing headache
- Seizure or convulsion
- Loss of consciousness
- Deteriorating conscious state
- Vomiting
- Increasingly restless, agitated or combative

Remember:

- In all cases, the basic principles of first aid (danger, response, airway, breathing, circulation) should be followed.
- Assessment for a spinal cord injury is critical.
- Do not attempt to move the player (other than required for airway support) unless trained to do so.
- Do not remove a helmet or any other equipment unless trained to do so safely.

If there are no Red Flags, identification of possible concussion should proceed to the following steps:

STEP 2: OBSERVABLE SIGNS

Visual clues that suggest possible concussion include:

- Lying motionless on the playing surface
- Slow to get up after a direct or indirect hit to the head
- Disorientation or confusion, or an inability to respond appropriately to questions
- Blank or vacant look
- Balance, gait difficulties, motor incoordination, stumbling, slow laboured movements
- Facial injury after head trauma

© Concussion in Sport Group 2017

STEP 3: SYMPTOMS

- Headache
- "Pressure in head"
- Balance problems
- Nausea or vomiting
- Drowsiness
- Dizziness
- Blurred vision
- Sensitivity to light
- Sensitivity to noise
- Fatigue or low energy
- "Don't feel right"
- More emotional
- More Irritable
- Sadness
- Nervous or anxious
- Neck Pain
- Difficulty concentrating
- Difficulty remembering
- Feeling slowed down
- Feeling like "in a fog"

STEP 4: MEMORY ASSESSMENT

(IN ATHLETES OLDER THAN 12 YEARS)

Failure to answer any of these questions (modified appropriately for each sport) correctly may suggest a concussion:

- "What venue are we at today?"
- "Which half is it now?"
- "Who scored last in this game?"
- "What team did you play last week/game?"
- "Did your team win the last game?"

Athletes with suspected concussion should:

- Not be left alone initially (at least for the first 1-2 hours).
- Not drink alcohol.
- Not use recreational/ prescription drugs.
- Not be sent home by themselves. They need to be with a responsible adult.
- Not drive a motor vehicle until cleared to do so by a healthcare professional.

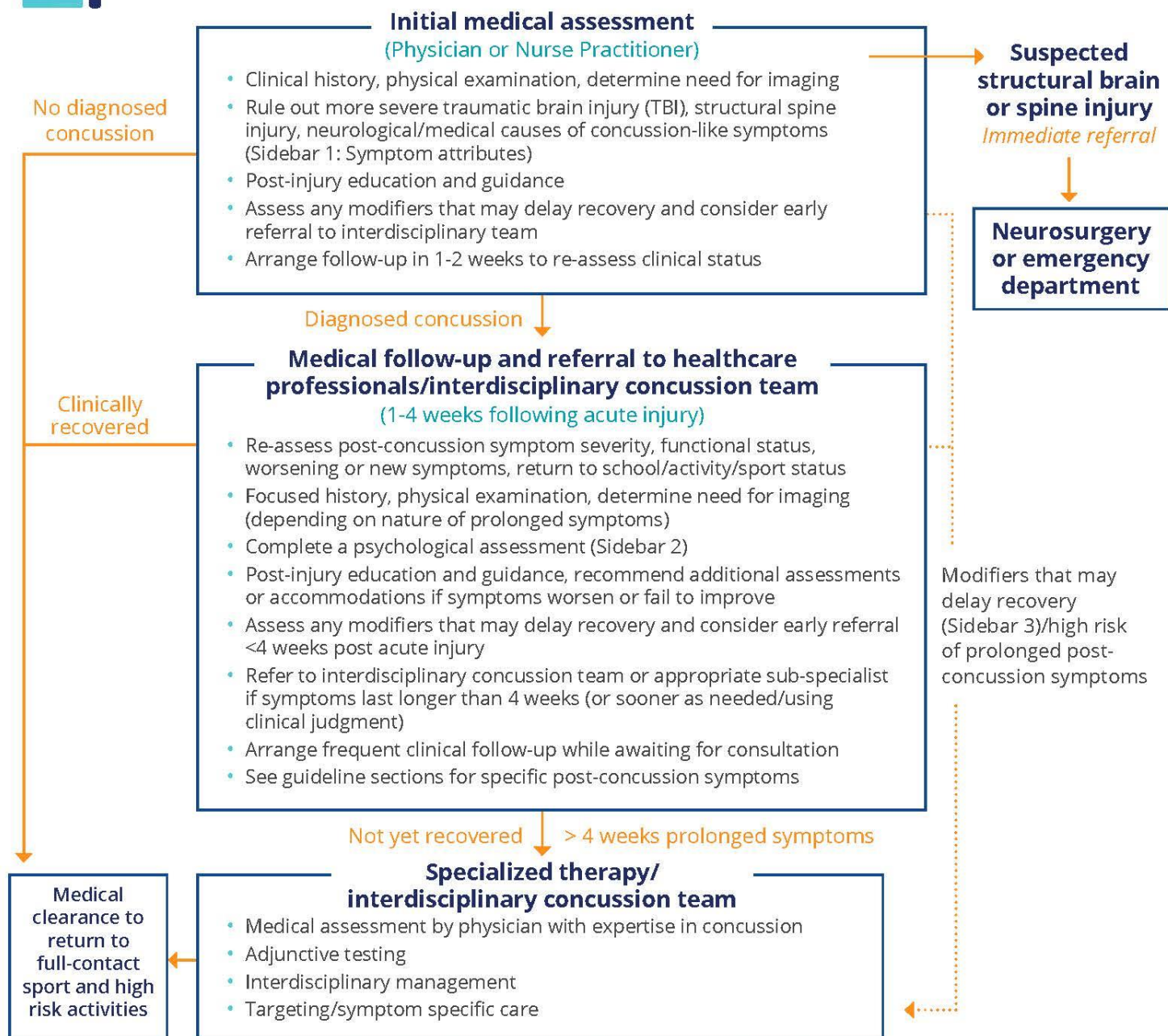
The CRT5 may be freely copied in its current form for distribution to individuals, teams, groups and organisations. Any revision and any reproduction in a digital form requires approval by the Concussion in Sport Group. It should not be altered in any way, rebranded or sold for commercial gain.

ANY ATHLETE WITH A SUSPECTED CONCUSSION SHOULD BE IMMEDIATELY REMOVED FROM PRACTICE OR PLAY AND SHOULD NOT RETURN TO ACTIVITY UNTIL ASSESSED MEDICALLY, EVEN IF THE SYMPTOMS RESOLVE

© Concussion in Sport Group 2017



TOOL 1.3: Manage Acute and Prolonged Concussion Symptoms Algorithm



Links to Recommendations:

- 1: Recognition and Directing to Care
- 2: Initial Medical Assessment and Management
- 3: Medical Follow-up
- 4: Medical Clearance: Full-contact sport or high-risk activity
- 5: Sport Concussion Considerations
- 6: Headache
- 7: Sleep
- 8: Mental Health
- 9: Cognition
- 10: Vision/vestibular/oculomotor
- 11: Fatigue
- 12: Return-to-School and Work

SIDEBAR 1

> Symptom Attributes

- Duration, onset, triggers, location
- Intensity and impact
- Perception of symptoms
- Impact on functioning
- Previous episodes
- Previous treatment and response

SIDEBAR 2

> Psychosocial Evaluation

- Support system
- Mental health history
- Comorbid conditions (chronic pain, mood/stress/personality disorder)
- Substance use disorder
- Absence from school or academic difficulties

SIDEBAR 3:

Modifiers That May Delay Recovery:

- Age (increases with age)
- Sex (female)
- Personal and family history of migraines
- History of learning or behavioural difficulties
- Personal and family history of mental health
- Family socioeconomic status/education
- High pre-injury symptom presentation

Adapted with permission from the [Ontario Neurotrauma Foundation Standards for Post-Concussion Care - Post Concussion Care Pathway](#)



TOOL 2.1: Physical Examination

All patients presenting with a suspected acute concussion in the emergency department or office setting should undergo a complete physical examination that includes a neurological examination and cervical spine examination, as well as examination for any other traumatic injury. Depending on the patient's presenting or prolonged symptoms, supplementary physical examination tests can be considered. This also depends on the healthcare professional's experience in performing and interpreting these objective tests.

Core physical examination: neurological and cervical spine examination

Neurological examination

- Glasgow Coma Scale (GCS) scoring
- Vital signs (resting heart rate, blood pressure)
- Screen for signs of orbital trauma or calvarial or basilar skull fracture signs (raccoon eyes, battle sign, hemotympanum)
- Screening for other trauma
- Cranial nerve examination (fundoscopy, pupillary function, visual acuity, visual fields, extra-ocular movements, facial sensation and motor function, hearing, palate symmetry, trapezius strength, tongue movements)
- Pronator drift, motor function, and tone/bulk
- Sensory function
- Reflexes
- Cerebellar testing (finger-to-nose, rapid alternating movements)
- Balance (tandem stance, one leg stance, Romberg)
- Gait (tandem gait)
- Cognitive screen (concentration, immediate, and delayed recall components of the SCAT5 and Child SCAT5 can be considered)

Cervical spine examination

- Palpation of cervical spine and posterior and anterior paraspinal musculature (tenderness or reactive muscle guarding)
- Range of motion

Supplemental tests

Healthcare professional experience and comfort with these tests may vary. If a child/adolescent develops prolonged post-concussion symptoms, consider referring to an interdisciplinary concussion team.

For patients with prolonged dizziness, vertigo, postural imbalance or motion sensitivity

- Dix-Hallpike test (to assess for benign paroxysmal positional vertigo)
- Head thrust test (to assess vestibulo-ocular reflex)
- Balance Error Scoring System (for balance)

For patients with prolonged blurred vision, difficulty focusing, headaches with visual stimulation, and motion sensitivity

- Cover/Uncover test
- Convergence
- Accommodation
- Horizontal and vertical saccades
- Smooth pursuits
- Head thrust testing

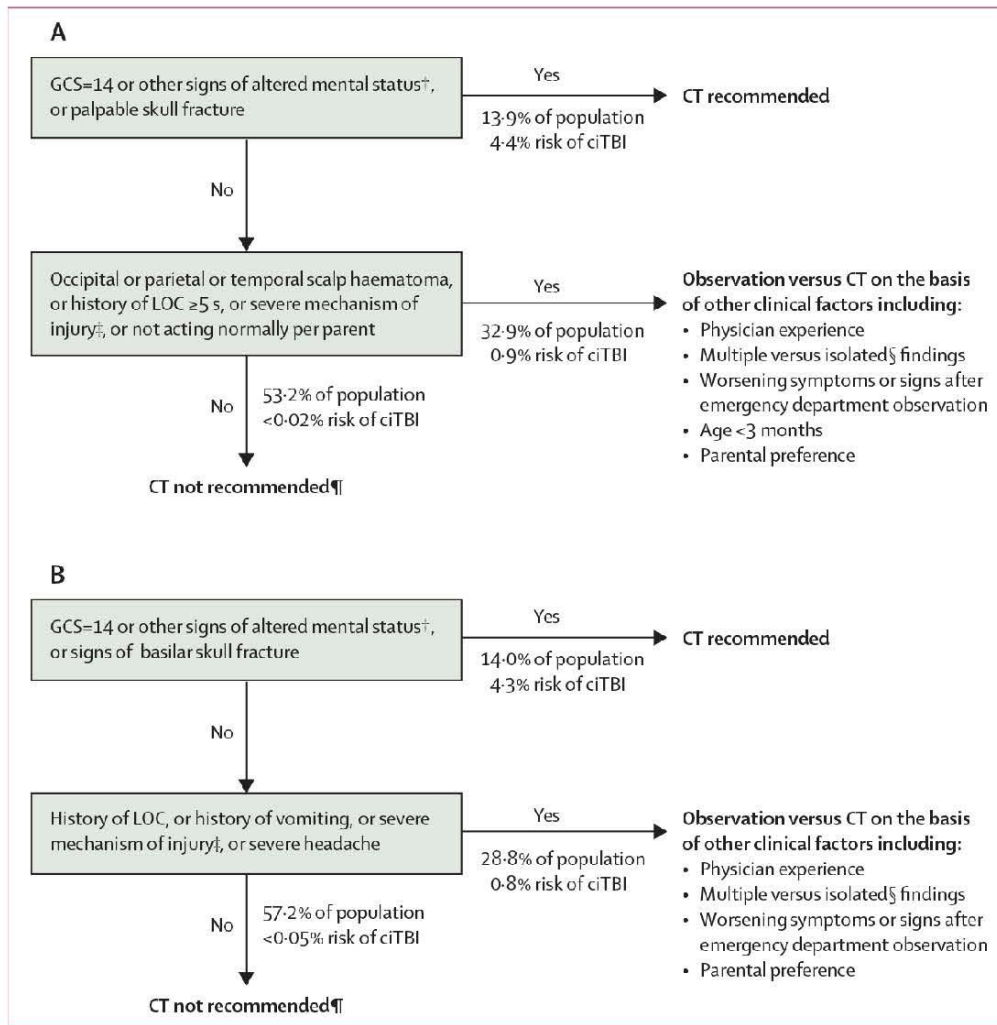
The healthcare professional should also note if performing these tests significantly recreates or exacerbates the patient's symptoms.

For patients with jaw/temporo-mandibular joint pain or discomfort

- Inspection of jaw alignment and occlusion
- Observation of active movements
- Palpation of the jaw and the temporo-mandibular joints
- Otoscopic examination



TOOL 2.2: PECARN Management Algorithm for Children after Head Trauma



Source: Kuppermann N, Holmes JF, Dayan PS, Hoyle JD Jr, Atabaki SM, Holubkov R, Nadel FM, Monroe D, Stanley RM, Borgialli DA, Badawy MK, Schunk JE, Quayle KS, Mahajan P, Lichenstein R, Lillis KA, Tunik MG, Jacobs ES, Callahan JM, Gorelick MH, Glass TF, Lee LK, Bachman MC, Cooper A, Powell EC, Gerardi MJ, Melville KA, Muizelaar JP, Wisner DH, Zuspan SJ, Dean JM, Wootton-Gorges SL; Pediatric Emergency Care Applied Research Network (PECARN). Identification of children at very low risk of clinically-important brain injuries after head trauma: a prospective cohort study. *Lancet*. 2009 Oct 3;374(9696):1160-70. Epub 2009 Sep 14. PubMed PMID: 19758692. Reproduced with permission from Elsevier.



TOOL 2.3: The Canadian Assessment of Tomography for Childhood Head injury 2 (CATCH2) rule

CT of the head is required for children with minor head injury* and any 1 of these findings:

- GCS score < 15 at 2 hours after injury
- Suspected open or depressed skull fracture
- History of worsening headache
- Irritability on examination
- Any sign of basal skull fracture†
- Large, boggy hematoma of the scalp
- Dangerous mechanism of injury‡
- ≥ 4 episodes of vomiting

Note: CT = computed tomography, GCS = Glasgow Coma Scale.

* Minor head injury is defined as injury within the past 24 hours associated with witnessed loss of consciousness, definite amnesia, witnessed disorientation, persistent vomiting (> 1 episode) or persistent irritability (in a child aged < 2 yr) in a patient with a GCS score of 13–15.

† Signs of basal skull fracture include hemotympanum, raccoon eyes, otorrhea or rhinorrhea of the cerebrospinal fluid, and Battle sign.

‡ Dangerous mechanism is a motor vehicle crash, a fall from elevation ≥ 3 ft (≥ 91 cm) or 5 stairs, or a fall from a bicycle with no helmet.

Reprinted from Osmond MH, Klassen TP, Wells GA, et al. Validation and refinement of a clinical decision rule for the use of computed tomography in children with minor head injury in the emergency department. CMAJ. 2018;190(27):E816–E822. <http://www.cmaj.ca/content/190/27/E816>. © Canadian Medical Association (2018). This work is protected by copyright and the making of this copy was with the permission of the Canadian Medical Association Journal (www.cmaj.ca) and Access Copyright. Any alteration of its content or further copying in any form whatsoever is strictly prohibited unless otherwise permitted by law.



TOOL 2.4: Algorithm for the Management of the Pediatric Patient ≥ 2 years with Minor Head Trauma

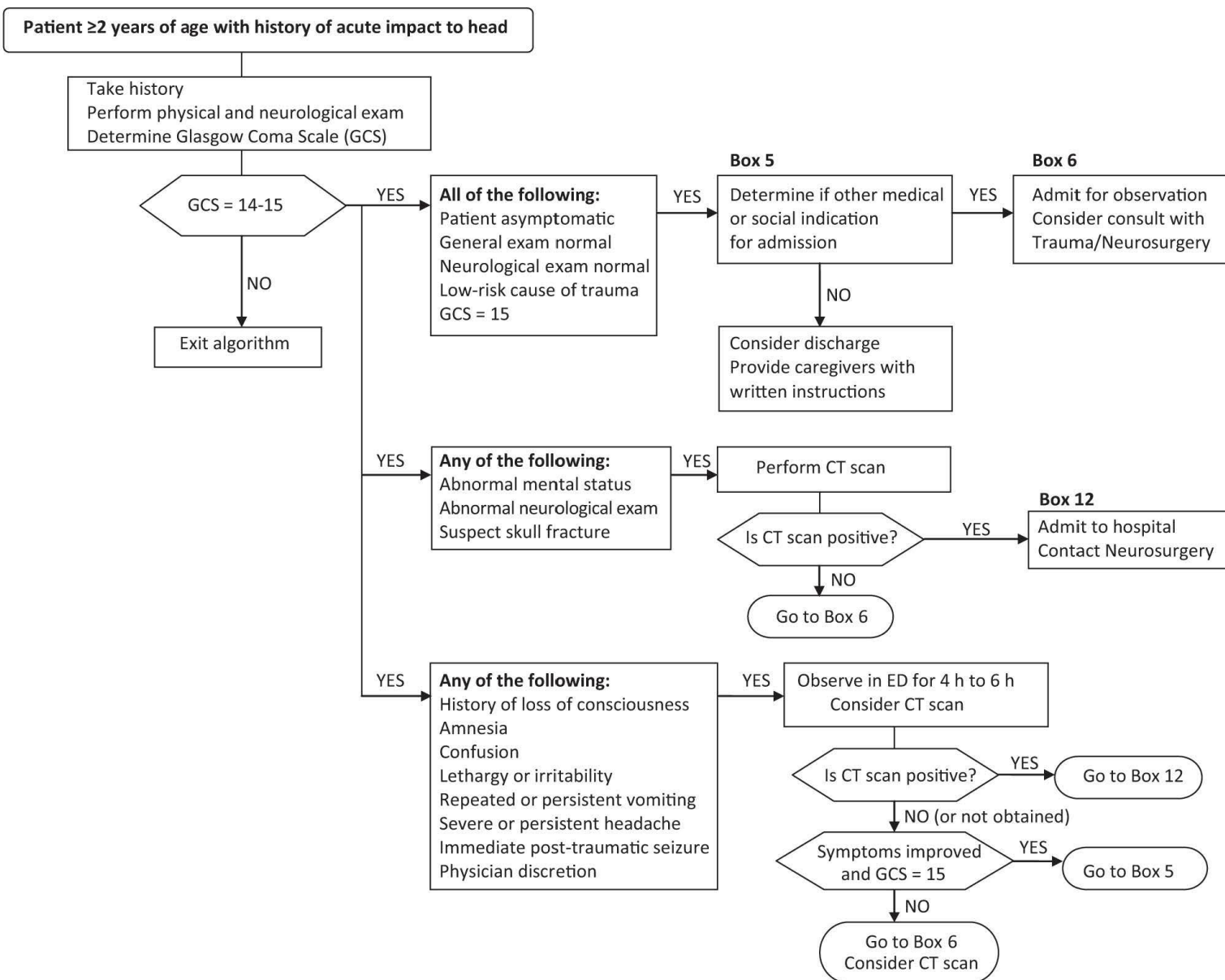


Figure 1. Algorithm for the management of the paediatric patient ≥ 2 years of age with minor head trauma.

CT Computed tomography; ED Emergency department

Source: Catherine A Farrell; Canadian Paediatric Society, Acute Care Committee. Management of the paediatric patient with acute head trauma. Paediatr Child Health 2013;18(5):253-8.



TOOL 2.5: “Four P’s” – Prioritize, Plan, Pace and Position

1

PRIORITIZE

What is important?

Decide: What is important? What are you able to do based on how you currently feel?

Break daily activities into:

- Urgent: must be done today.
- Important: must be done in next few days.
- For later: must be done this week or month.
- Don't need to/can't do: can it wait until you are feeling better?
- Help needed: can someone do it for you?

3

PACE

How are you going to do it?

- Take breaks if you don't feel well.
- After a concussion, your brain is less able to do tasks that were once automatic (e.g. tasks you have a lot of practice doing). It may take more time to complete tasks as you recover.
- Complete the task over stages (you may not be able to finish everything all at once).
- Plan tasks throughout the day so you aren't doing too much at one time.

2

PLAN

What are you going to do?

- Organize what you need to do.
- Do tasks that use more energy at times in the day when you feel best.
- Make rest breaks part of your routine.
- Planning saves mental energy and helps you avoid trying to remember what to do and when to do it during the day.

4

POSITION

Where are you going to do it?

- Noisy, busy and distracting environments make it hard to concentrate and will use up more energy.
- Think about your environment and how you stand/sit in it.
- Your environment can make a difference in how you feel.

Source: [Holland Bloorview Kids Rehabilitation Hospital. Concussion & You: A Handbook for Parents and Kids. \(Page 9\)](#)



TOOL 2.6: Post-Concussion Information Sheet

What is a concussion?

A concussion is a brain injury that affects how the brain works. A concussion can't be seen on x-rays or brain scans.

What causes a concussion?

Any blow or hit to the head, face, neck, or body that causes sudden shaking of the head can cause a concussion. Concussions can happen from falling, during sports like hockey, soccer, or volleyball, or during motor vehicle collisions.

When should a concussion be suspected?

A concussion should be suspected in any child/adolescent who takes a blow to the head, face, neck, or body and shows ANY of the signs or symptoms of a concussion.

What are the symptoms of a concussion?

Symptoms of concussion describe how someone feels after they are injured. A child/adolescent does not have to be knocked out (black out or unconscious) to have a concussion. Some symptoms may not appear until the next day.

Common symptoms of a concussion are:

- Headaches or head pressure
- Dizziness
- Nausea and vomiting
- Blurred or fuzzy vision
- Sensitivity to light or sound
- Balance problems
- Feeling slow, tired or having no energy
- Not thinking clearly
- Easily upset or angered
- Sadness
- Nervous or anxious
- Feeling more emotional
- Sleeping more or less
- Having a hard time falling asleep
- Difficulty working on a computer
- Difficulty reading or remembering

What are the visible (can be seen) signs of a concussion?

Signs of concussion describe how a child/adolescent looks or acts when they are injured. Common signs of a concussion are:

- Lying still on the ground or ice
- Slow to get up
- Confusion or can't answer questions
- Emotional lability
- Blank stare
- Difficulty standing or walking
- Injury to the face or holding their head

What should a child/adolescent do if they think they or a friend has a concussion?

All children/adolescents who may have a concussion should stop the activity they are doing right away and see a medical doctor or nurse practitioner. All children/adolescents with a concussion should get permission from a doctor or nurse practitioner before returning to full contact sport or high-risk activities.

When should a child/adolescent with a concussion go back to see a doctor or nurse?

Children/adolescents with a concussion should see the doctor or nurse if they have any of the following signs or symptoms:

- Increased confusion
- Worsening headache
- Vomiting more than once
- Seizures
- Not waking up
- Trouble walking
- Difficulty talking
- Strange behaviour

What can a child/adolescent do to help recover from a concussion?

It is important that all children/adolescents with a concussion give their brain time to heal. Here are some tips:

- Return to activities that do not worsen symptoms
- Take a break if symptoms get worse
- Get regular amounts of sleep
- Avoid activities that may involve contact or falling
- Avoid activities that make symptoms worse
- Drink water and eat balanced meals
- Spend time with friends and family
- Avoid drugs and alcohol
- Go to all medical appointments
- If school is more difficult or if they feel sadder or more nervous than normal, they should tell someone they are comfortable with. This might be a trusted friend, family member, teacher, guidance counsellor, doctor, nurse, or Elder.

When can a child/adolescent return-to-school, activities, and sports after a concussion?

Begin by resting for the first 1-2 days. After this short rest, the child/adolescent should start activities (physical and thinking) that do not make symptoms worse or bring on new symptoms. These gentle activities are encouraged 1-2 days after a concussion, even if the child/adolescent still has symptoms. Avoid any activity where the child/adolescent could fall or hit their head.

It is important that all children/adolescents with a concussion make a gradual (step-by-step) return to school/activity/sport. Use the steps below to guide this gradual process. Each step should take about one day. If symptoms get worse, go back to the last step. Try it again until the child/adolescent can do it without bringing on new symptoms or making symptoms worse. It is important to get a doctor's note before returning to full contact sport or high-risk activities.

Return-to-activity/sport steps:

Steps	Activity	Example
1	Physical and cognitive activities that do not make the child/adolescent feel worse	Walking at home or in school
2	Light physical activity	Jogging or stationary cycling at slow to medium speed. No weight training.
3	Sport-specific exercise	Running or skating drills. No drills with risk of head injury.
4	Non-contact activities	Practice without body contact. Gym class activities without risk of head injury.
5	Full-contact activities	Full activities/sports practices after doing full-time school and getting a doctor's note
6	Return to all activities and sports	Normal full-contact game play

Return-to-school steps:

Steps	Activity	Example
1	Activities at home that do not make the child/adolescent feel worse	Reading, texting, screen time and other activities that do not worsen symptoms. Start at 5-15 minutes at a time.
2	School activities	Homework, reading or other activities outside of the classroom
3	Return-to-school part-time	Getting back to school for a few hours or half days
4	Return-to-school full-time	Gradual return to full days at school

How long does it take a child/adolescent to recover after a concussion?

Most children/adolescents with a concussion will recover in 1-4 weeks, but it can take longer for some people.

What can happen if a child/adolescent goes back to sports too early after a concussion?

If children/adolescents with a concussion go back to sports too early after a concussion, they can have another injury. Having another injury can lead to worse symptoms that last longer, or result in a more severe or fatal brain injury.

How can children/adolescents help prevent concussions and their consequences?

To prevent concussions, children/adolescents should follow the rules of their sport, respect their opponents, and avoid head contact. Youth should always wear helmets during outdoor activities such as skating or riding bikes, snowmobiles, or all-terrain vehicles. It is important for children/adolescents to tell parents, teachers, or coaches if they think anyone may have a concussion.

Content from this tool was adapted with permission from the [Parachute Canadian Guideline on Concussion in Sport](#)



TOOL 2.7: Strategies to Promote Good Sleep and Alertness

Healthy habits to promote good sleep and alertness

Make sure the child/adolescent:

- Maintains the same sleep and wake times during the week and on weekends throughout the year, including in the summer, to the best of their ability. If this is challenging to maintain on weekends and holidays, try to keep the child/adolescent within one hour of weekday bed and wake times.
- Has a fixed bedtime routine. A warm bath about one hour before bed may help to facilitate sleep through relaxation. (Note: taking a warm bath too close to bedtime may raise body temperature, which can delay sleep.) If taking a warm bath or shower close to sleep time is unavoidable, it can be followed by a cool (not cold) rinse or shower to prevent the body temperature from rising.
- Turns off the computer and electronic devices including cell phones, at least 30 minutes before bedtime. This helps to ensure that the light emitted from these devices does not affect the ability to fall asleep or maintain sleep, and, that the cognitive or thinking activities involved with screens don't affect the ability to quiet the mind in preparation for sleep.
- Gets increased sleep and naps in the first few hours/days after a concussion. During this acute period, the child/adolescent should be allowed to sleep as much as needed throughout the day and night. Consult a physician or emergency department if the child/adolescent is not easily awoken in the first few hours or days after a concussion. After this acute period, those who have night-time sleep issues should avoid naps (unless it is part of the younger child's typical routine) to promote night-time sleep and gradual return-to-activity.
- Limits naps to once a day. If the child/adolescent is sleepy during the day and cannot avoid napping, then a nap should take place before 3 pm and for no longer than 30 minutes.
- Naps in bed, and not in another room or in front on the TV.

Nutrition, exercise, and lifestyle

Make sure the child/adolescent:

- Avoids caffeine (coffee, tea, chocolate, some over-the-counter medications) within 4-6 hours of bedtime.

- Avoids energy drinks and alcohol altogether.
- Avoids eating heavy meals late in the evening.
- Avoids sugar 4 hours before bedtime. Try a bedtime snack containing proteins.
- Has a balanced diet. Foods such as Goji berries, walnuts, almonds, pineapple, bananas and oranges all contain substantial amounts of melatonin, a hormone produced in the brain that promotes sleep.
- Has an adequate amount of magnesium and zinc in the diet or by supplement. Zinc is known to regulate sleep and magnesium is also directly involved in sleep and relaxation. Magnesium and Zinc-rich foods include:
 - Dark leafy greens
 - Seeds and nuts, including sunflower and sesame seeds, cashews and almonds
 - Squash, broccoli, and other vegetables (magnesium)
 - Potatoes (zinc)
 - Legumes
 - Dairy products
 - Meat
 - Unprocessed whole grains.
- Does 30-60 minutes of vigorous exercise a day, when tolerated and medically indicated, and at least 2 hours before bedtime. Exercise during the 2 hours before bedtime can delay sleep while regular exercise earlier in the day can promote sleep.
- Gets some natural light during the day, especially in the morning.
- Gets 15-30 minutes of quiet time after periods of cognitive activity, if he/she has significant cognitive fatigue (not sleepiness) during the day. Ideally, quiet time should be in an environment with natural light and no electronic devices. This can also promote night-time sleep.
- Avoids loud music with a strong beat before bedtime. If the child/adolescent is used to listening to music before bed, make sure that it is music that promotes relaxation.

Sleeping environment

Make sure the child/adolescent:

- Has a dark, cool and comfortable sleeping area.
- Removes all sources of light in the bedroom while sleeping.
- Opens the curtains and has natural light immediately upon awakening.
- Keeps the bedroom clean, tidy and quiet. Neutral or natural sounds can help to block out distracting sounds.
- Reserves the bed and bedroom for sleep, and does other activities (reading, watching TV, using the internet, playing games) in another room. Ideally, there should be no electronic equipment in the bedroom. If this is unavoidable, make sure that all computers, tablets, cell phones, etc. are turned off or are in —“sleep” mode.
- Turns any digital clocks with numbers that light up away from the bed during sleep.

Adapted with permission from the authors: C. Wiseman-Hakes (U of Toronto, Canada), M-C. Ouellet (U Laval) & S. Beaulieu-Bonneau (U Laval).



TOOL 6.1: Post-Concussion Headache Algorithm

Medical follow-up and referral to healthcare professionals/interdisciplinary concussion team (1-4 weeks following acute injury)

- Focused headache history, physical examination, determine need for imaging
- Post-injury education and guidance on use of over the counter analgesics
- Headache and medication diary
- Refer to healthcare professionals/interdisciplinary concussion clinic if symptoms last longer than 4 weeks post-concussion (or sooner as needed/using clinical judgment), or if the child has modifiers that may delay recovery
- Consider trial of conservative management or prophylactic or abortive headache medication

Consider early referral (< 4 weeks) if child/adolescent has modifiers that may delay recovery/high risk of prolonged post-concussion symptoms

Not yet recovered Headache > 4 weeks post-concussion

Healthcare professionals/ interdisciplinary concussion team

- Medical assessment by physician with expertise in concussion
- Adjunctive testing (graded aerobic exercise testing)
- Interdisciplinary management of post-concussion headache subtypes

Migraine, tension, cluster headache, other

Neurologist

Headaches with prolonged visual stimulation

Neurologist,
Physiotherapist,
Neurooptometrist

Physiological or exercise-induced headache

Physiotherapist,
Athletic Therapist

Cervicogenic headaches

Neurologist,
Physiotherapist,
Chiropractor



TOOL 6.2: General Considerations Regarding Pharmacotherapy

- Address significant psycho-social stressors before starting treatment ([Heads Up Checkup: Mental Health and Behavioral Risk Screening System](#)).
- Review current medications, including over-the-counter medicines and supplements, before starting treatment. If possible, minimize or withdraw agents that may exacerbate or maintain symptoms. Screen for overmedication headaches.
- Ideally a psychiatrist would be involved in the prescribing of psychotropic medication but an experienced family doctor or paediatrician, may initiate and monitor treatment. In severe cases, a psychiatrist should be consulted.
- Change only one medication at a time.
- Target drug therapy to specific symptoms (example: dysphoria, anxiety, mood swings, irritability, fatigue, sleep, headache and pain), and monitor during the course of treatment.
- Consider waiting until 12 weeks before starting a mood-altering medication unless the clinical scenario dictates otherwise.
- Choose therapies that minimize the impact of adverse effects on awakening, cognition, sleep and motor coordination, as well as on seizure threshold-domains in which children/adolescents with concussion may already be compromised.
- Start at the lowest effective dose and titrate slowly upwards, monitoring tolerability and clinical response, and also aiming for adequate dose and duration. Treatment often fails because either are insufficient. At times, you may have to prescribe the maximum tolerated doses.
- Aim to use a single agent to alleviate several symptoms. However, as individual symptoms may not show a coupled response to treatment, you may have to try a combination of strategies.
- Offer limited quantities of medications to those at a higher risk of suicide.
- Continue successful pharmacotherapy for at least 6 months, preferably 9 to 12 months for SSRIs, before tapering off on a trial basis.
- Use a specific SSRI as first-line treatment for mood and anxiety syndromes. Avoid using benzodiazepines as first-line therapy for anxiety. Avoid opiates.
- Follow-up regularly.

Adapted from Silver JM, Arciniegas DB, Yudofsky SC. Psychopharmacology. In: Silver JM, Arciniegas DB, Yudofsky SC, eds. Adapted with permission from the Textbook of Traumatic Brain Injury, (Copyright ©2005). American Psychiatric Association. All Rights Reserved.



TOOL 6.3: Approved Medications for Pediatric Indications

Medications may be used as indicated or as off-label use for symptoms related to features of concussion (i.e., migraine headache, muscular discomfort, sleep, cognitive, and mood disorders) or for patients with an acute or prolonged concussion who also have co-existing medical diagnoses that may have been prescribed medications for other indications.

Use clinical judgment and discretion at all times when prescribing medication.

Drug	Health Canada Approval	FDA Approval	Dosage#
Headache or Muscular Strain			
> Acetaminophen	Treatment of mild/moderate pain and fever. All ages.	All ages for mild to moderate pain and fever	10-15 mg/kg/dose orally/rectal every 4h as needed (maximum 75 mg/kg/day or 4,000 mg/day)
> Ibuprofen	Pediatric patients for mild to moderate pain. Fever in pediatric patients.	Mild to moderate pain in patient's ≥ 6 months old. Reduction in fever in patient's ≥ 6 months old. Juvenile arthritis in pediatric patients.	5-10 mg/kg/dose orally every 6-8h as needed (max 600 mg/dose or 40 mg/kg/day)
> Naproxen	Children ≥ 2 years of age: 1) osteoarthritis, ankylosing spondylitis, juvenile rheumatoid arthritis 2) aches/pains and mild to moderate pain due to sprains/strains 3) primary dysmenorrhea	> 2 years of age for analgesia, inflammatory disease	5 mg/kg/dose orally twice daily. Max 500 mg/dose, 1,000 mg/day (usual adult dose: 250-500 mg)
Migraine Headache			
> Amitriptyline	None	None	Chronic pain: 0.1 mg/kg, increase as needed to 0.5-2 mg/kg (off label dosing)
> Diclofenac powder for oral suspension (Cambia®)	No indication <18 years of age. Acute treatment of migraine attacks with or without aura in adults 18 years of age or older.	No indication <18 years of age. Acute treatment of migraine attacks with or without aura in adults 18 years of age or older.	Adult: 50 mg q24 hrs prn (max 15 doses/ month). *Note: the safety of taking a second dose has not been studied, as per the manufacturer.
> Triptan (example: rizatriptan, sumatriptan)	Almotriptan: acute treatment of migraine headache with or without aura in patients 12-17 years of age. Naratriptan: None Rizatriptan: None Sumatriptan: None Zolmitriptan: None	Almotriptan: acute treatment of migraine headache with or without aura in patients 12-17 years of age. Naratriptan: None Rizatriptan: > 6 years of age for acute treatment of migraine with or without aura Sumatriptan: None Zolmitriptan: None	Almotriptan: 6.25-12.5 mg, may repeat dose after 2 hours if needed (max 25 mg/day) Rizatriptan: ≤ 40 kg: 5 mg/24 hrs, ≥ 40 kg: 10 mg/24 hrs

Tool 6.3: Approved Medications for Pediatric Indications (cont'd)

Drug	Health Canada Approval	FDA Approval	Dosage#
> Beta-blockers- Propranolol	migraine prophylaxis	Adult-approved indication for migraine	0.5-4 mg/kg/day
> Topiramate		Adult- approved prevention of migraine headaches	2-16 years: initial dose: 1-3 mg/kg/day orally, increase every 1-2 weeks by 1-3 mg/kg/day divided twice per day. Maintenance dose 5-9 mg/kg/day divided twice per day. ≥ 17 yrs. 50 mg daily. Increase each week by 50 mg/day. Max dose 600 mg/day (Off label dosing)
Sleep Disorders			
> Trazodone	None	None	
> Zopiclone	None	None	
> Magnesium Oxide	Classified as a natural health product. For hypomagnesemia/dietary supplement. No age restrictions.	Magnesium supplement. No age restriction.	20-40 mg/kg/day
> Melatonin	Licensed Natural Health product –no Health Canada monograph	None	0.5-3 mg every night at bedtime (Max. 12 mg)
> Zinc	Licensed Natural Health Product – no Health Canada monograph	Treatment and prevention of zinc deficiency states.	Recommended dietary allowance: infants 5 mg/day, 1-10 years: 10 mg/day, > 11 years: 12-15 mg/day
> Tryptophan	None	None	
Mood Disorders			
> Fluoxetine	none	Major depressive disorder (≥ 8 years) Obsessive compulsive disorder (≥ 7 years)	Major depressive disorder: 10-20 mg/day Obsessive compulsive disorder: 10-30 mg/day. Up to 60 mg/day in higher weight children/ adolescents
> Sertraline	None	Obsessive-compulsive disorder (≥ 6 years)	6-12 years: 25 mg daily, increase as needed to a max of 200 mg/day 13-17 years: 50 mg/day, increase as needed to 200 mg
> Fluvoxamine	None	Obsessive-compulsive disorder (≥ 8 years)	8-17 years: 25 mg daily, increase as needed to a max of 200 mg in 8-11 years and 300 mg/day in adolescents
> Paroxetine	None	None	

Tool 6.3: Approved Medications for Pediatric Indications (cont'd)

Drug	Health Canada Approval	FDA Approval	Dosage#
> Citalopram	None	None	
> Escitalopram	None	Major depressive disorder ≥ 12 years	10 mg daily, increase as needed to a max of 20 mg
> Duloxetine	None	Generalized Anxiety Disorder: 7-17 years	Starting dose: 30mg daily Recommended dose: 30mg daily; max dose: 120mg daily
> Venlafaxine	None	None	
> Mirtazapine	None	None	
> Lurasidone	Depressive episodes associated with bipolar I disorder: 13-17 years Schizophrenia: 15-17 years	Depressive episodes associated with bipolar I disorder: 10-17 years Schizophrenia: 13-17 years	Bipolar I depression starting dose: 20mg po daily; recommended dose 20mg-80mg daily. Schizophrenia starting dose: 40mg daily; recommended dose: 40mg-80mg daily
> Prochlorperazine	≥ 2 years for or > 9 kg 1) psychotic disorders (agitation, confusion, delusion, tension, and anxiety) 2) nausea and vomiting 3) relief of excessive anxiety associated with psychoneurotic or somatic conditions	≥ 2 years or children > 9 kg for 1) nonsurgical nausea and vomiting 2) psychosis	Antiemetic: 0.4 mg/kg/day Psychosis: 2.5 mg orally increase as needed to 20 mg/day (off label dosing)
Cognitive Problems			
> Methylphenidate, extended release (Concerta®, Biphentin®)	Treatment of ADHD in children 6 years of age and older.	Treatment of ADHD in children 6 years of age and older for Concerta® only; Biphentin® not marketed in the USA.	<i>Concerta®</i> Children and adolescents: Initial - 18mg once a day. Increase daily dose by 18 mg at weekly intervals. Maximum of 54 mg in children and 72 mg in adolescents. <i>Biphentin®</i> Children and adolescents: Initial – 10-20 mg once daily. Increase daily dose by 10 mg/day at weekly intervals. Typical maximum of 60 mg/day. However, manufacturer states in some children, higher doses up to a maximum of 1 mg/kg/day may be necessary.

Drug	Health Canada Approval	FDA Approval	Dosage#
> Concerta (OROS Methyphenidate Hydrochloride)	ADHD	ADHD	Children and adolescents between 6-18 years old. Starting dose: 18mg po qAM, with increases weekly as needed and as tolerated to a max of 54mg.
>Dextroamphetamine (Dexedrine®) Dextroamphetamine, sustained release (Dexedrine Spansules®)	Treatment of ADHD and in the adjunctive treatment of narcolepsy in children 6 years of age and older.	Treatment of ADHD and in the adjunctive treatment of narcolepsy in children 6 years of age and older.	<i>ADHD</i> Children 6 years of age or older: Initial - 5 mg once or twice daily. Increase the daily dose by 5 mg at weekly intervals. *Dexedrine Spansules® may be used for once-a-day dosing where appropriate. Typical maximum of 40 mg/day, but may go up to 60 mg/day if needed. <i>Narcolepsy</i> Children 6-12 years: Initial- 5 mg daily. Increase daily dose by 5 mg at weekly intervals. Maximum of 60 mg/day. Adolescents 12 years or older: Initial - 10 mg daily. Increase daily dose by 10 mg at weekly intervals. Maximum of 60 mg/day.
> Lisdexamphetamine (Vyvanse®)	Treatment of ADHD in children 6 years of age and older.	Treatment of ADHD in children 6 years of age and older.	Children and adolescents: Initial – 20-30 mg once a day in the morning. Increase daily dose by 10-20 mg at weekly intervals. Max of 60 mg/day (Canada). Max of 70 mg/day (US).
> Clonidine (Catapres®)	No indication <18 years of age. No approved indications for ADHD at any age. Used off-label in children 6 years of age and older for ADHD treatment as monotherapy or adjunctive to stimulant medications.	Extended-release clonidine (Kapvay®) approved for ADHD treatment in children 6 years of age and older. *note: extended-release clonidine is not available in Canada.	Off label dosing: Children less than 45 kg: Initial – 0.05 mg at bedtime. Increase every 3-7 days in 0.05 mg/day increments given as 0.05 mg BID, then TID, then QID. Maximum daily dose for 27-40.5 kg = 0.2 mg/day. Maximum daily dose for 40.5-45 kg = 0.3 mg/day. Children and adolescents greater than 45 kg: Initial – 0.1 mg at bedtime. Increase every 3-7 days in 0.1 mg/day increments given as 0.1 mg BID, then TID, then QID. Max daily dose of 0.4 mg/day.

Drug	Health Canada Approval	FDA Approval	Dosage#
> Amphetamine Mixed Salts, extended-release (Adderall XR®)	Treatment of ADHD in children 6 years of age and older.	Treatment of ADHD in children 6 years of age and older.	Children 6-12 years of age: Initial – 5-10 mg daily in the morning. Increase daily dose by 5-10 mg at weekly intervals. Maximum of 30 mg/day. Adolescents 13-17 years of age: Initial – 10 mg once a day in the morning. Increase daily dose by 5-10 mg at weekly intervals. Maximum of 30 mg/day.
> Atomoxetine (Strattera®)	Treatment of ADHD in children 6 years of age and older.	Treatment of ADHD in children 6 years of age and older.	Children up to 70 kg: Initial: 0.5 mg/kg/day for 7-14 days. If tolerated, increase to 0.8 mg/kg/day for 7-14 days, then increase to 1.2 mg/kg/day given once daily or divided BID. Maximum 1.4 mg/kg/day or 100 mg, whichever is less. Adolescents greater than 70 kg: Initial – 40 mg/day for 7-14 days. If tolerated, increase to 60 mg/day for 7-14 days, then increase to 80 mg/day given once daily or divided BID. Maximum 100 mg/day.
> Guanfacine, extended-release (Intuniv XR®)	Monotherapy or as adjunctive to stimulant medications for the treatment of ADHD in children 6 years of age and older.	Monotherapy or as adjunctive to stimulant medications for the treatment of ADHD in children 6 years of age and older.	Children 6-12 years of age: Initial – 1 mg once daily. Increase daily dose by 1 mg at weekly intervals. Maximum of 4 mg/day. Adolescents 13-17 years of age: Initial – 1 mg once daily. Increase daily dose by 1 mg at weekly intervals. Maximum of 7 mg/day for monotherapy, or 4 mg/day for adjunctive therapy to stimulants.

Tool 6.3: Approved Medications for Pediatric Indications (cont'd)

Drug	Health Canada Approval	FDA Approval	Dosage#
> Amantadine	No indications less than 18 years of age for ADHD.	No indication less than 18 years of age for ADHD.	(off label use in children 5 years of age and older as an adjunctive agent to stimulants for ADHD treatment, and to target irritability and hyperactivity in ASD) ADHD Children 5 years of age and older: Initial – 50 mg/day. Increase daily dose by 50 mg at 4-7 day intervals. Typical range: 50-150 mg/day in divided doses 1-3 times daily. Maximum of 100 mg/day in children less than 30 kg. Maximum of 150 mg/day in children and adolescents 30 kg and greater.

Abbreviations: Attention deficit hyperactivity disorder (ADHD), autism spectrum disorder (ASD); Per os (po); bis in die (two times a day- BID); hour (h); IV (intravenous); omne in die -once a day (od); every morning (qAM); ter in die- three times a day (TID); as needed dosing (prn).

Dosages from Sick Kids 2013/2014 Drug Handbook and Formulary, Lexicomp Pediatric Dosage Handbook 19th Edition, CHEO Pediatric Doses of Commonly Prescribed Medications 2011, <https://www.accessdata.fda.gov/scripts/cder/daf/>, Clinical Handbook of Psychotropic Drugs for Children and Adolescents, 4th ed. 2019.



TOOL 7.1: Prolonged Post-concussion Sleep Disturbances Algorithm

Medical follow-up

(Sleep disturbances >1-4 weeks)

- Screen for medical conditions, current medication use, comorbid psychopathology, and risk factors for sleep disturbances (Sidebar 1)
- Reinforce the importance of sleep hygiene and physical and cognitive activity
- Assess return to school and return to sport status
- Consider treating with cognitive behavioural therapy (CBT) or refer to interdisciplinary concussion team if symptoms last longer than 4 weeks (or sooner as needed/using clinical judgment)

Sleep disturbances persist > 4 weeks post-concussion

Cognitive behavioural therapy or referral to healthcare professionals/ interdisciplinary concussion team

1. Cognitive behaviour therapy (CBT): treatment of choice
2. Daily supplements: magnesium, melatonin, and zinc
3. If CBT is not available: monitor behavioural interventions weekly, consider referral to an interdisciplinary concussion clinic or occupational therapist

Sleep disturbances persist > 6 weeks post-concussion and interventions at 4-weeks have been unsuccessful

Referral to healthcare professionals/ interdisciplinary concussion team

Refer to a sleep specialist for consultation

Particularly if sleep hygiene can't be optimized and poor sleep quality is impacting ability to return to school or ability to recondition.

Ideal specialist: a specialist who has experience with mTBI and polysomnography.

Pharmacological treatment

Consider prescribing medication on a short-term basis if sleep has not improved > 6 weeks post-injury. Ensure that medications do not result in dependency and that patient has minimal adverse effects. The aim is to establish a more routine sleep pattern. (Sidebar 2)

SIDEBAR 1

Factors that may influence the child/adolescent's sleep/wake cycle

> Medical conditions

- Endocrine dysfunction
- Metabolic dysfunction
- Obesity
- Enlarged tonsils
- Obstruction
- Sleep-related breathing disorders
- Early morning headaches

> Current medication use

Verify if patient's prescribed or non-prescribed medications impact sleep:

- Inadequate medication
- Dosage
- Timing of administration
- Screen for stimulants

> Comorbid psychopathology

- Mood and anxiety disorders
- Post-traumatic stress disorder (PTSD)
- Query nightmares and/or night terrors

> Unhealthy habits

- Lack of exercise
- Variable sleep-wake schedule
- Excessive napping
- Excessive time spent in bed
- Exercising close to bedtime
- Screen time
- Use of nicotine, caffeine, energy drinks, processed foods, processed sugars, alcohol and drugs

SIDEBAR 2

Potential medication options – short-term basis only

- Trazodone 12.5 mg
- Amitriptyline 5.0 – 10.0 mg



TOOL 7.2: Factors that may Influence the Child/Adolescent's Sleep/Wake Cycle

Examples of factors and considerations

Medical Conditions	• Endocrine dysfunction • Metabolic • Obesity • Enlarged tonsils • Obstruction • Sleep-related breathing disorders (possibly sleep apnea) • Early morning headaches
Current Medication Use	Verify if prescribed or non-prescribed medications that are being taken have an impact on sleep and screen for stimulants. Factors to consider: • Type of medication (adequate/inadequate type) • Dosage • Timing of administration
Comorbid Psychopathology	• Mood and anxiety disorders (Domain 8: Mental health considerations) • Post-traumatic stress disorder (PTSD) • Query nightmares and/or night terrors
Unhealthy Habits	• Lack of exercise • Variable sleep-wake schedule • Excessive napping • Excessive time spent in bed • Exercising close to bedtime • Screen time (hours per day) • Use of nicotine, caffeine, and/or energy drinks • Diet: Consumption of processed foods, processed sugars, alcohol, and/or drugs

Adapted from Guidelines for Concussion/Mild Traumatic Brain Injury and Persistent Symptoms, Third Edition



TOOL 8.1: Post-concussion Mental Health Considerations Algorithm

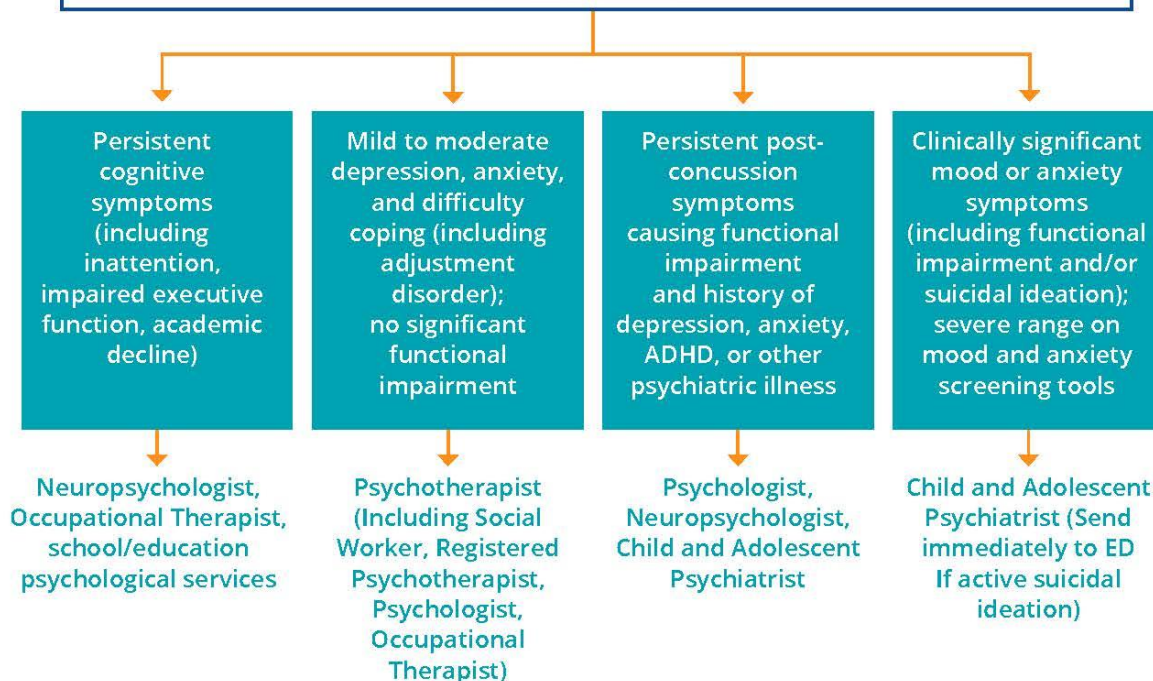
Medical follow-up and referral to healthcare professionals/interdisciplinary concussion team (1-4 weeks following acute injury)

- Focused clinical history, physical examination, determine need for imaging
- Screen for possible complicating factors that may impede recovery e.g. previous mental illness, family history of mental illness, migraine, current stress level, and any other modifiers that may delay recovery. Consider early referral to specialist or interdisciplinary team.
- Screen for mood, anxiety, and cognitive symptoms
- Send immediately to the emergency department (ED) if active suicidal ideation
- Review pre-injury mental health status (including pre-injury symptoms or diagnoses of depression, anxiety disorders, ADHD, and behavior disturbances)
- Review previous school history (attendance, learning, behavior) using information from school records if possible
- Post-injury education and guidance on symptom management (including advice regarding pacing of activities and general recovery expectations within the context of pre-existing circumstances)
- Refer to healthcare professionals/interdisciplinary concussion team if symptoms last longer than 4 weeks (or sooner as needed/using clinical judgment)

Consider early referral (< 4 weeks) if child/adolescent has modifiers that may delay recovery/high risk of prolonged post-concussion symptoms

Not yet recovered Symptoms lasting > 4 weeks post-concussion

Specialist or interdisciplinary concussion team

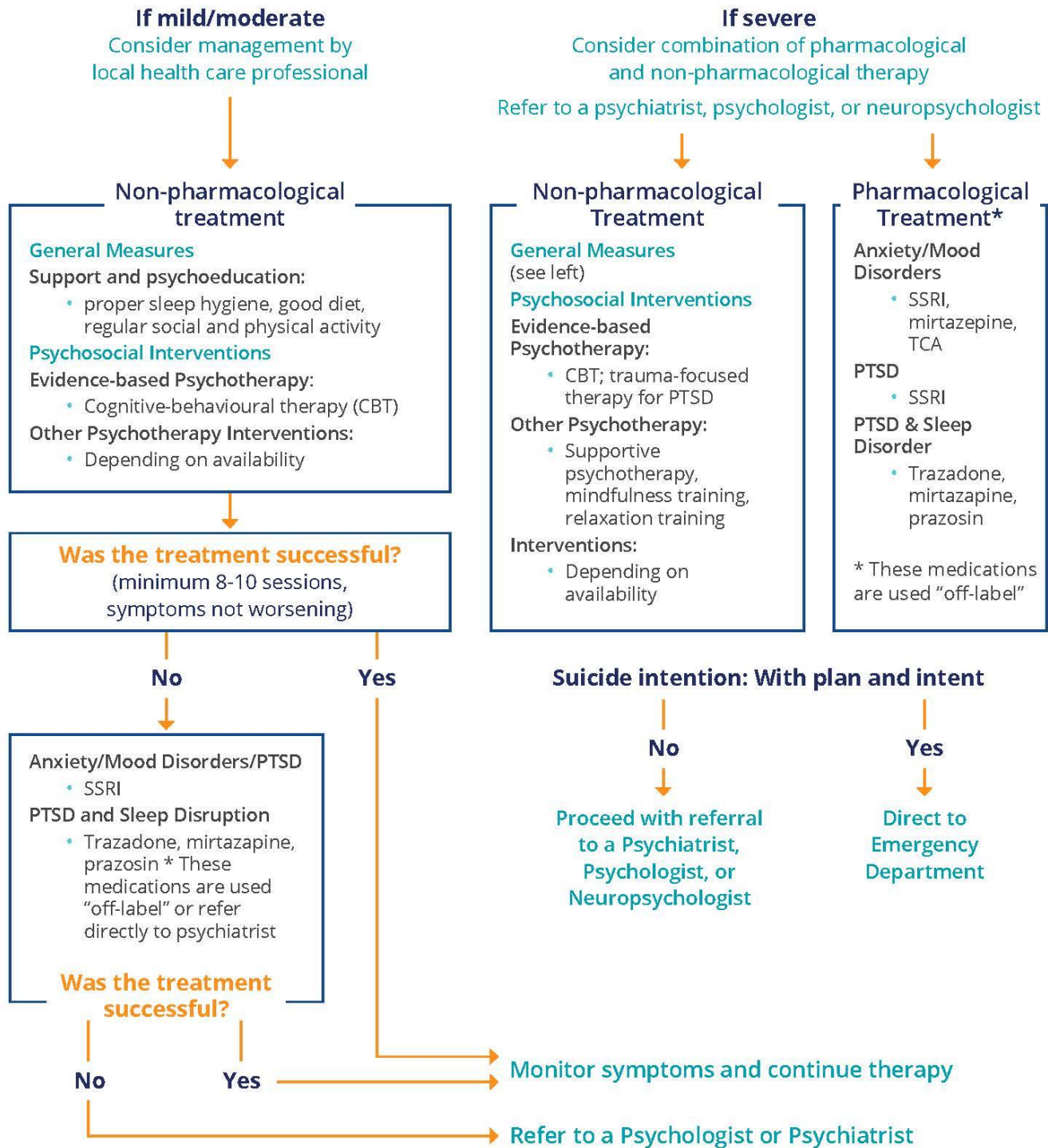


Acronyms: attention deficit hyperactivity disorder (ADHD)

Living Guideline for Diagnosing and Managing Pediatric Concussion



TOOL 8.2: Management of Prolonged Mental Health Disorders Algorithm



Adapted from *Guideline for Concussion/Mild Traumatic Brain Injury and Persistent Symptoms*, Third Edition
Acronyms: Post-traumatic stress disorder (PTSD); Selective serotonin reuptake inhibitors (SSRI); Tricyclic antidepressants (TCA)



TOOL 10.1: Post-concussion Vision, Vestibular, and Oculomotor Disturbances Algorithm

Medical follow-up and referral to healthcare professionals/interdisciplinary concussion team

(1-4 weeks following acute injury)

- Focused vision and vestibular history, physical examination, determine need for imaging
- Post-injury education and guidance on symptom management
- Refer to healthcare professionals/interdisciplinary concussion team if symptoms last longer than 4 weeks (or sooner as needed/using clinical judgment), or if the child has modifiers that may delay recovery
- Consider particle re-positioning for BPPV (e.g. Epley)

Consider early referral (< 4 weeks) if child/adolescent has modifiers that may delay recovery/ high risk of prolonged post-concussion symptoms

Not yet recovered

Symptoms lasting >4 weeks post-concussion

Healthcare professionals/interdisciplinary concussion team

- Medical assessment by physician with expertise in concussion
- Adjunctive testing (graded aerobic exercise testing, formal vestibular testing, automated visual field testing)
- Interdisciplinary management of vision and vestibular disorders

BPPV, vestibular hypofunction, balance

Physiotherapist*

Central vestibulopathy SSCD

Otolaryngologist/
Neuro-Otologist,
Neuro-Ophthalmologist,
Physiotherapist*

Vision or oculomotor disorder

Neuro-Ophthalmologist,
Neuro-Optometrist,
Physiotherapist*

Cervicogenic dizziness

Physiotherapist*,
Chiropractor*

Acronyms: paroxysmal positional vertigo (BPPV); superior semicircular canal dehiscence syndrome (SSCD)

*denotes health care professional with competency-based training in vestibular or visual system rehabilitation



TOOL 12.1: Concussion Implications and Interventions for the Classroom

Area of concern after concussion	Possible classroom behaviour that indicates difficulty	Proactive solution for student
> Physical	Headaches in school	Allow for rest breaks; hydration; quieter location.
	Dizziness	Provide calm environment without multiple distractions.
	Fatigue: Sleeping more or less than usual	Allow for rest breaks; plan daily schedule with a variety of classes with different levels of difficulty interspersed (e.g. difficult vs less difficult).
	Vision problems with sensitivity to lights	Avoid fluorescent lights when possible; decrease brightness on computers, smart phones, tablets; limit time on computers; electronic gaming devices.
	Hearing issues with noise (gyms, concerts, music causing irritability)	Avoid loud activities, including gym classes, loud music from headphones, dances or parties.
> Cognitive-communication	Unable to concentrate	Provide information in smaller chunks; assign less material for task completion.
	Forgetting recently learned information or conversations	Allow for written and verbal cues; use note taker or provide written notes for reference.
	Slow to process information (verbally or written)	Allow additional time for response; provide alternative methods for test taking; breakdown complex directions into simple steps; decrease length of assignments or do not give any.
	Asks for repeats	Provide written and verbal cues and notes; post schedule and assignments.
	Word finding/naming (vocabulary) difficulty	Pre-teach new vocabulary; cue using categorization and association.
	Poor social interactions with others (easy to anger, rude, interrupts with irrelevant information)	Do not punish unacceptable behaviours; work on social skills one on one.

Area of concern after concussion	Possible classroom behaviour that indicates difficulty	Proactive solution for student
> Cognitive-communication	Confused about recent activities/ experiences	Use technology (tablets, smart phones) to record cues as activities occur; use day organizer for reference.
	Unable to attend to task completion	Break tasks into smaller pieces so each portion is successfully completed.
> Social/emotional/behavioural	More irritable in the classroom	Provide clear schedule for the day; recognize beginning of irritable behaviour and provide alternative activity.
	Feeling anxious or tense	Provide time for rest during the day; provide clear schedule of activities for the day; less required work.
	Feeling depressed	Point out strengths and successes during each day.
	Easily overwhelmed by school requirements or activities	Diminish schedule to reasonable load that can be successfully accomplished; add activities only when success is demonstrated; decrease or eliminate homework.

Adapted from DePompei, R.& Blosser J. *Pediatric traumatic brain injury: proactive interventions*. (2019). Plural Press: San Diego.



TOOL 12.2: Template for Concussion Team's Letter to the Child/Adolescent's School

Name:

Date:

School:

Address:

Re: Concussed student name

DOB:

Dear _____,

This letter is to update the school regarding _____ (Student's name) recovery from a concussion sustained on _____ (Date). We take this opportunity to let you know what to expect during the recovery period.

Current best practice guidelines on concussion management emphasize the importance of recovery and return to function, and encourages students to return-to-school as soon as possible, with accommodations, even though they may still be symptomatic.

Please be aware of signs and symptoms the student may display: (Check those relevant to this student)

- | | |
|--|---|
| ☐ Decreased short term memory | ☐ Decreased attention span |
| ☐ Slower processing speed | ☐ Irritability |
| ☐ Fatigue | ☐ Headaches |
| ☐ Photo/audio sensitivity | ☐ Other _____ |

In the classroom, students with these symptoms may exhibit difficulty paying attention, difficulty following lessons, and sensitivity to noisy/busy classrooms and environments (e.g. assemblies, shop classes).

We have advised your student to pace him/herself with regards to their return-to-school and to use accommodations at school as needed. The following accommodations are recommended: (Check the recommendations for this student)

- ☐ Allowed to take short breaks from the classroom; access to a quiet workspace
- ☐ Class notes provided to assist focus and reduce writing/copying demands
- ☐ Allowed to photograph the teacher's board notes with a smartphone
- ☐ Extended deadlines; modified demands for homework/assignments
- ☐ Access to technology as needed (e.g. audiobooks, Google read and write)
- ☐ Chunking of tests/exams; allowed extra time and quiet, alternative setting
- ☐ Close monitoring and communication with the student regarding his/her progress
- ☐ Rest periods during the day as needed
- ☐ It is recommended that all non-essential missed workload expectations be forgiven to allow student to prioritize essential knowledge building areas and focus on current curriculum rather than missed work
- ☐ Other _____

Thank you for your assistance in supporting _____ (Student's name). The parents and the concussion team will monitor the student and work with the school, as accommodations need to be adjusted to support the student's recovery.

Please do not hesitate to contact us with any questions or concerns regarding this information, or if we can be of further assistance.

Sincerely,

Adapted with permission from the authors: Sinclair Elder AJ, Kadel R, O'Keefe EK. Headin' for Healin' Teacher's Letter. Colorado Springs, CO: University of Colorado Colorado.

Living Guideline for Diagnosing and Managing Pediatric Concussion



TOOL 12.3: Template for Physician's Letter to the Child/Adolescent's School

Name _____

Date: _____

School _____

Address _____

Re: Concussed student name _____

DOB: _____

Dear _____,

_____ (Student's name) was diagnosed with a concussion on _____ (Date), and has now been medically cleared to return-to-school.

Current best practice guidelines on concussion management emphasize the importance of recovery and return to function, and encourages students to return-to-school as soon as possible, with accommodations, even though they may still be symptomatic.

Please identify a "point person" within the school environment (example: guidance counsellor, homeroom teacher, vice principal) who can help the student implement a return-to-school plan, communicate it to teachers/school staff involved in the student's activities, and modify it as needed. The attached document outlines accommodations that may be helpful while your student is reintegrating back into the school environment.

Feel free to contact me with additional questions.

Sincerely,



TOOL 12.4: Sample Letter/Email from School to Parents

Dear _____ (Parents' names)

We are happy to hear that your child is feeling well enough to start the return-to-learn process after his/her concussion. To make sure teachers and staff are prepared, we would like your insight on the following symptoms. Please check the answers that best fit your child.

Fatigue

My child ☐ tires easily ☐ has the normal amount of energy.

My child has the most energy in the ☐ morning ☐ afternoon ☐ evening.

Behaviour

My child ☐ is easily frustrated ☐ isn't easily frustrated.

My child has been acting ☐ the same ☐ different compared to before concussion.

Memory

My child's memory seems ☐ fine ☐ impaired.

Cognition

My child seems to be able to understand complex thoughts and ideas. ☐ Yes ☐ No

My child is able to read for ☐ less than ½ hour ☐ ½ to 1 hour ☐ more than 1 hour.

My child can handle different technologies (example: TV, computers). ☐ Yes ☐ No

My child can complete some homework. ☐ Yes ☐ No

Stamina

My child makes it through a day without a period of rest. ☐ Yes ☐ No

Social

My child is becoming isolated or has different friends than before the concussion. ☐ Yes ☐ No

My child can handle noisy/busy environments. ☐ Yes ☐ No

Awareness

My child feels like there is nothing wrong with him/her after the concussion. ☐ Yes ☐ No

My child understands that there have been changes and would like help. ☐ Yes ☐ No

Please elaborate on any other changes you've noticed in your child. We want to be ready to manage your child's return-to-learn process and make accommodations to ensure success.

Sincerely,

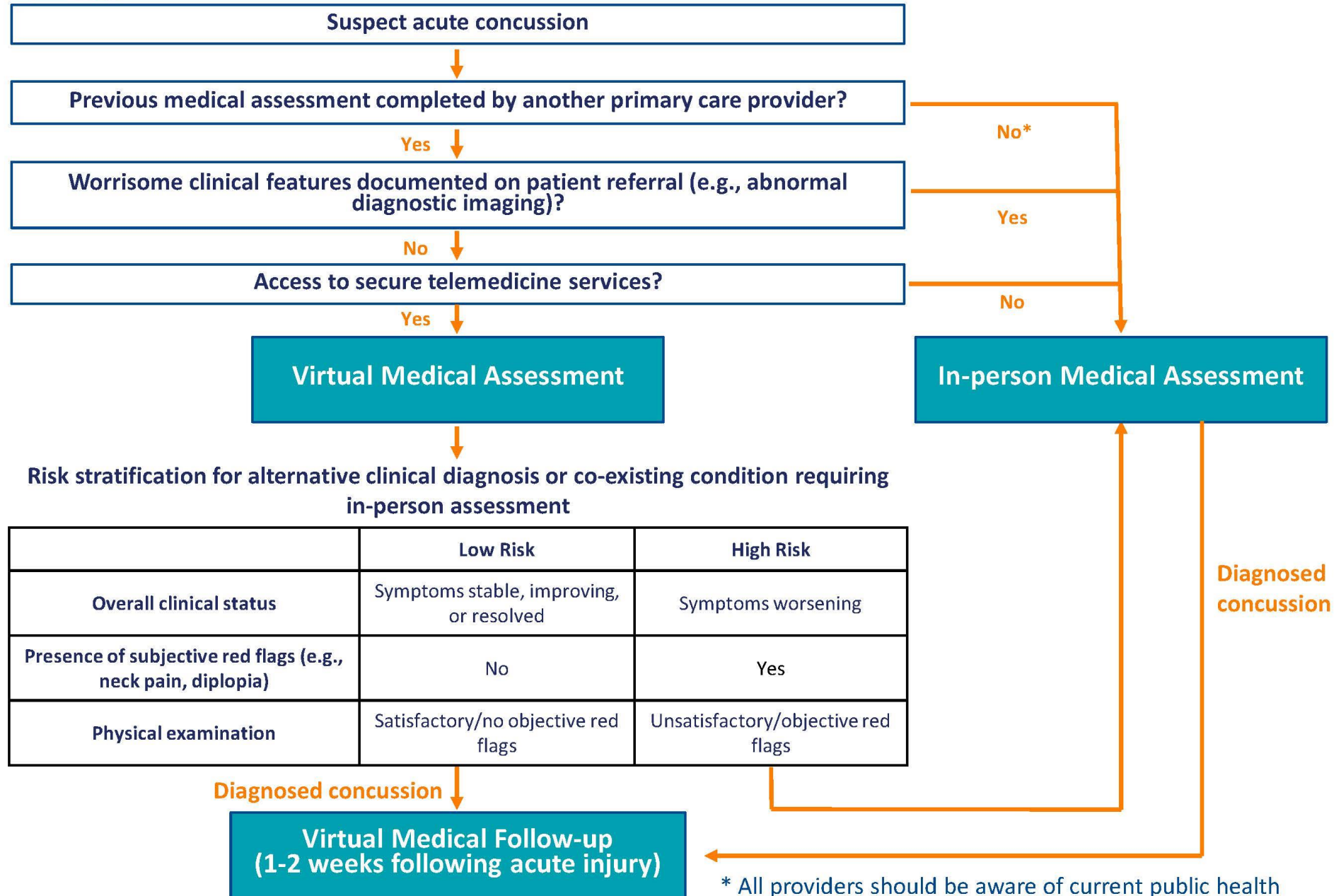
_____ (school contact person's name)

Telephone/email _____

Source: [Developed by Vermont's concussion task force, used with permission from the Brain Injury Association of Vermont.](#)



Tool 15.1 Considerations for Telemedicine and Virtual Care Algorithm



Adapted with permission from: Ellis M, Mendez, Russell K. (2020). Preliminary clinical algorithm to optimize delivery of remote pediatric concussion care in Canada's North. *International Journal of Circumpolar Health* (published online October 22, 2020) <https://doi.org/10.1080/22423982.2020.1832390>



Tool 15.2: Considerations for a virtual physical examination for medical assessment and follow-up of concussion patients

All patients presenting with a suspected acute concussion should undergo a complete physical examination that includes a neurological examination and cervical spine examination, as well as examination for any other traumatic injury. Depending on the patient's presenting or prolonged symptoms, supplementary physical examination tests can be considered. This also depends on the healthcare professional's experience in performing and interpreting these objective tests.

See [Tool 2.1: Physical Examination](#) for a comprehensive list of tests for the neurological and cervical spine examination.

Due to conditions such as the visual quality of in-person videoconferencing and environmental factors (e.g., lighting, examination room size and configuration) some of the components of the virtual physical examination recommended here may not be able to be performed during all patient assessments. **Providers should use their clinical discretion when performing and interpreting components of the virtual physical examination.**

Mental status and speech:

Mental status: Document if patient is alert and oriented to date and place.

- Alternatively document Glasgow Coma Scale score.

Speech: Speech may be assessed during clinical interview or on examination.

- Abnormal: document any evidence of dysarthria, word finding difficulty, apraxia, aphasia etc.

Modified cranial nerve exam:

Extraocular movements: Ask the patient to position themselves close to the camera. Ask the patient to look directly at the camera. Observe ocular alignment in the primary position and assess for any evidence of ptosis or nystagmus. Ask the patient to keep their head still and hold their finger up at eye level approximately half an arm's length away from their face. While following their finger with their eyes, ask the patient to slowly move their finger to the right and left and then trace an "H" allowing the examiner to evaluate the six cardinal positions of gaze. Ask the patient if they experience any diplopia during gaze in any direction. Then ask the patient to visually track their finger by moving it slowly back and forth in the horizontal direction with the head stationary, making sure not to go more than about 30 degrees from neutral to avoid eliciting end-gaze nystagmus. This can also be repeated in the vertical plane.

- *Abnormal:* Note any abnormal alignment of the eyes in the primary position or any restriction in eye movement during testing. Note any patient-reported diplopia during testing.

Smooth Pursuits: Ask the patient to position themselves close to the camera. Ask the patient to look directly at the camera. Ask the patient to visually track his/her finger by moving it slowly back and forth in the horizontal direction with the head stationary, making sure not to go more than about 30 degrees from neutral to avoid eliciting end-gaze nystagmus. The patient may be cued to move closer to or farther from the camera or vary the plane of their finger to enable optimal viewing of the patient's eyes. This may also be repeated in vertical visual plane.

- **Abnormal:** Abnormal findings include sustained beats of nystagmus, staccatic (i.e., jerking) eye motion, loss of conjugate vision, corrective (catch-up or back-up) saccades, loss of visual fixation OR symptom provocation (dizziness, nausea or headache).

Facial symmetry: Ask the patient to look directly at the camera. Ask the patient to smile. Ask the patient to elevate their eyebrows and then close their eyes tightly.

- **Abnormal:** Note any facial asymmetry at rest or during testing.

Facial sensation: Ask the patient to take the index finger of each of their hands and lightly touch their face along the ipsilateral V1, V2, and V3 distributions of the trigeminal nerve. Ask the patient to indicate whether sense of touch is perceived in all distributions and is equal when comparing both sides.

- **Abnormal:** Note any evidence of decreased sensation.

Movement of palate and tongue: Ask the patient to position themselves close to the camera. Ask the patient to open their mouth and say “ah”. Ask the patient to stick out their tongue, move it to the right and left, and retract it.

- **Abnormal:** Note any asymmetry or abnormal findings. Note any restricted movement of the jaw/mouth opening.

Motor/coordination:

Pronator drift: In a standing position, ask the patient to raise their arms forward to shoulder level (90 degrees of shoulder flexion) and position their hands with palms facing upwards. Ask the patient to close their eyes and maintain this position.

- **Abnormal:** Note any pronation of the hand or downward drift of the arm.

Rapid alternating hand movements: In a standing position, ask the patient to place one hand palm upwards in front of them and place the opposite hand palm down on top of the other hand. Ask the patient to rapidly flip the top hand quickly back and forth from a palm up and down position. Ask the patient to switch hands and repeat the same movements.

- **Abnormal:** Note any differences in speed or coordination between sides.

Cervical spine

In a standing or seated position ask the patient to slowly flex their neck forward, extend their neck backward, and bend their neck to the left and right as far as comfortably possible. Ask the patient to slowly rotate their neck to the left and right as far as comfortably possible. Ask the patient whether they experience any pain during any movements.

- **Abnormal:** Note any restricted range of motion or any reported pain.

Balance

Balance tests may be performed near a supporting caregiver or object (e.g., chair/wall).

Feet together stance: With the patient positioned so their entire body is visible to the examiner, ask the patient to stand with their feet together and their hands on their hips. Ask the patient to close their eyes and hold the position for 20 seconds.

- **Abnormal:** Note any instability, stumbling out of position or eye opening.

Tandem stance: With the patient positioned so their entire body is visible to the examiner and near a supporting caregiver or object (e.g., chair/wall), ask the patient to stand with their

dominant foot slightly in front of their non-dominant foot with their hands on their hips. Ask the patient to close their eyes and hold the position for 20 seconds.

- *Abnormal:* Note any instability, stumbling out of position or eye opening.

Tandem gait (eyes open): With the patient positioned so their entire body is visible and positioned perpendicular to the examiner, ask the patient to walk in the forward direction heel-to-toe with their eyes open for 5 paces.

- *Abnormal:* Note any instability or stumbling out of position

Tandem gait (eyes closed): With the patient positioned so their entire body is visible to the examiner, ask the patient to walk in the forward direction heel-to-toe with their eyes closed for 5 paces.

- *Abnormal:* Note any instability, stumbling out of position or eye opening.
- *Note:* Consider if tandem gait is normal.

Memory:

Explain to the patient that you are going to test their memory. Provide the patient with a list of five items. Ask the patient to immediately repeat the items back in any order. Record the number of correctly repeated items out of five (immediate recall). After a set time period (e.g., five minutes) has elapsed, ask the patient if they can remember the five items provided. Record the number of correctly repeated items out of five (delayed recall).

Vestibulo-ocular:

Horizontal saccades: Ask the patient to position themselves close to the camera. Ask the patient to hold up their index fingers and place them at half arm length away from the face and 30 cm apart from each other. While keeping their head still, ask the patient to look quickly and accurately back and forth between each index finger for a total of 10-15 repetitions. Ask the patient whether performing these eye movements worsen or elicit any concussion symptoms (e.g., dizziness, headache)

- *Abnormal:* Note evidence of over- or undershooting of the target or saccadic corrections. Note slowed or delayed initiation of saccades.
- *Symptomatic:* Note whether the movements elicit or worsen any concussion-like symptoms (e.g., dizziness, headache)

Gaze stabilization: Ask the patient to hold their thumb up at the level of the eyes and at arm length distance from the face. Ask the patient to fix their eyes on their thumb and quickly but comfortably shake their head back and forth within 15-20 degrees of rotation for 10 repetitions.

- *Symptomatic:* Note whether the movements elicit or worsen any concussion-like symptoms (e.g., dizziness, nausea)
- *Note:* Do not perform test unless patient has full and painless range of motion of the cervical spine.

Reference

Modified from: Ellis M, Mendez, Russell K. (2020). Preliminary clinical algorithm to optimize delivery of remote pediatric concussion care in Canada's North. *International Journal of Circumpolar Health* (published online October 22, 2020)
<https://doi.org/10.1080/22423982.2020.1832390>